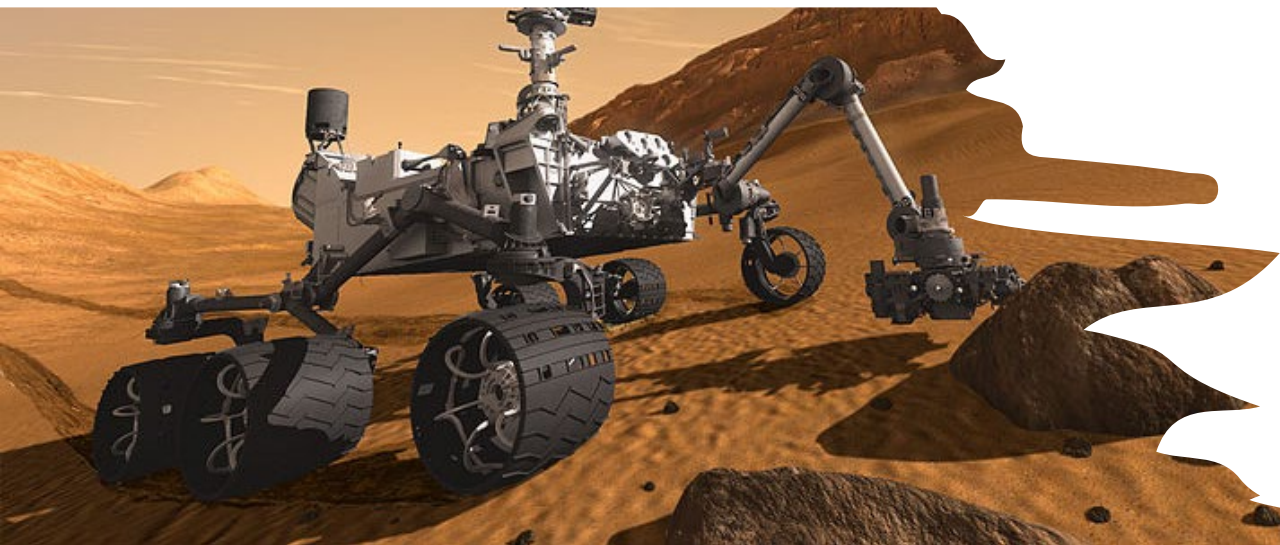
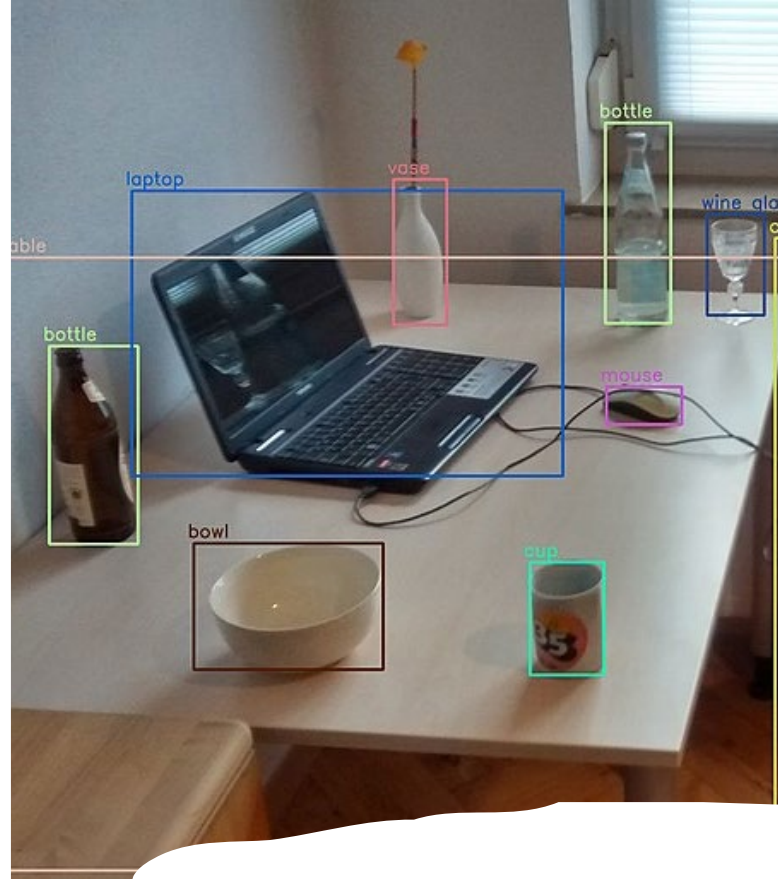
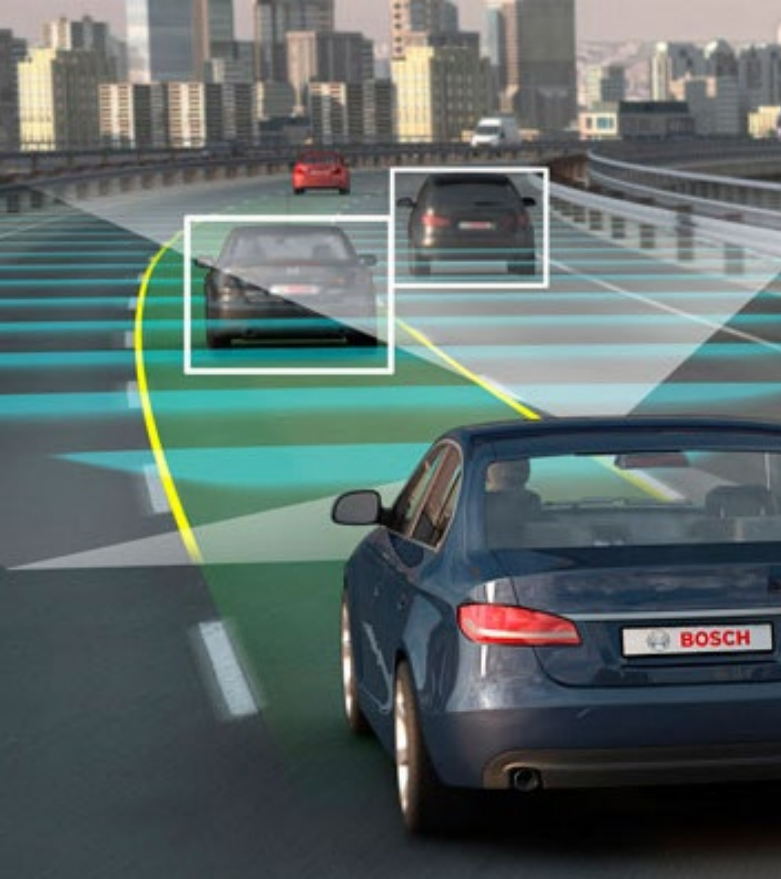


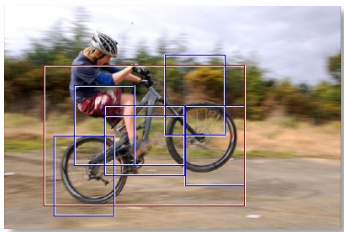
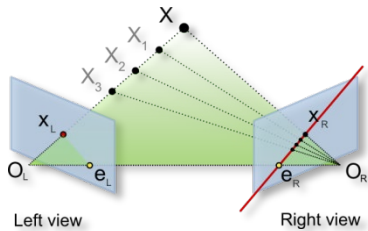
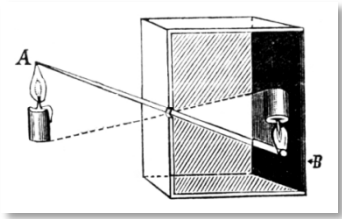


DT8055

Fundamentals of Computer Vision with Deep Learning

L2: Image perception, representation

Course Overview



1. Low-level vision: *primitive* operations

- Image acquisition and representation, transformations, enhancements, convolution and filtering, edges and lines...

2. Mid-level vision: *attribute* extraction

~~— Segmentation, depth, motion, 3D... (not covered)~~

- Feature extraction, classification, image matching...
- Machine Learning (ML), Convolutional Neural Nets (CNN)

3. High-level vision: *making sense* of the image

- Object / category / activity... detection and recognition

Acknowledgements

Many thanks for the materials to:

- CS5670 (Intro to CV, Cornell Tech: Noah Snavely)
- CSE576 (Computer Vision, University of Washington: Steve Seitz, Richard Szeliski, Linda Shapiro)
- <https://www.imageprocessingplace.com> (Gonzalez & Woods)
- <http://szeliski.org/Book> (Richard Szeliski)

and their contribution to having prepared this course in a finite time, thus keeping sanity of the teacher under control

Today

1. Visual perception, camera factors
2. Image sensing, sampling and quantization
3. Spatial and intensity resolution
4. RGB and gray-scale image acquisition and representation
5. Other image acquisition sources (non-visible light)
6. Color spaces to represent images
7. Histogram

Readings for today

- Szeliski, Chapter 2.3 (before 2.3.1) and Chapter 2.3.2 (color)
- Gonzalez & Woods, Ch 2 until interpolation (not included), Ch 6.1, 6.2 (color models)

Visual Perception

Camera Factors

Visual Perception: Eye Sensing Elements

- **Rods** (sensitive to the amount of light, work with very low light level)
- **Cones** (sensitive to color, do NOT work with low light level)

FIGURE 2.1
Simplified
diagram of a
cross section of
the human eye.

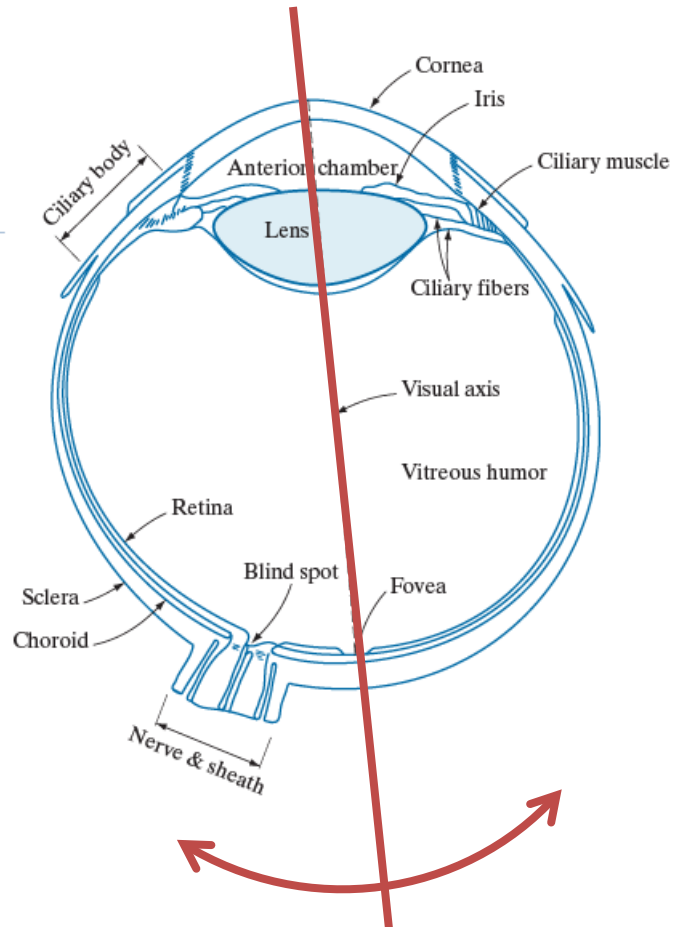
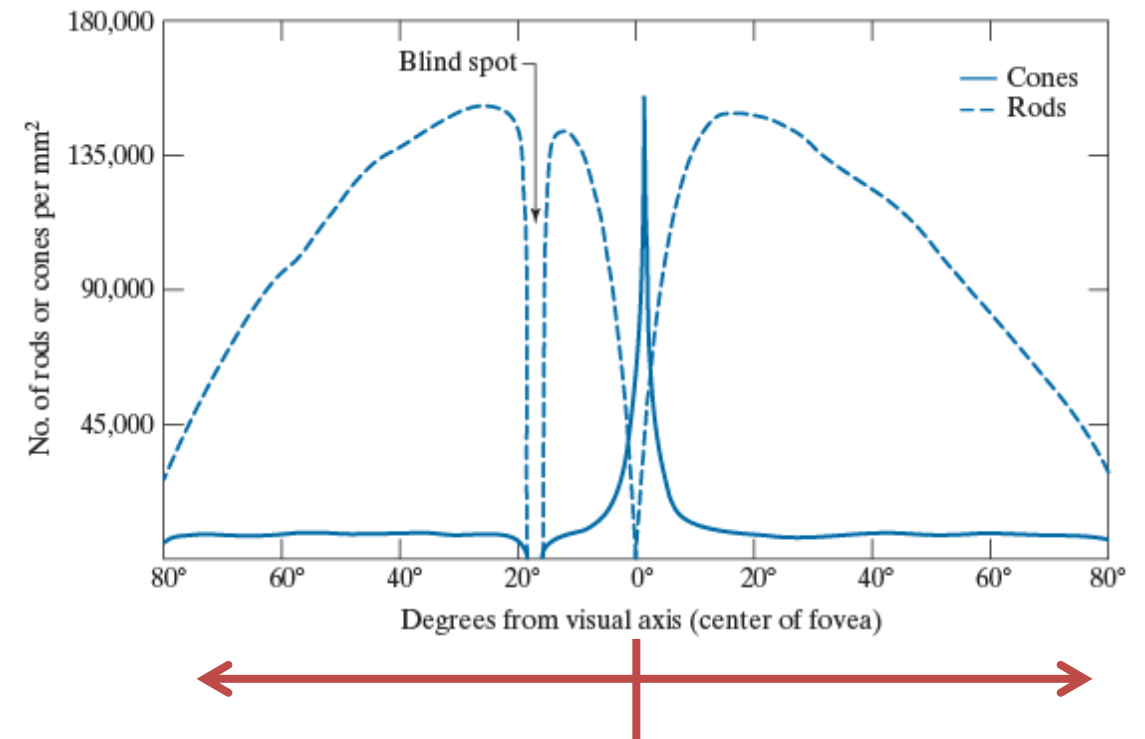


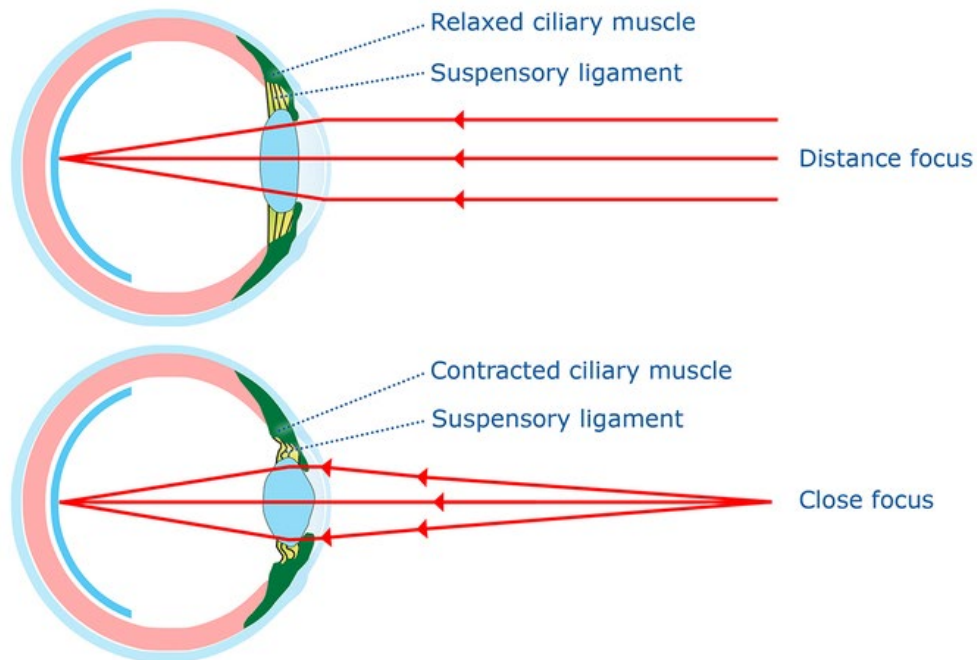
FIGURE 2.2
Distribution of
rods and cones in
the retina.



Visual Perception: Lens Focal Distance

- **Focal distance:** distance (mm) between the "sensor" and the "optical center"
- Eye typical range: 14-17mm changed by **accommodation** due to contraction/dilation of ciliary muscles (14mm very close objects... 17mm for far away objects >3m)

How the eye focuses light

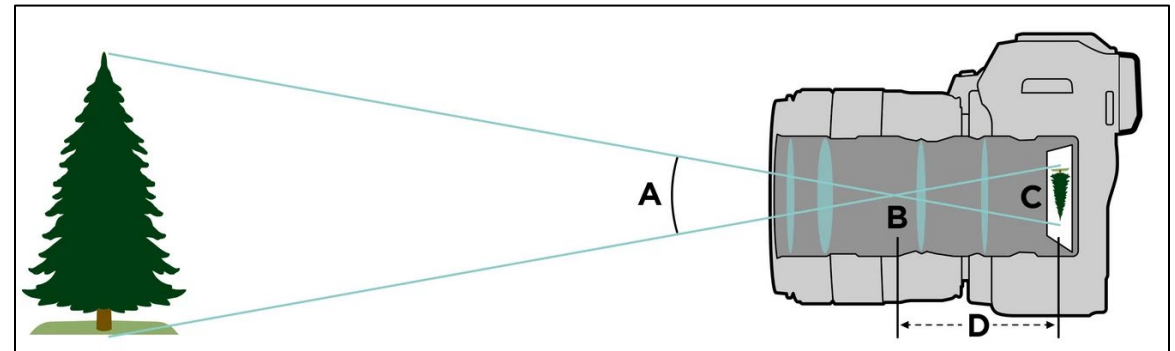
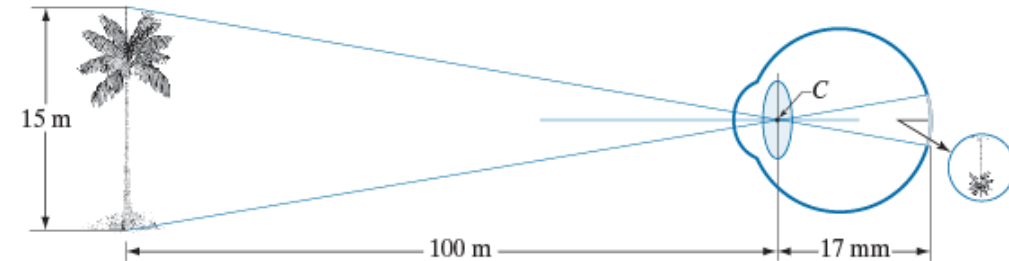


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$$\frac{15}{100} = \frac{h}{17}$$

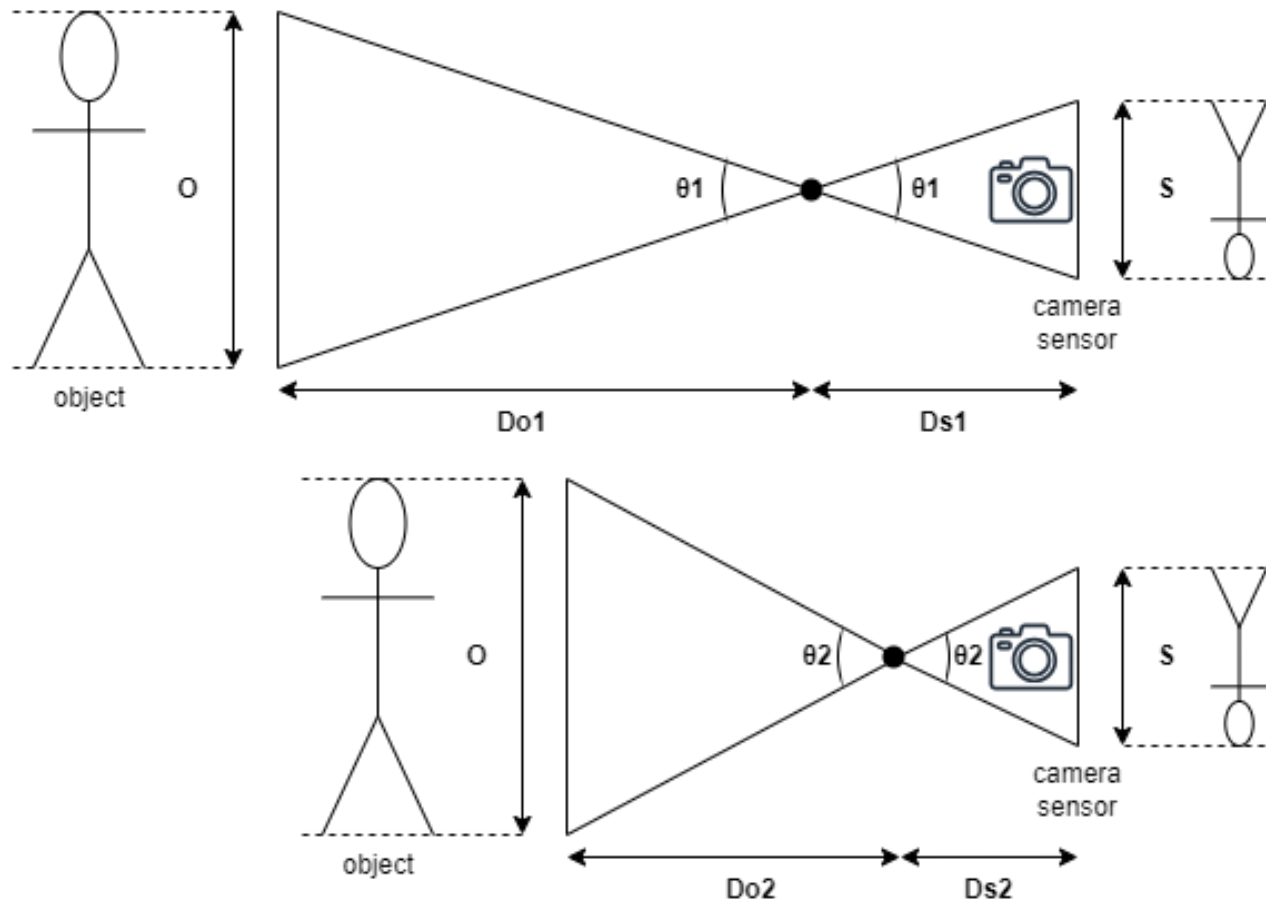
$$h = 2.5\text{mm}$$

FIGURE 2.3
Graphical representation of the eye looking at a palm tree. Point C is the focal center of the lens.



Visual Perception: Focal Distance & View Angle

- There are lenses with fixed or variable focal lens (zoom lenses)
- Far/close objects: increase/decrease focal distance (view angle FoV reduces/ increases)



FAR OBJECT, $D_{o1} > D_{o2}$

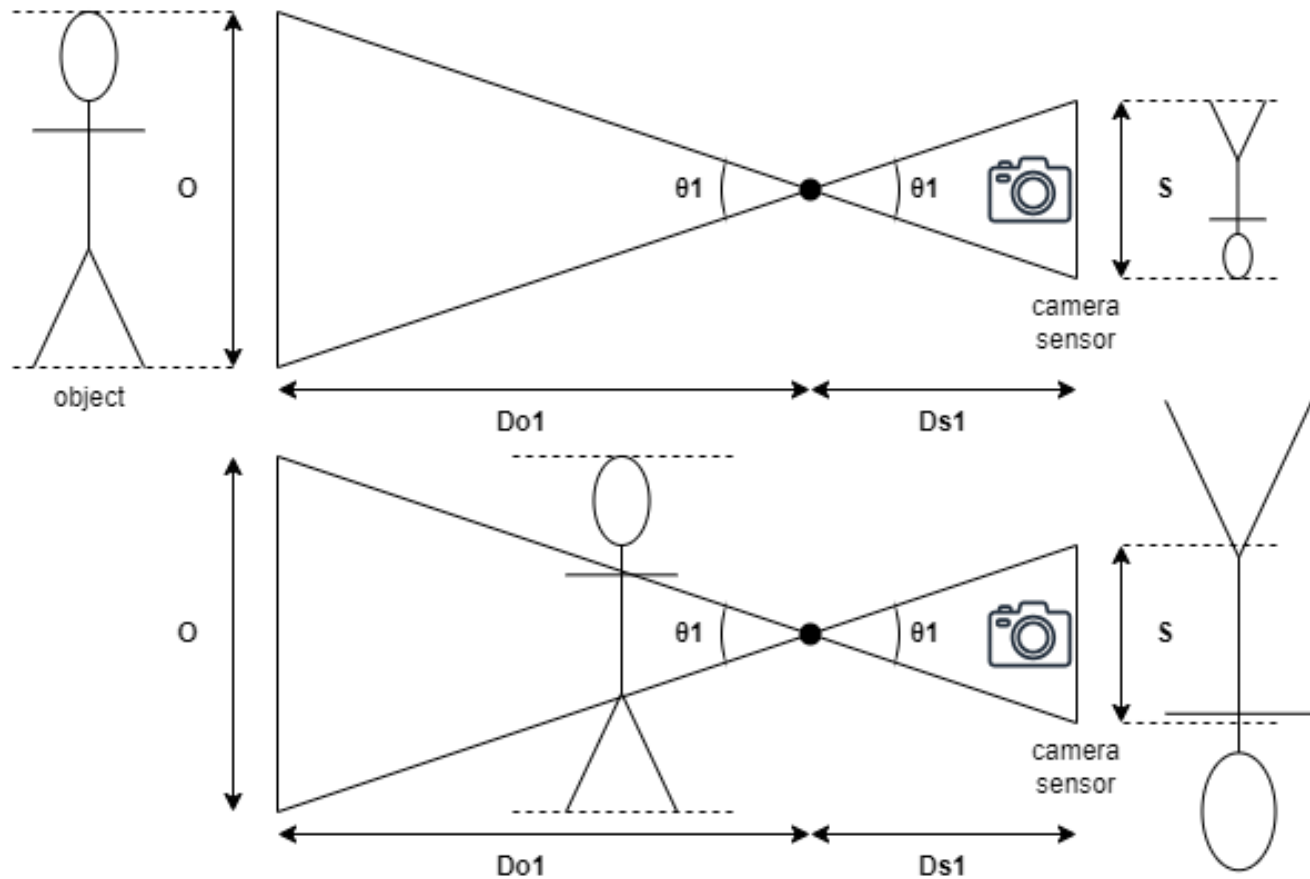
- Focal distance $D_{s1} > D_{s2}$
- View angle (FoV) $\theta_1 < \theta_2$

S = size of the sensor
(always constant)

O = size of the object
(constant in this example)

Visual Perception: Focal Distance & View Angle

- There are lenses with fixed or variable focal lens (zoom lenses)
- Far/close objects: increase/decrease focal distance (view angle FoV reduces/ increases)

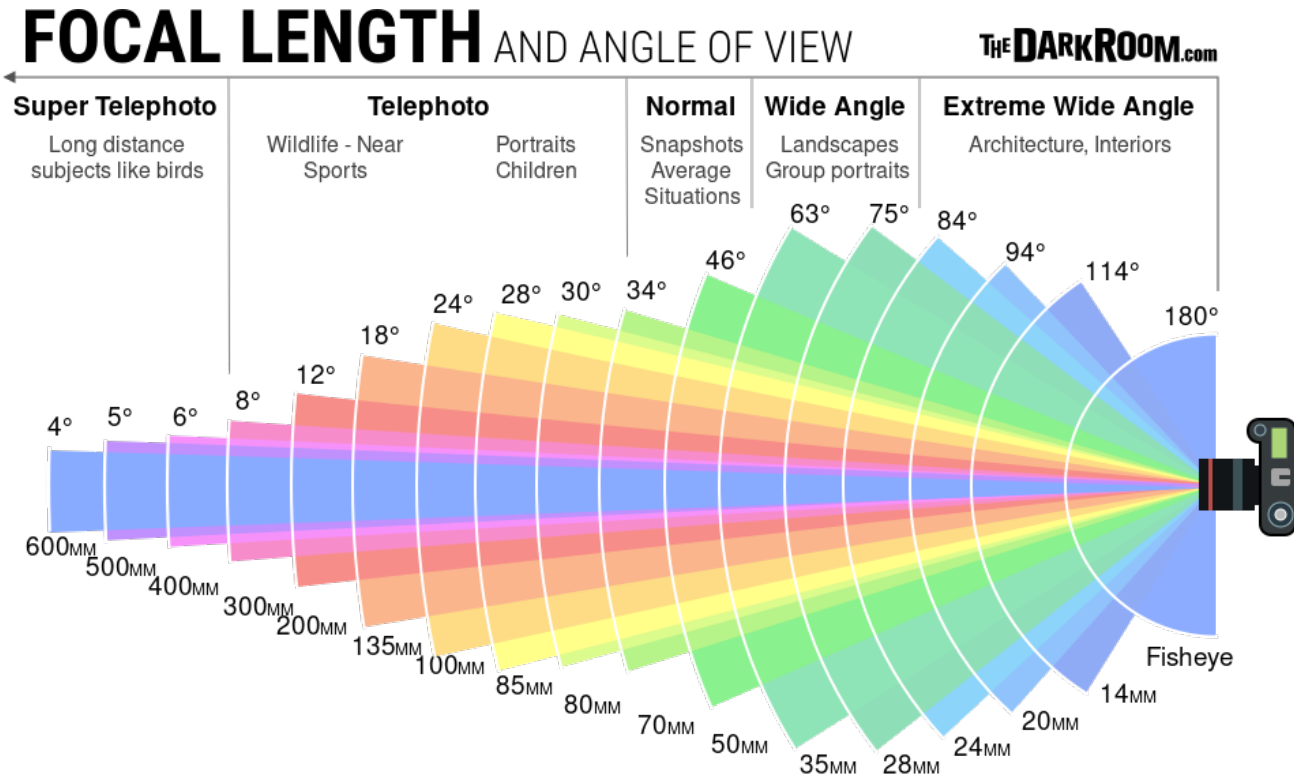


**OBJECT MOVES CLOSER
WITHOUT CHANGING THE
FOCAL DISTANCE**

- Does not fit the sensor

Visual Perception: Focal Distance & View Angle

- There are lenses with fixed or variable focal lens (zoom lenses)
- Far/close objects: **increase/decrease** focal distance (view angle FoV **reduces/ increases**)
- Human-eye equivalent: 35mm (APS-C lens) with sharp FoV of ~55 degrees



Fish eye



Visual Perception: Other Camera Factors

Lens Aperture (f-number): opening of the lens

- Affects **depth-of-field** (large aperture: small dof)

Shutter speed (exposure time)

- Affects **blur** if the objects moves fast

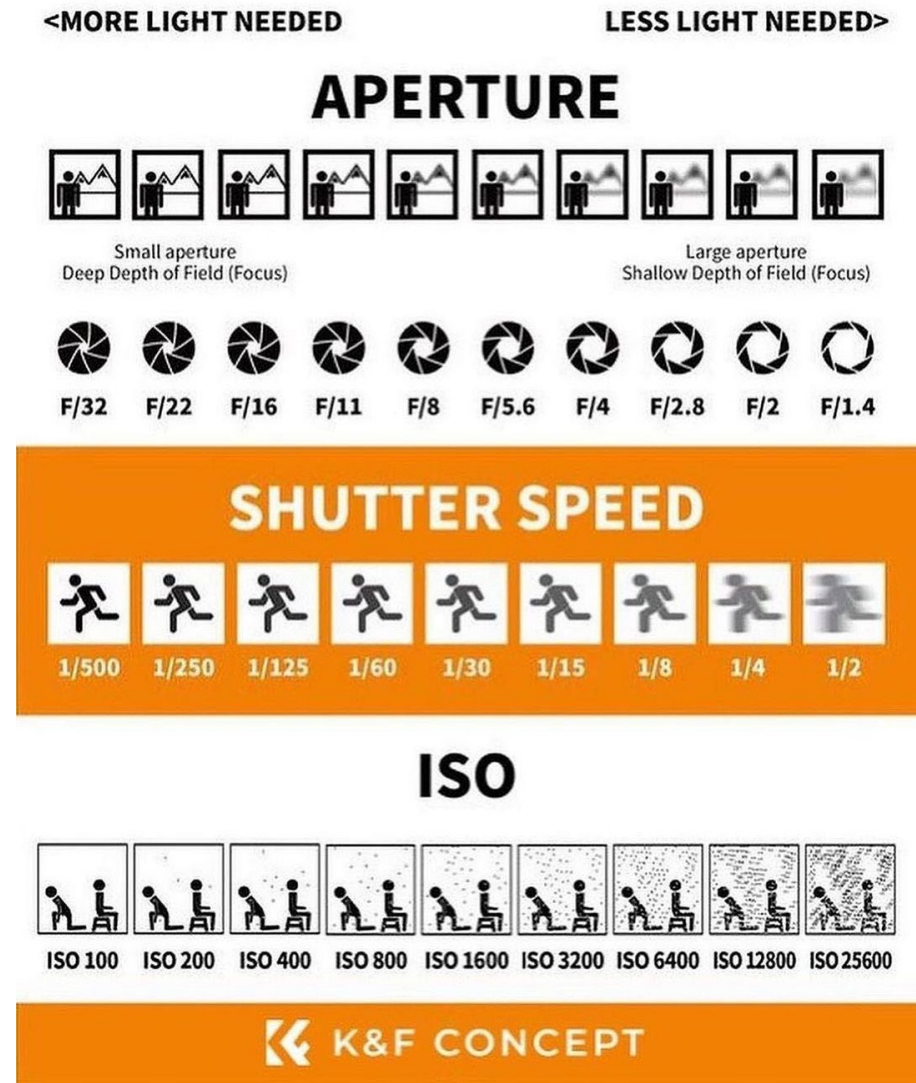
Both aperture & shutter speed affects the **amount of light** entering the camera

Analog gain (ISO)

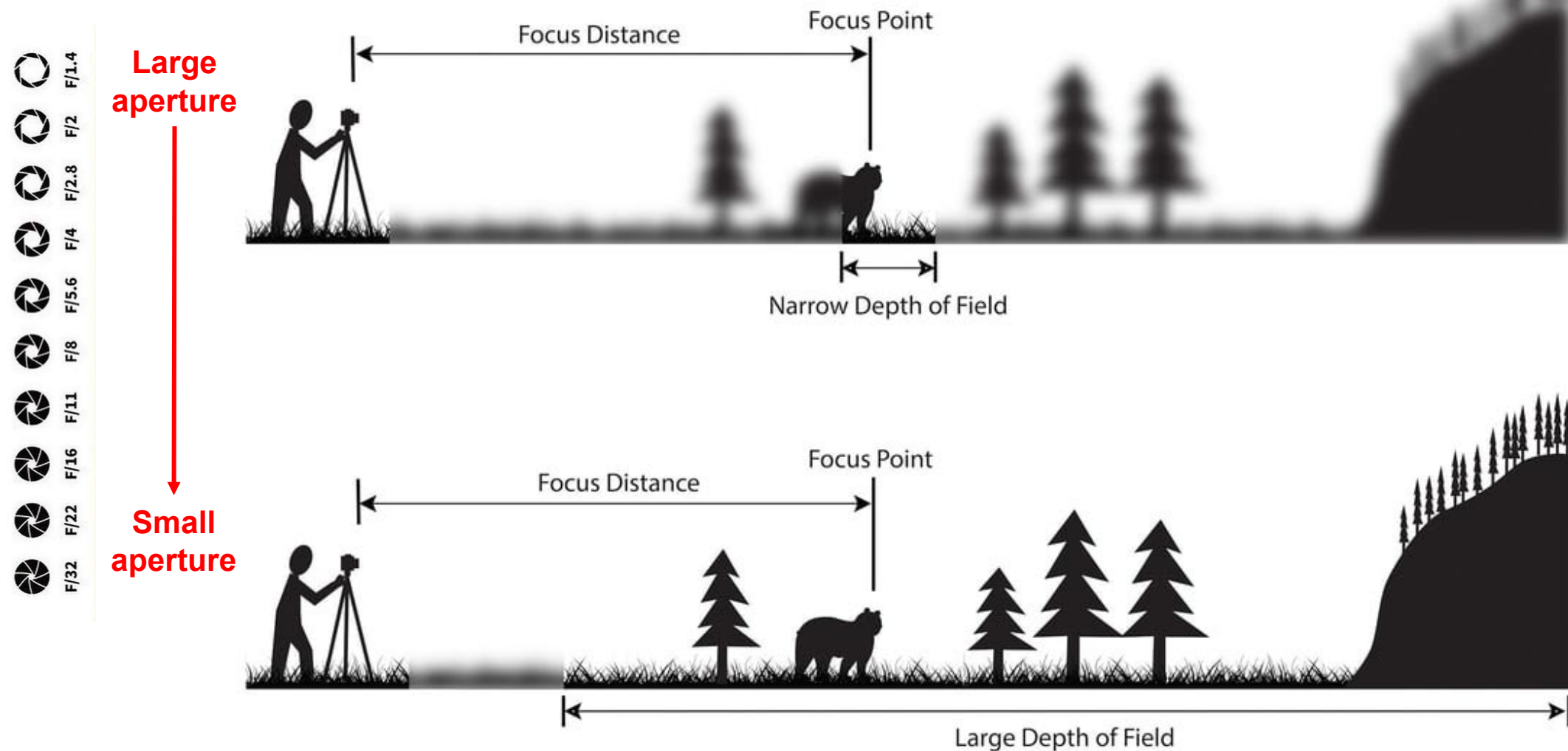
- Can amplify **noise**

Sensor size (chip)

- Large: expensive, space
- Small: noise



Depth of Field (DoF)



Shutter Speed

1/30 (0.033 seconds)

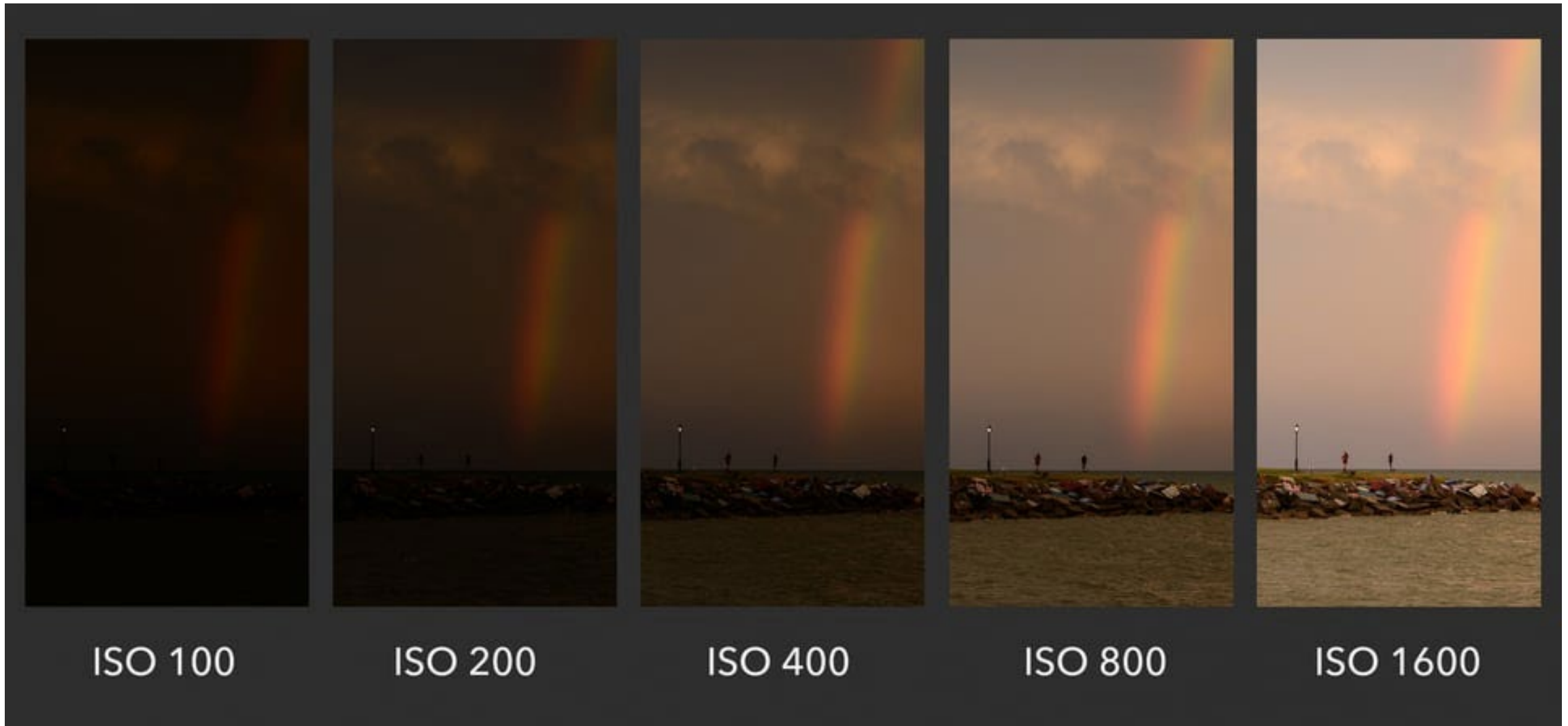


1/500 (0.002 seconds)



- How illumination is kept "equal"? (hint: blur of the background)

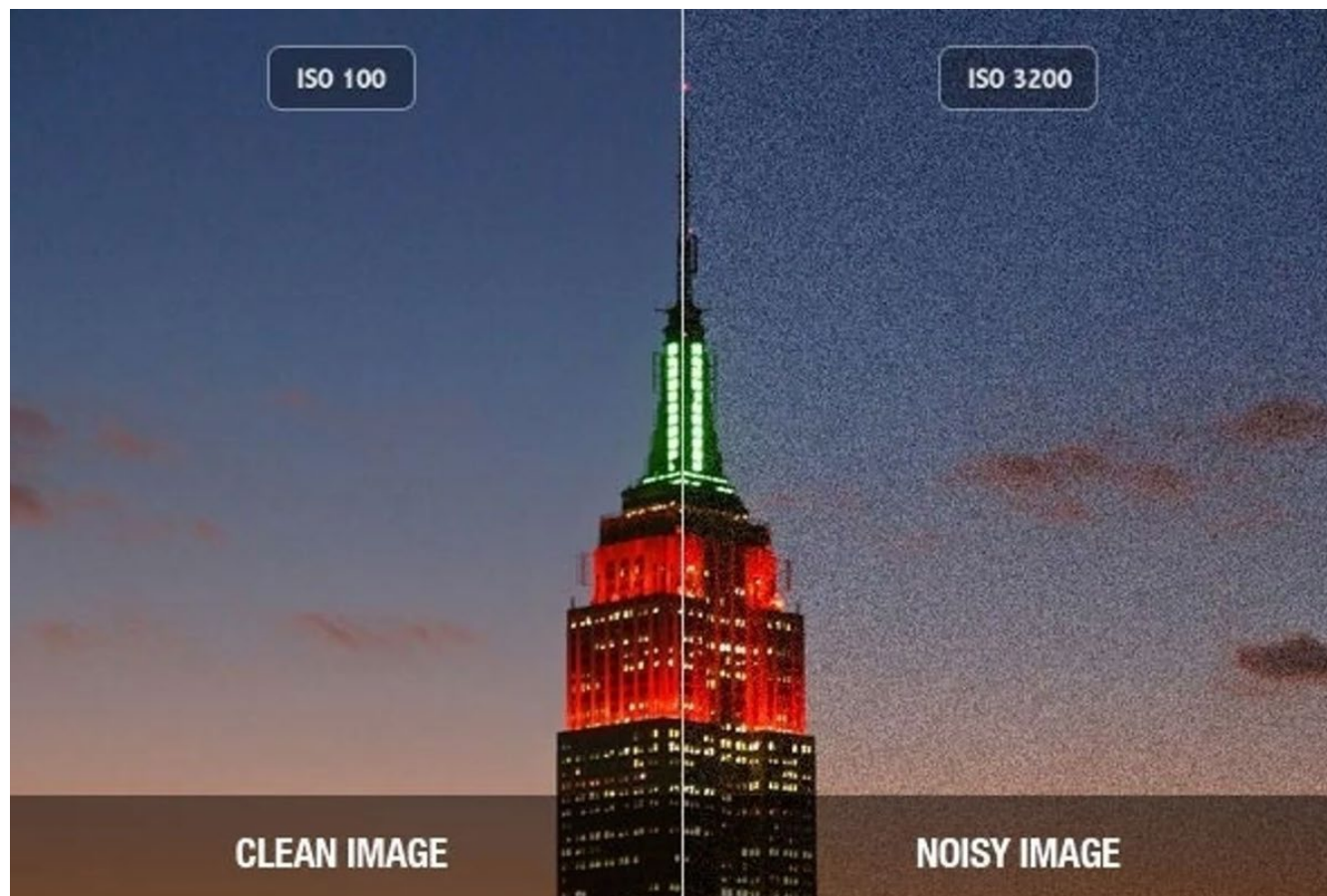
ISO number



ISO number

Analog gain of the sensor

- Amplify the signal at the output of the sensor
- But also amplifies **noise**
- How illumination is kept "equal"?



Visual Perception: Other Camera Factors

Lens Aperture (f-number): opening of the lens

- Affects depth-of-field (large aperture: small dof)

Shutter speed (exposure time)

- Affects blur if the objects moves fast

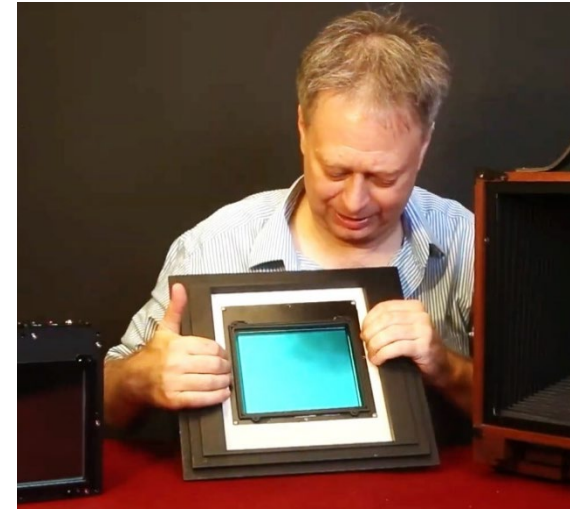
Both aperture & shutter speed affects the amount of light entering the camera

Analog gain (ISO)

- Can amplify noise

Sensor size (chip)

- Large: expensive, space
- Small: noise



**World
largest
sensor
14x12cm**

**Smartphone
sensor**

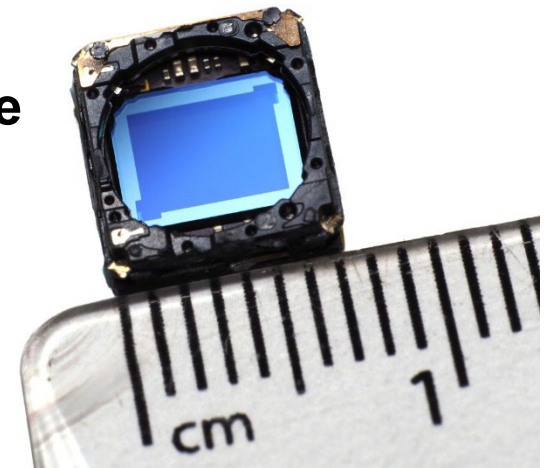


Image Acquisition, Sampling & Quantization

The Electromagnetic Spectrum

FIGURE 2.11
Graphical
representation of
one wavelength.

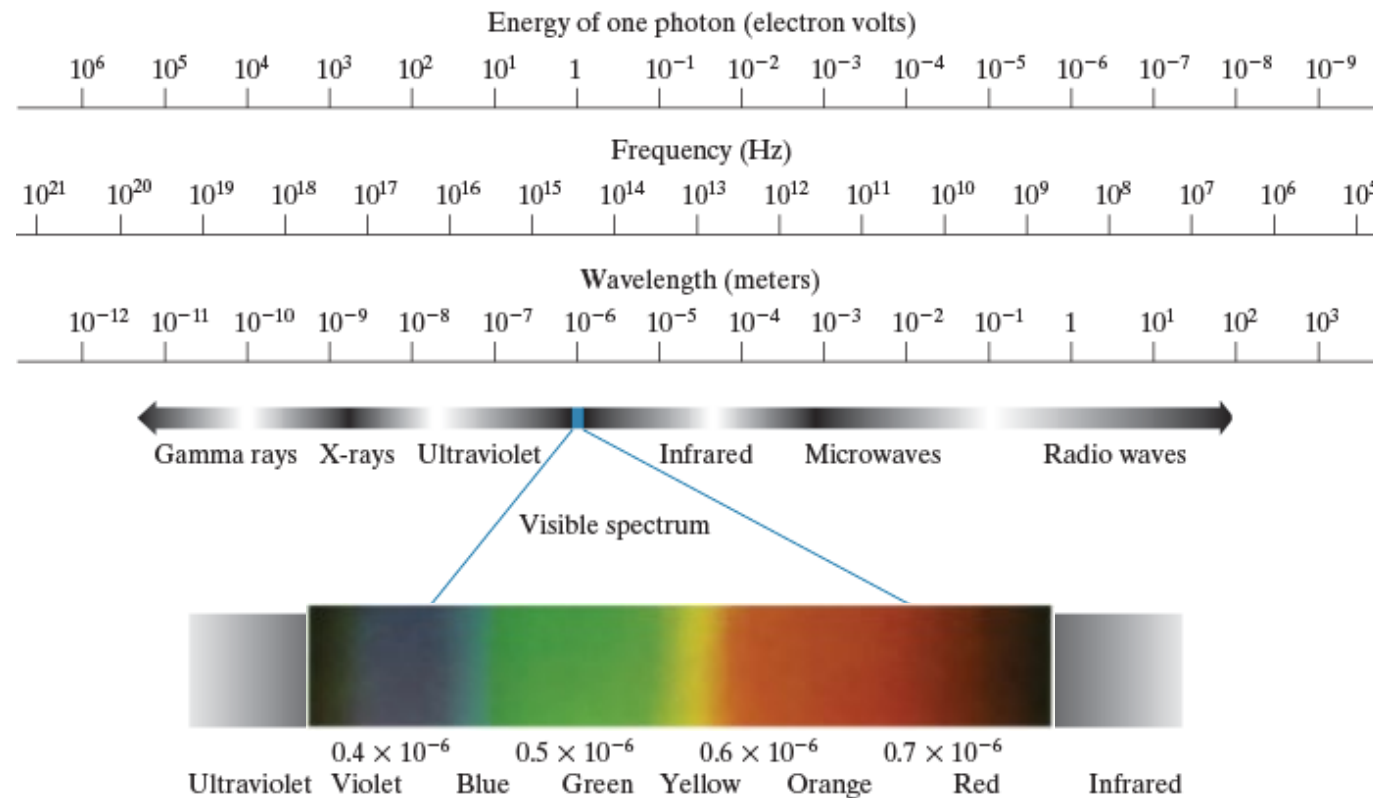
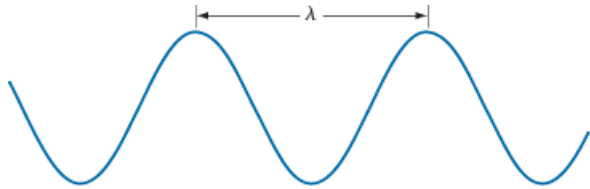
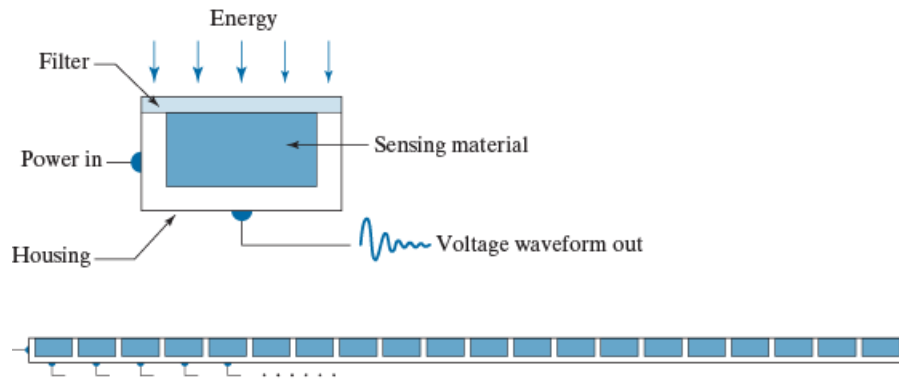


FIGURE 2.10 The electromagnetic spectrum. The visible spectrum is shown zoomed to facilitate explanations, but note that it encompasses a very narrow range of the total EM spectrum.

Image Sensing

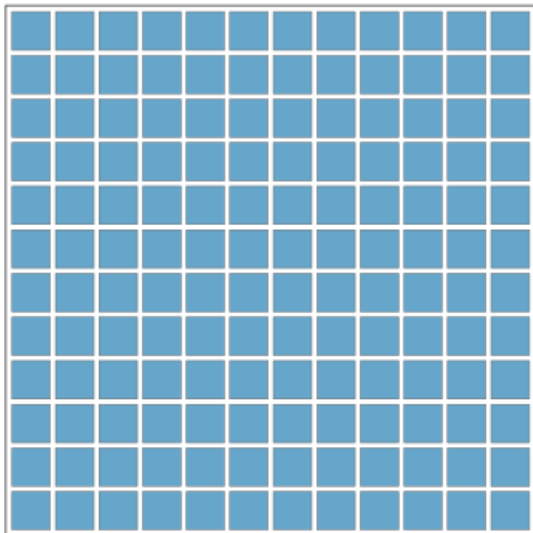
- Acquisition elements



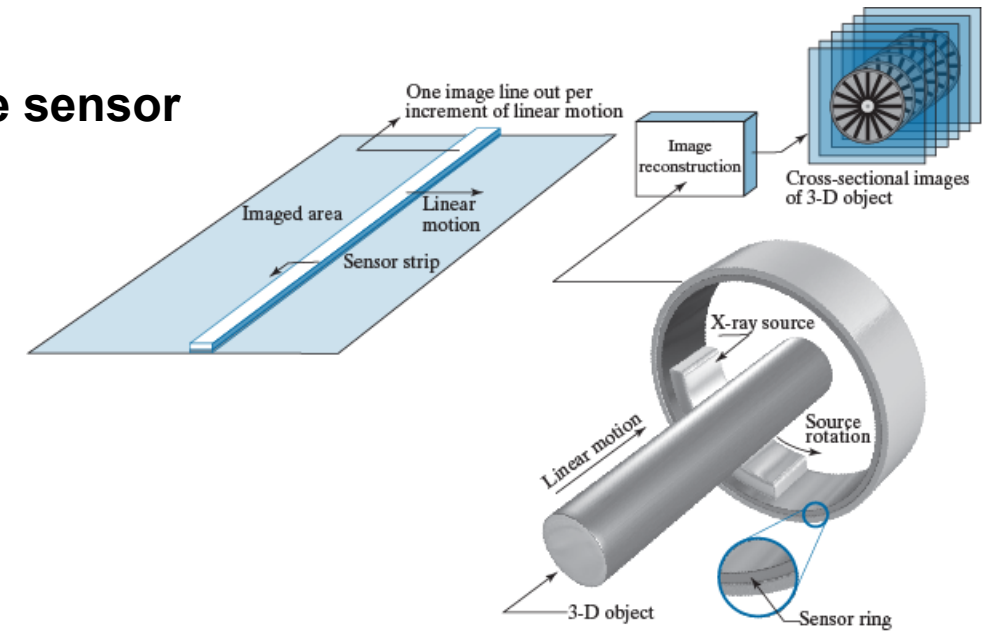
a
b
c

FIGURE 2.12

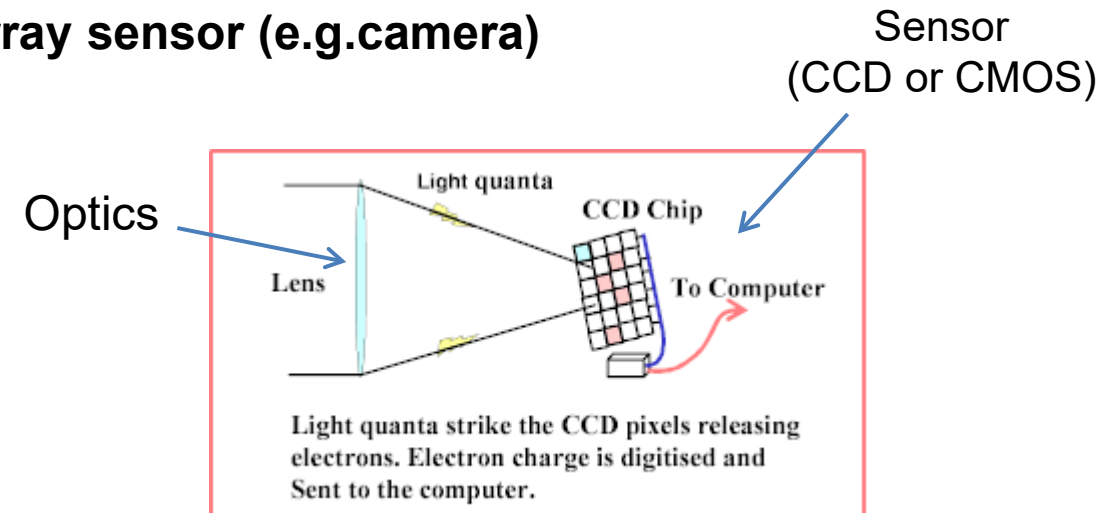
(a) Single sensing element.
(b) Line sensor.
(c) Array sensor.



Line sensor



Array sensor (e.g. camera)



Digital Image Formation

- The input “energy” can be any measurable property
 - Electromagnetic radiation (visible light, x-ray, radio waves...)
 - Sound
 - ...
- Variations in the property produces variation in the output voltage
- The output voltage is mapped onto a digital image that we can “see” (typically in the 0-255 range)

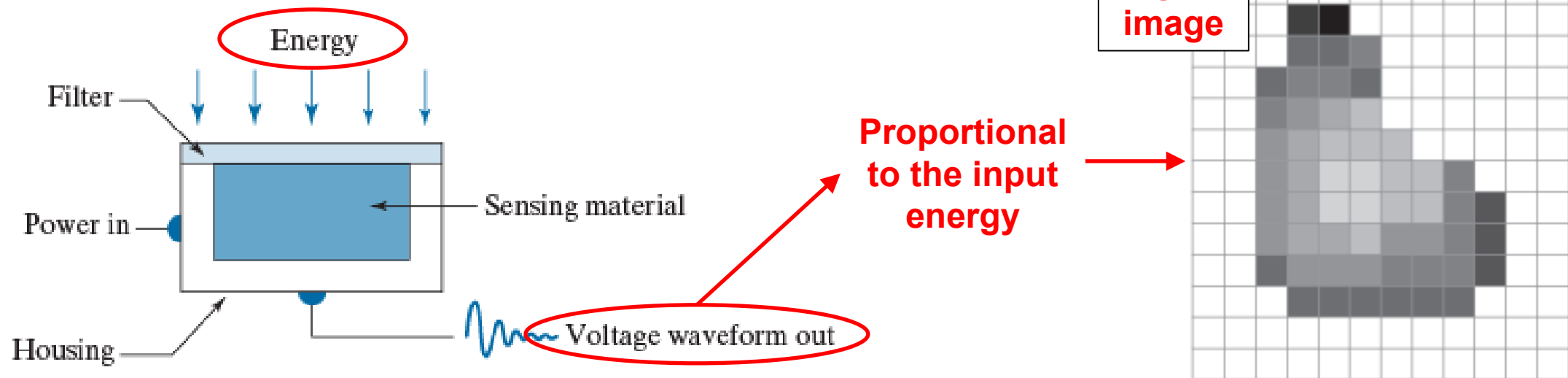
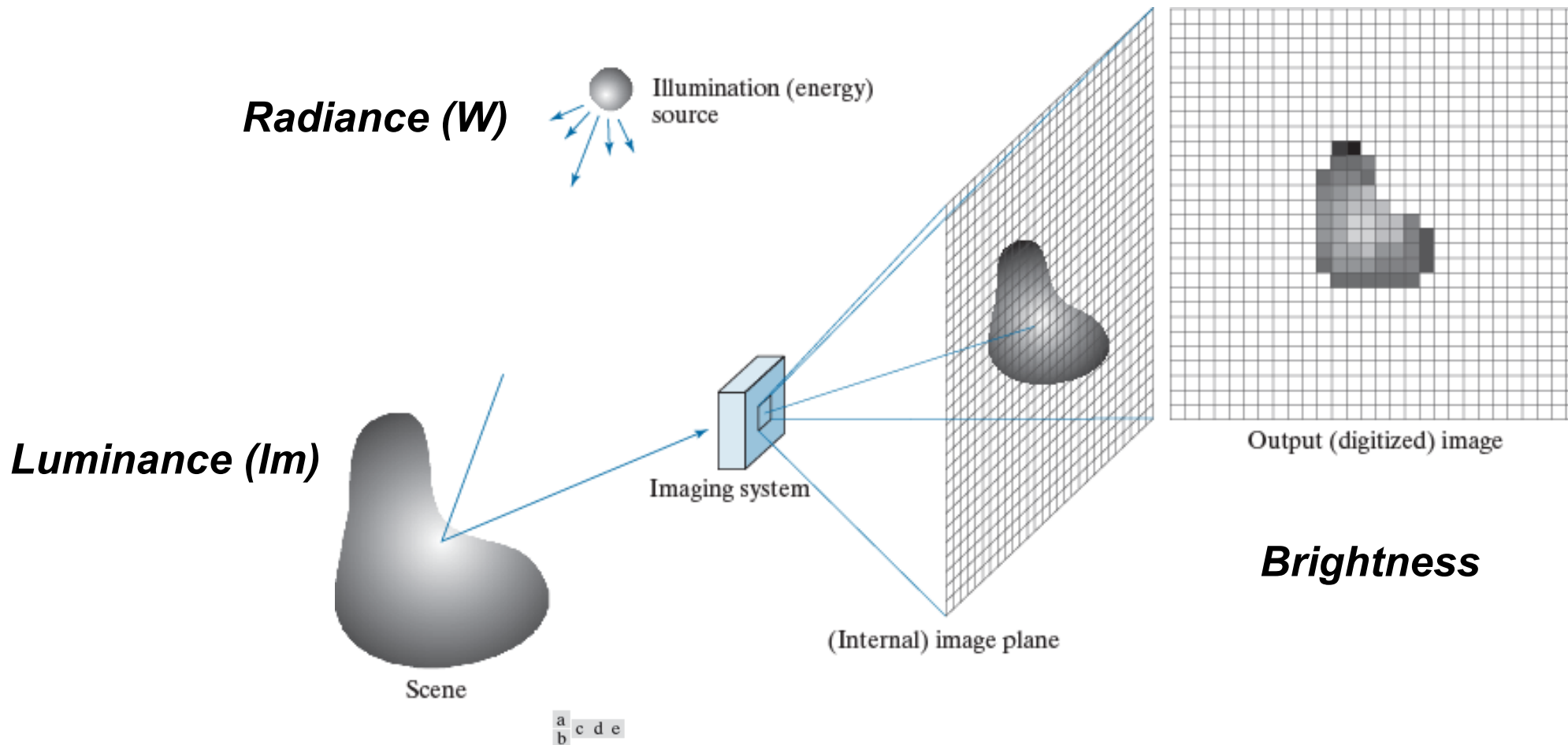


Image Acquisition

- Projection into a sensing element/image plane



a
b c d e

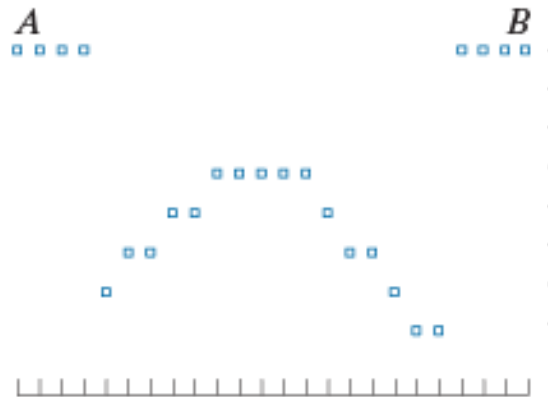
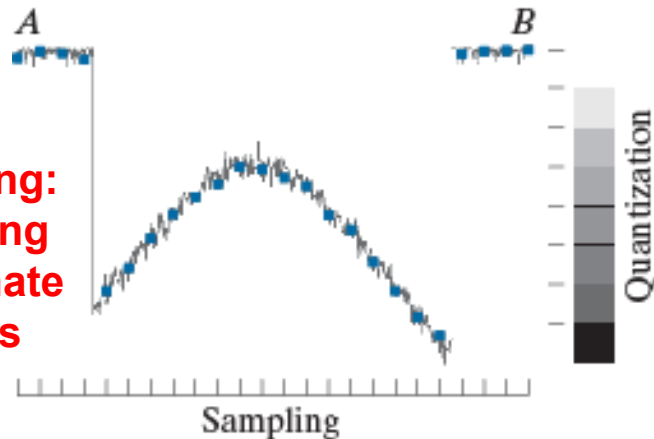
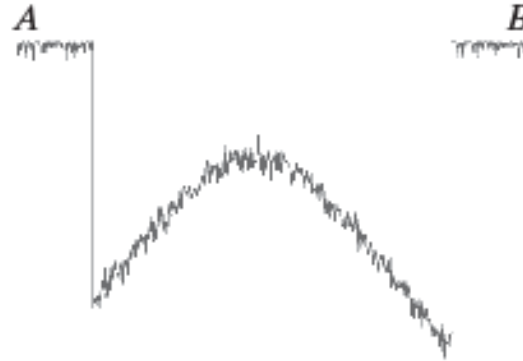
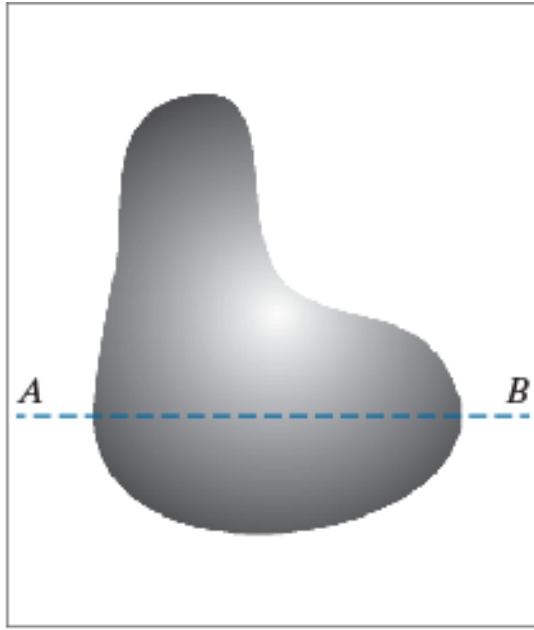
FIGURE 2.15 An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Sampling and Quantization

a b
c d

FIGURE 2.16

(a) Continuous image. (b) A scan line showing intensity variations along line AB in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).

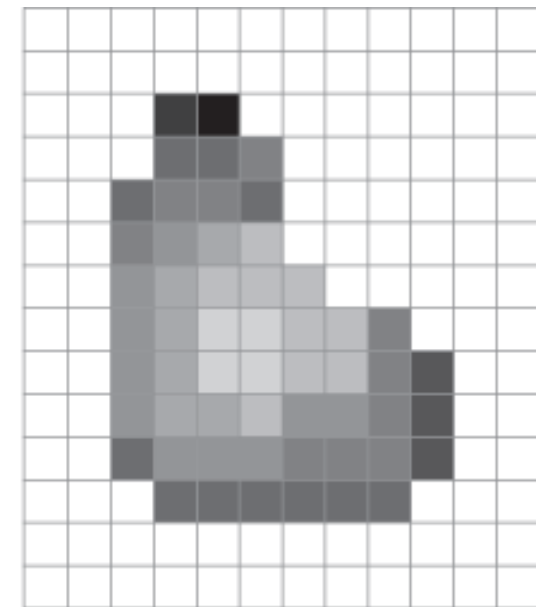
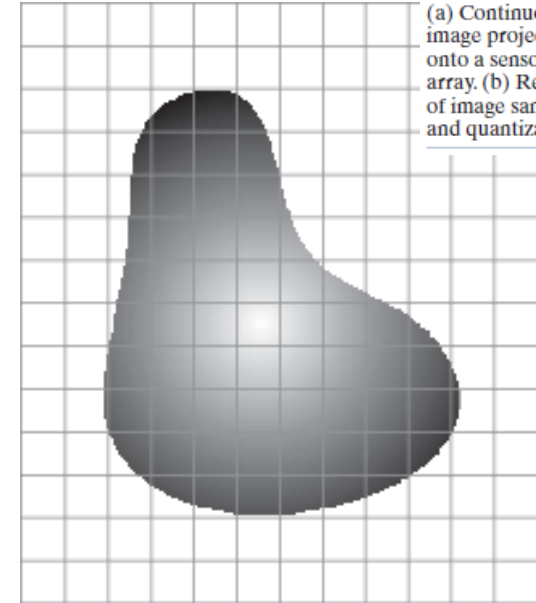


Quantization:
digitizing
amplitude
values

a b

FIGURE 2.17

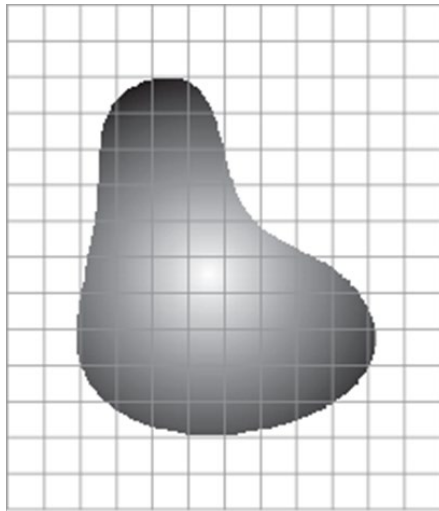
(a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.



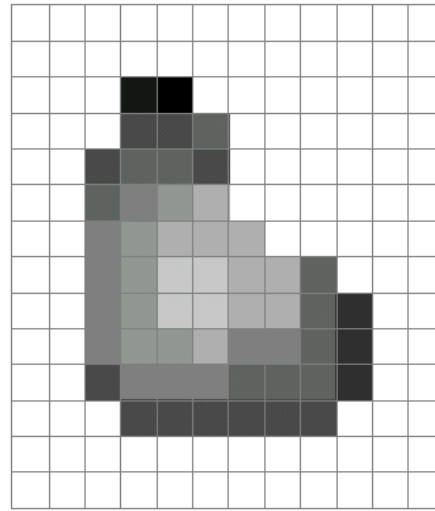
Sampling:
digitizing
coordinate
values

A Digital Image

- A grid (matrix) of intensity values called pixels




 Sampling
 Quantization

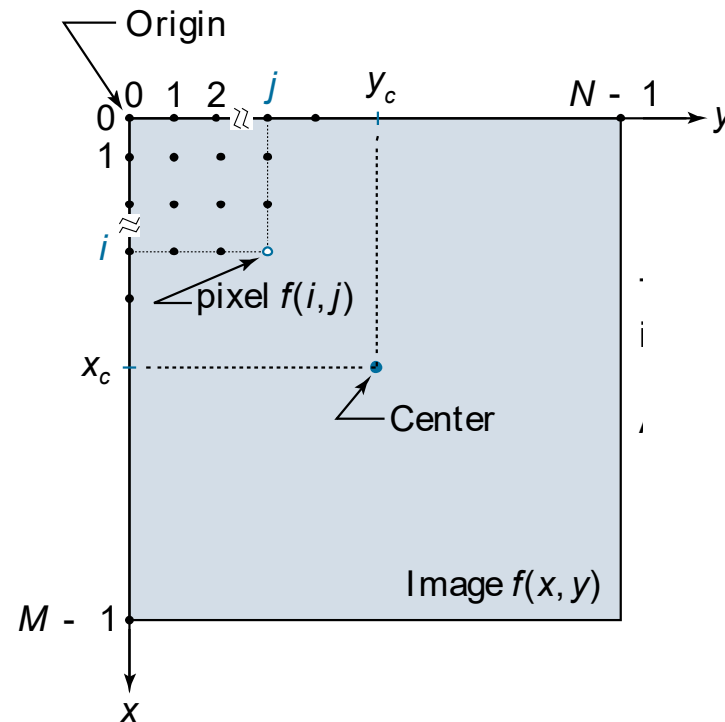
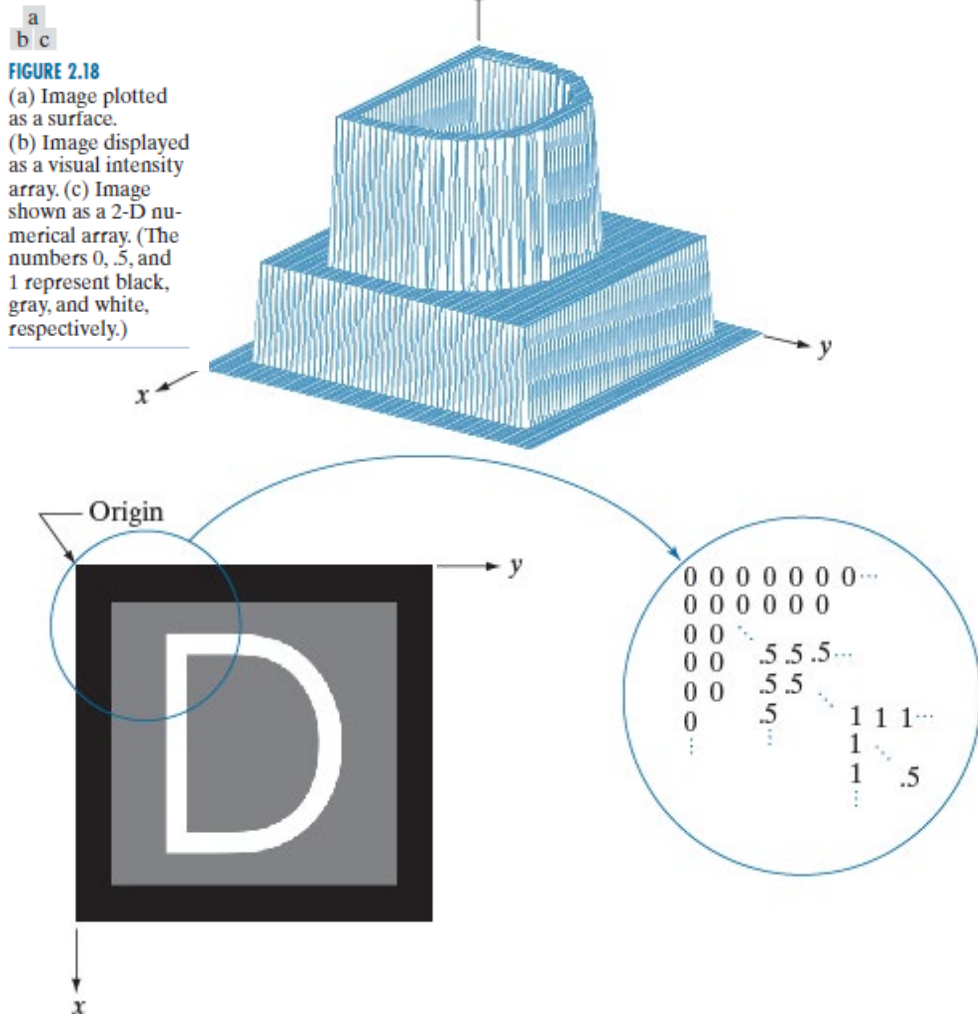


=

255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	20	0	255	255	255	255	255	255	255
255	255	255	75	75	75	255	255	255	255	255	255
255	255	75	95	95	75	255	255	255	255	255	255
255	255	96	127	145	175	255	255	255	255	255	255
255	255	127	145	175	175	175	255	255	255	255	255
255	255	127	145	200	200	175	175	95	255	255	255
255	255	127	145	200	200	175	175	95	47	255	255
255	255	127	145	145	175	127	127	95	47	255	255
255	255	74	127	127	127	95	95	95	47	255	255
255	255	255	74	74	74	74	74	74	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255	255

- The term digital denotes:
 - A finite number of elements (M rows x N columns)
 - A discrete number of intensity levels per pixel, e.g. 0, 1, 2, ..., 255
 (common to use one byte per value: 0 = black, 255 = white, or normalized as 0 = black, 1 = white)

Sampling and Quantization



- **Origin** at $(0,0)$ or $(1,1)$ depending on the software
- Image **center** may not belong to the image depending on M, N being odd or even
- (x,y) sometimes referred to as (r,c) or (i,j)
- (x,y) direction may also change with the software, so $x=c, y=r$ instead

A Digital Image as a Function

A digital image can be considered a function $f(x,y)$ with the particularities that:

- f, x, y can only take discrete and finite values
- f =intensity level, with $f=0,1,...,255$ (if using $k=8$ bits)
- (x,y) spatial coordinates, with $x=0,1,...,M-1$ and $y=0,1,...,N-1$

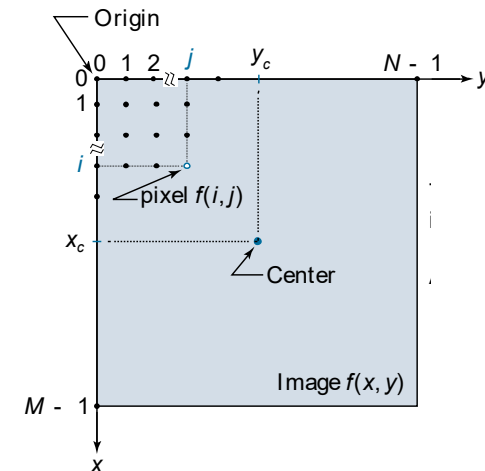
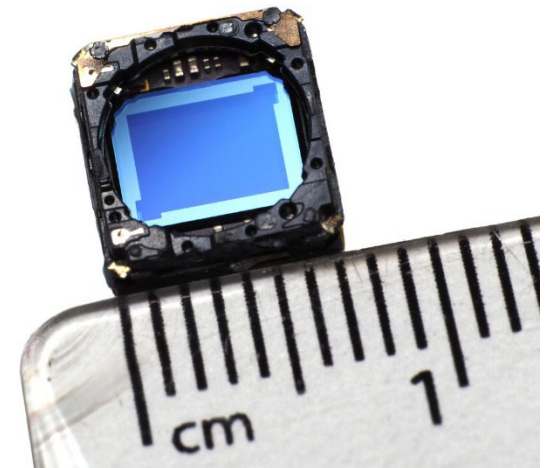
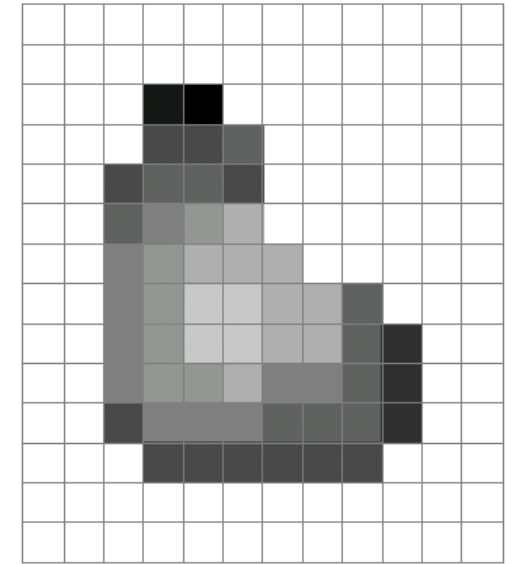


FIGURE 2.19
Coordinate convention used to represent digital images. Because coordinate values are integers, there is a one-to-one correspondence between x and y and the rows (r) and columns (c) of a matrix.

Spatial and Intensity Resolution

Spatial Resolution

- How many pixels are used to sample the image (# of rows M and columns N), will tell the **smallest discernible detail in the image**
- Depends on the sensing element employed. Usually given in **dpi (dots per inch)**, e.g. 500dpi means 500 pixels per inch (one inch is approx 25.4mm)
- If the sensor has a size of 18x13 mm, then the image size produced by such sensor is 354x256 pixels

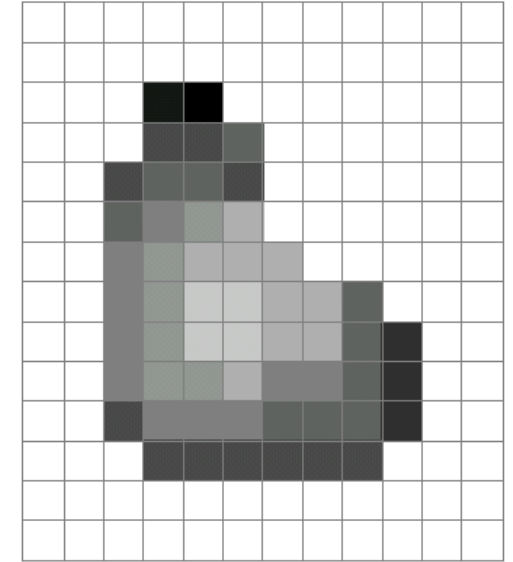
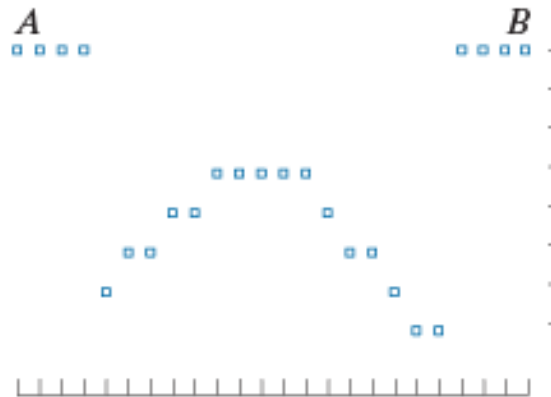
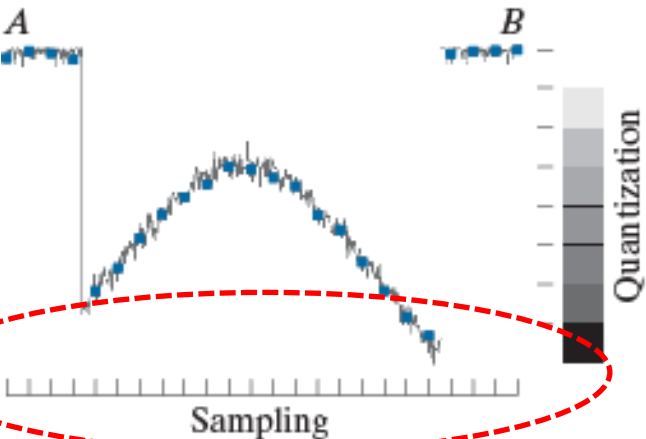
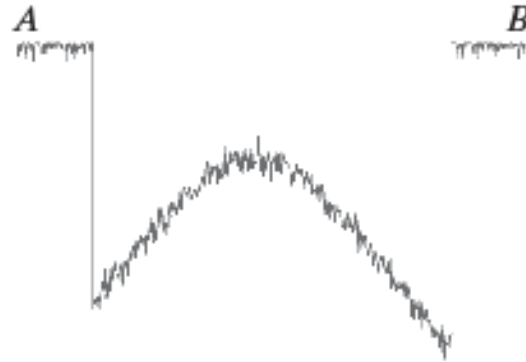
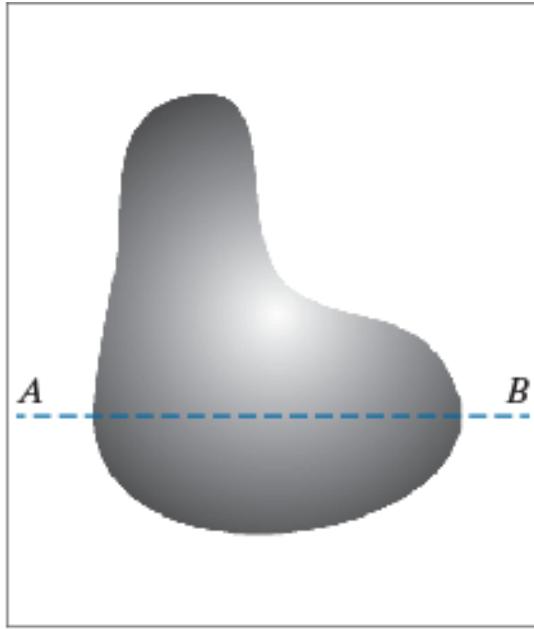


Spatial Resolution

a b
c d

FIGURE 2.16

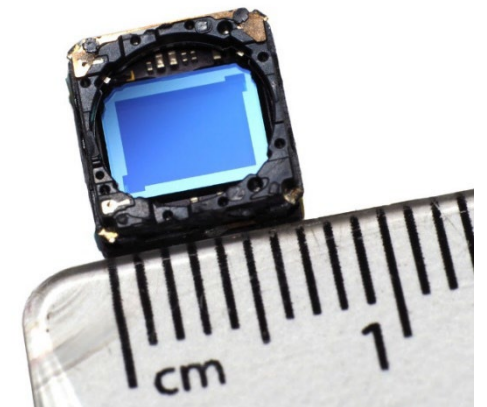
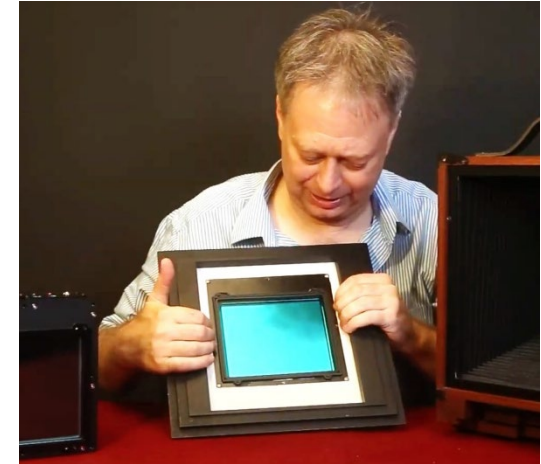
(a) Continuous image. (b) A scan line showing intensity variations along line AB in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).



Increasing Spatial Resolution?

- Increase **sensor size**
 - Not feasible in many cases (space available, cost)
 - Increase **pixel density** in the sensor, i.e. integrate more pixels in the same space
 - Smartphones:
 - Sensing area typically limited to 16 to 30 mm² (4x4 to 5.5x5.5 mm)
 - Pixel area typically 1-2 μm (12700-25400 dpi)
 - 16 mm² area, 2 μm (4 μm^2) pixel: 4Mpix
 - 16 mm² area, 1 μm (1 μm^2) pixel: 16Mpix...
- ...but pixel area is $\frac{1}{4}$ so receives $\frac{1}{4}$ of light!**
- More sensitive to noise!**

14x12cm



Spatial Resolution

Sensor Name	Full Frame	APS-C	1/2.3"	1/3.2"
Sensor Area	8.6 cm ²	3.73 cm ²	0.28 cm ²	0.15 cm ²
Image	~2.93x2.93cm 	~1.93x1.93cm 	~5.3x5.3mm 	~3.9x3.9mm 
Example				

SENSOR SIZE (if square)

IMAGE SIZE (examples)

Full frame sensor gathers 50x more light than phone sensor

24MPx: 6000 x 4000 (10.1Mb) max resolution
Sensor size: 23.5 x 15.6 mm
(Nikon D3300)

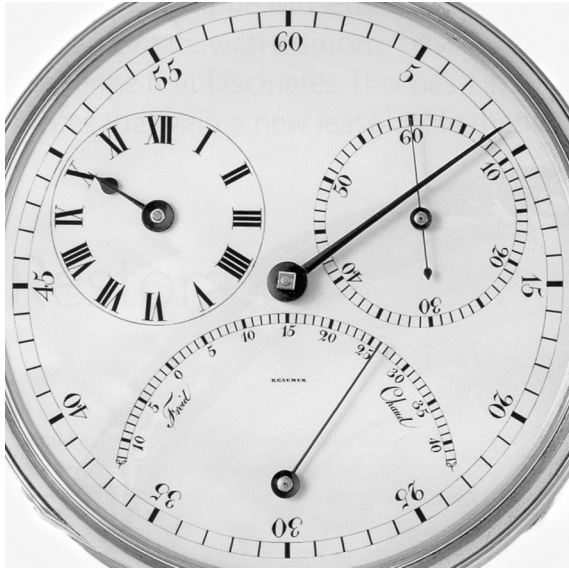
64MPx: 9248 x 6944 (10.5Mb) max resolution
16MPx: 4624 x 3472 (3.8 Mb) low resolution
Pixel size: 0.8um (micrometers)
(Samsung A52s)

Effects of changing spatial resolution

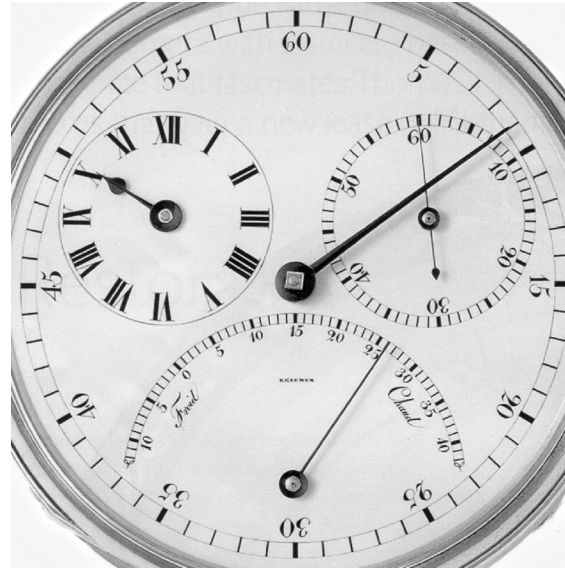
a b
c d

FIGURE 2.23
Effects of
reducing spatial
resolution. The
images shown
are at:
(a) 930 dpi,
(b) 300 dpi,
(c) 150 dpi, and
(d) 72 dpi.

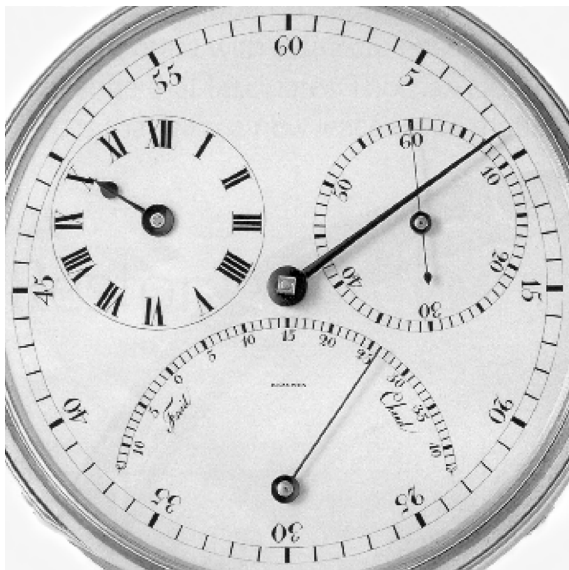
**2136 x
2140**



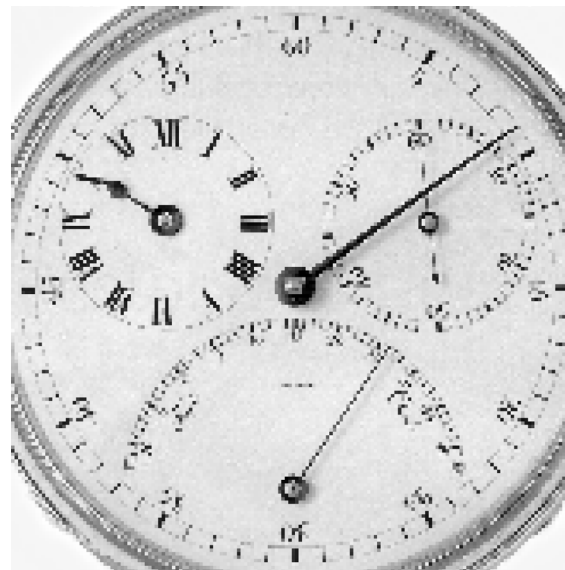
**689 x
690**



**344 x
345**



**165 x
166**



Reasons of a different resolution:

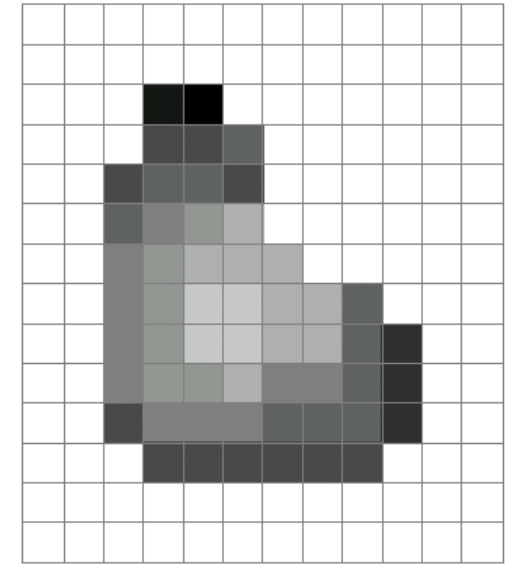
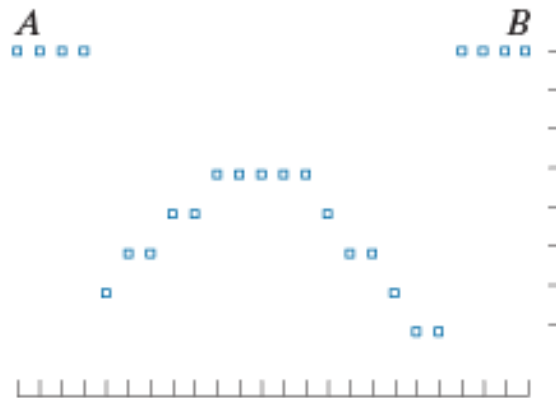
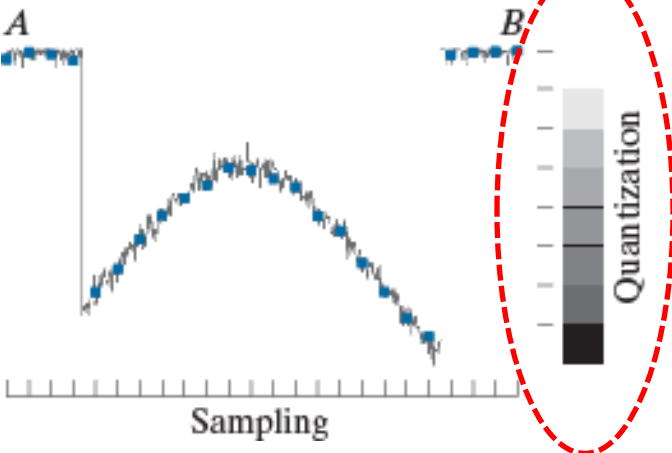
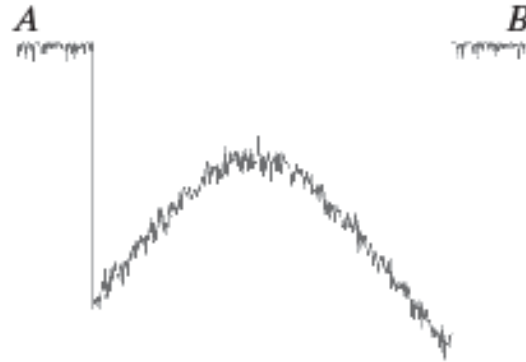
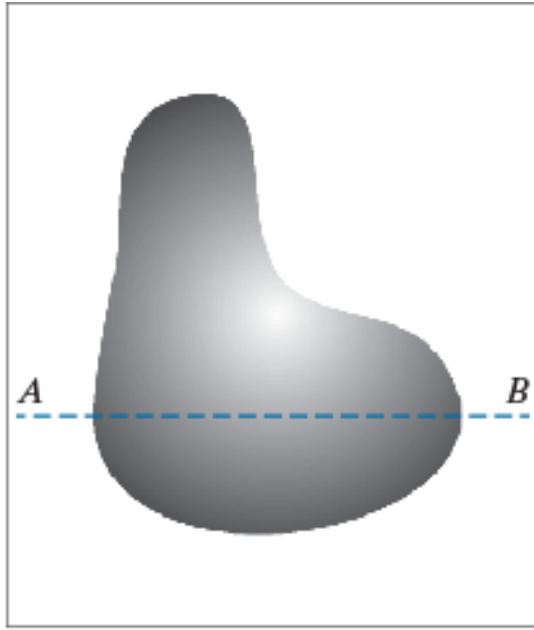
- Sensors with different resolution, or
- Image resizing (not seen)

Intensity Resolution

a b
c d

FIGURE 2.16

(a) Continuous image. (b) A scan line showing intensity variations along line AB in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).



Intensity Resolution

- How many levels are used to quantize the value of each pixel, will tell the **smallest discernible change in intensity**
- Usually an integer power of 2 (due to binary representation)
 - k bits, **$L=2^k$ intensity levels** per pixel
 - Pixel values then discretized to an integer between 0 and L-1

Typical: k=8 bits, 256 intensity levels

-> *humans cannot discern much more than 256 levels!*

Intensity Resolution

k bits, **$L=2^k$ intensity levels** per pixel

k=8 bits, 256 intensity levels

– 256 integers in $[0, 255]$, i.e. 0, 1, 2, ..., 255

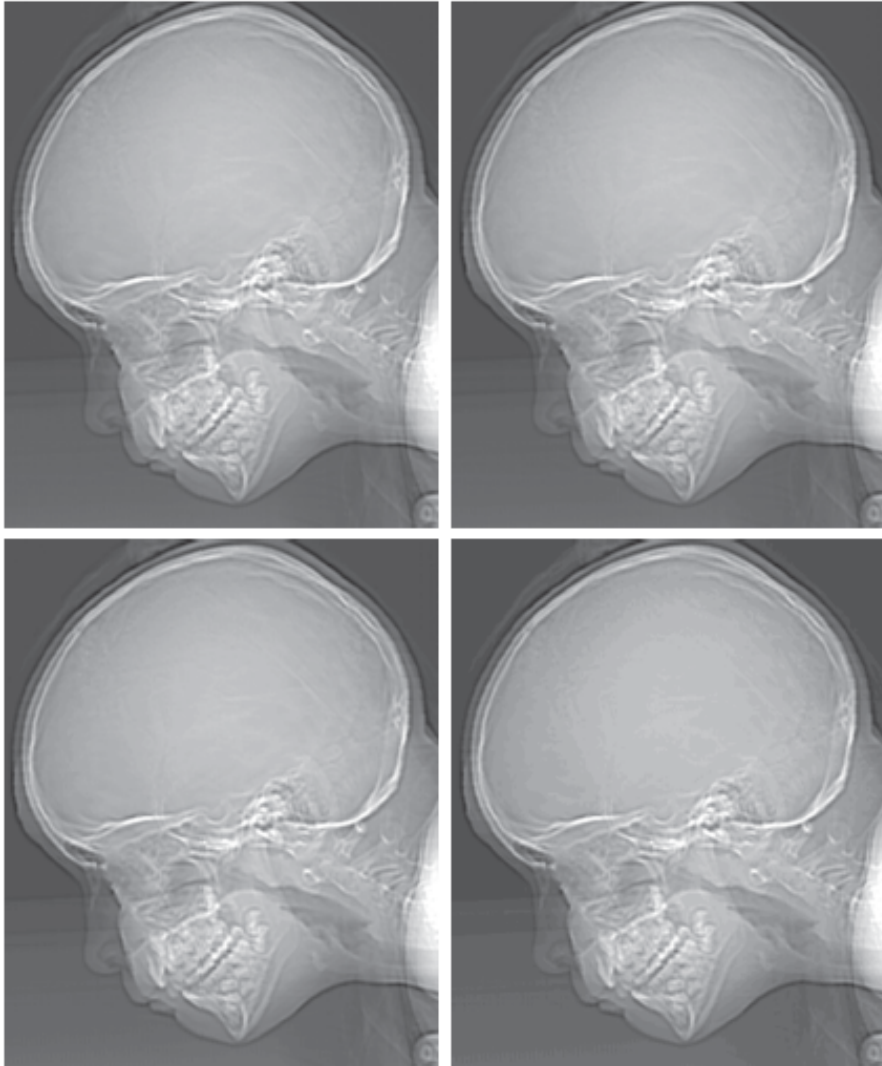
– Or 256 non-integers in $[0, 1]$, i.e. 0, $1/255$, $2/255$, ..., 1 (normalized)

- Integer representation in $[0, 255]$ used in **disk storage**
- Float representation in $[0, 255]$ or $[0, 1]$ to allow **decimals**, since image operations oftentimes result in non-integer values
- Then back to integer representation (usually rounding) when storing the image again in disk

Effects of changing intensity resolution

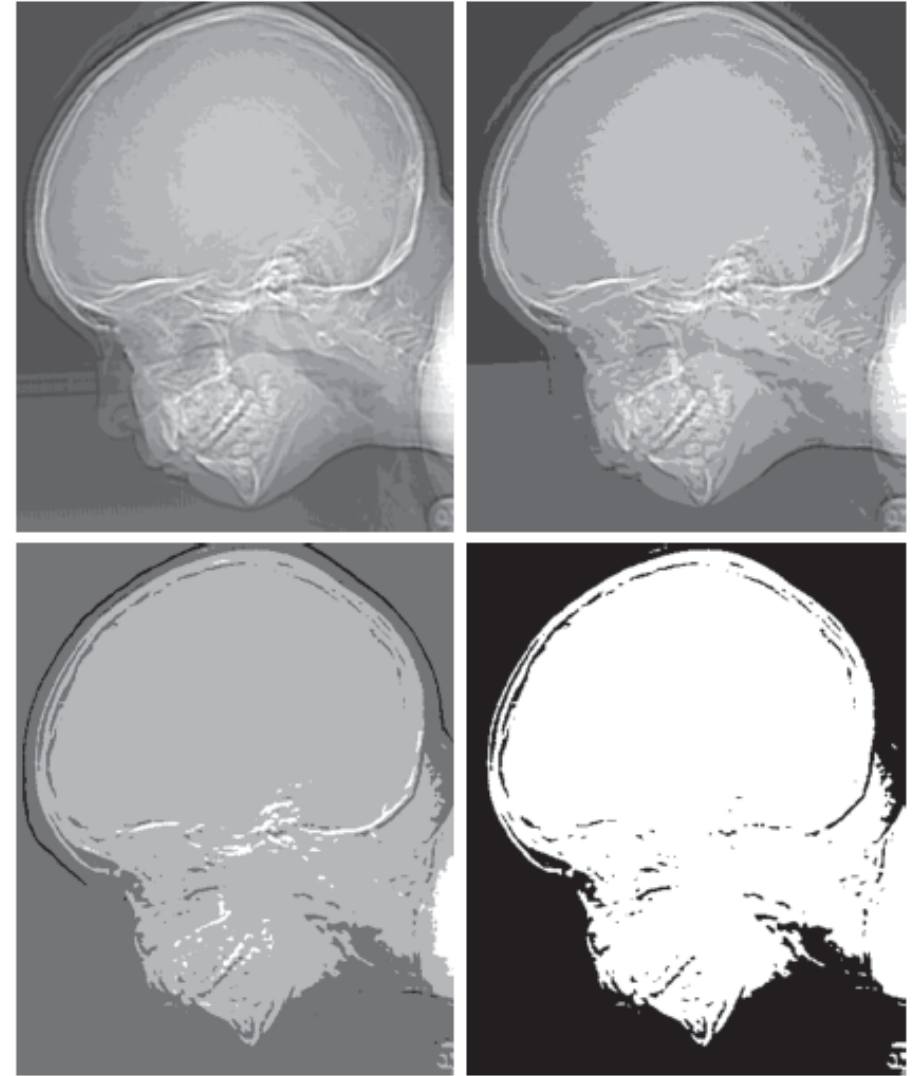
a b
c d

FIGURE 2.24
(a) 774×640 ,
256-level image.
(b)-(d) Image
displayed in 128,
64, and 32 inten-
sity levels, while
keeping the
spatial resolution
constant.
(Original image
courtesy of the
Dr. David R.
Pickens,
Department of
Radiology &
Radiological
Sciences,
Vanderbilt
University
Medical Center.)



e f
g h

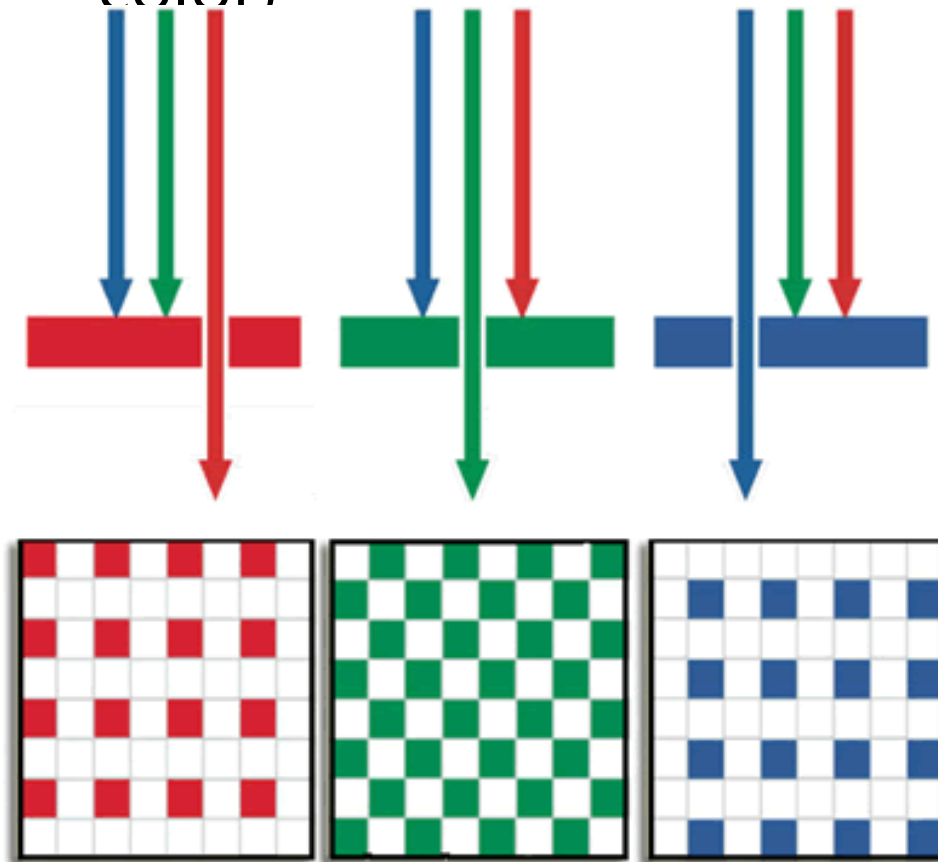
FIGURE 2.24
(Continued)
(e)-(h) Image
displayed in 16, 8,
4, and 2 intensity
levels.



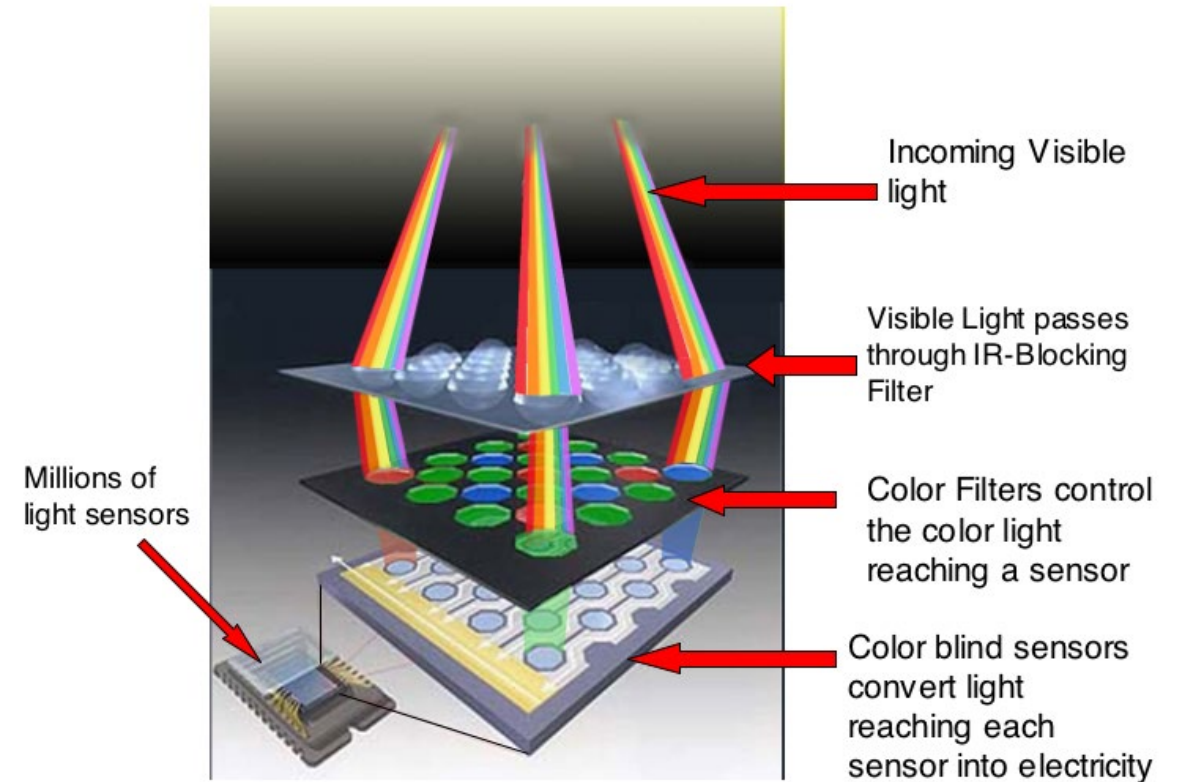
RGB and Gray-scale Image Acquisition and Representation

Image Acquisition in Visible Range

- Three color sensor – light is split into R,G,B to three different sensor elements (each pixel has a color filter that only lets in one color)



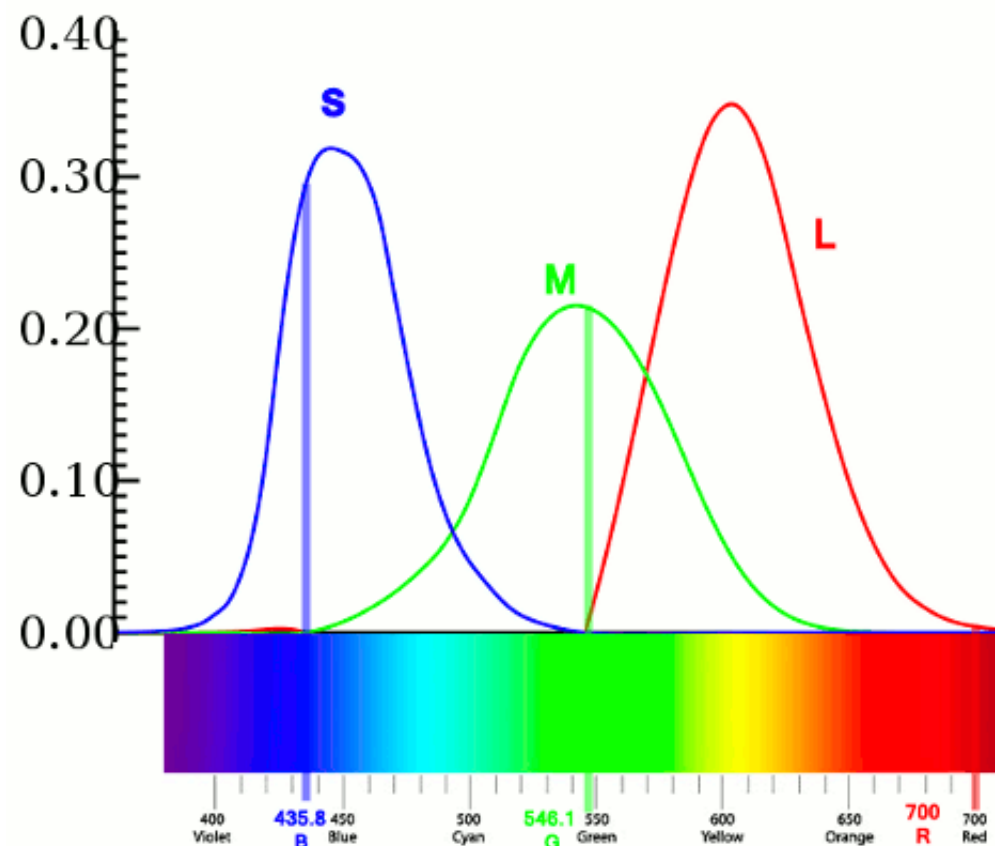
RGB Inside the Camera



Why twice as many photosites with green filters? (Bayer filter) -> see RGB to I equation, G has >double influence than RB

Why RGB

- Closely matches the physical working of the human eye
- Cone cells in three varieties, mostly sensitive to R, G and B
- We perceive colors by combination of these three



Tri-stimulus or tri-chromatic nature of the human visual system

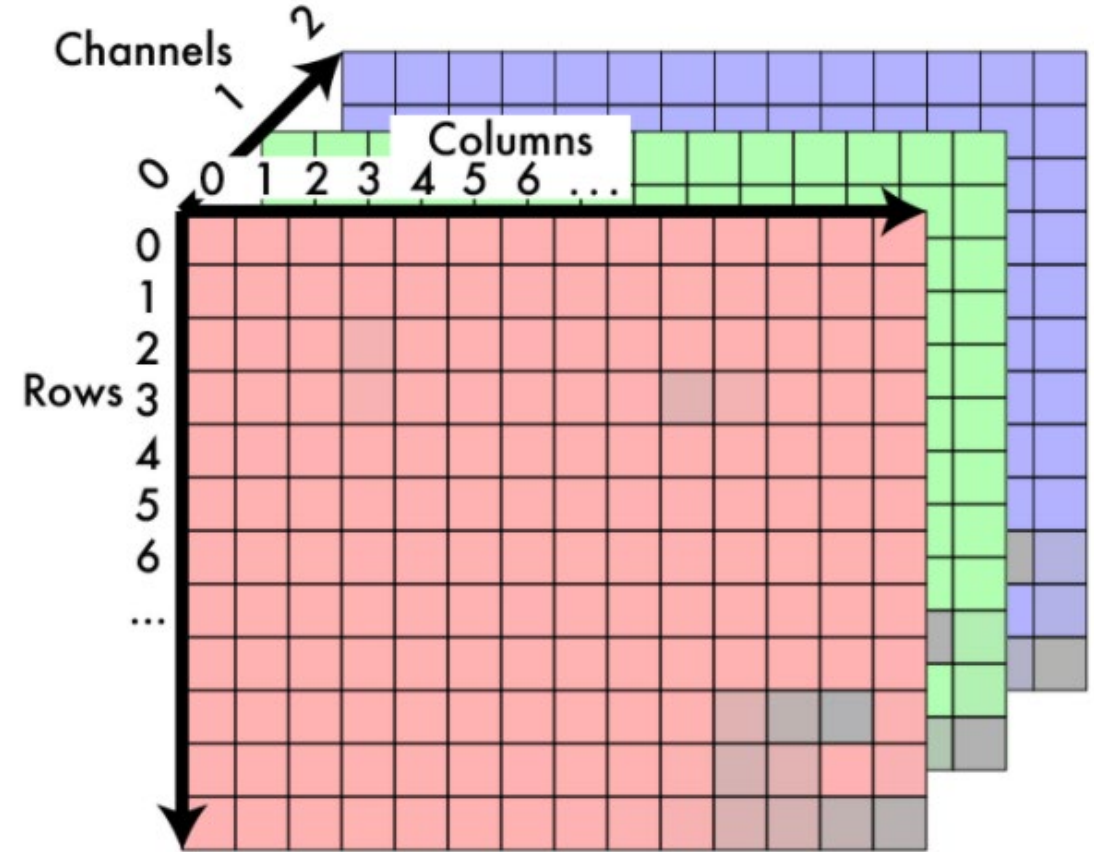
RGB images (3 channels)

what we see



What we get out of the camera

RGB images (3 channels)



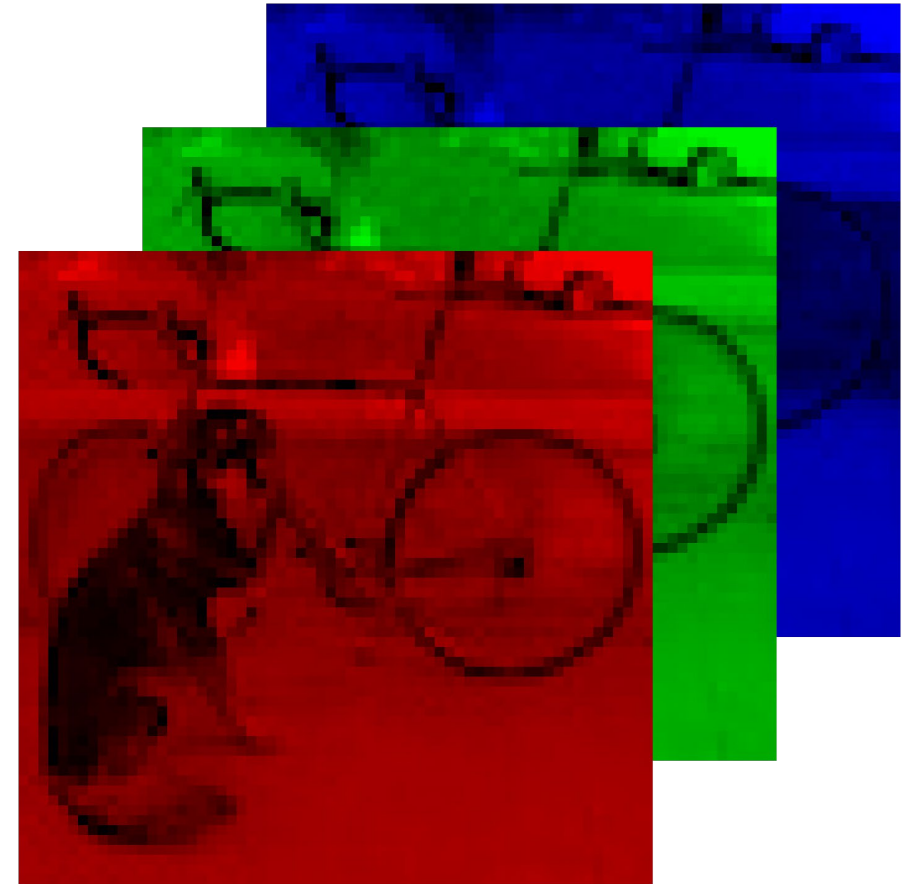
- One pixel value is a vector $[R, G, B]$
- E.g. $I(80, 24) = [78, 121, 18]$

↑
Position in image

↑
RGB values

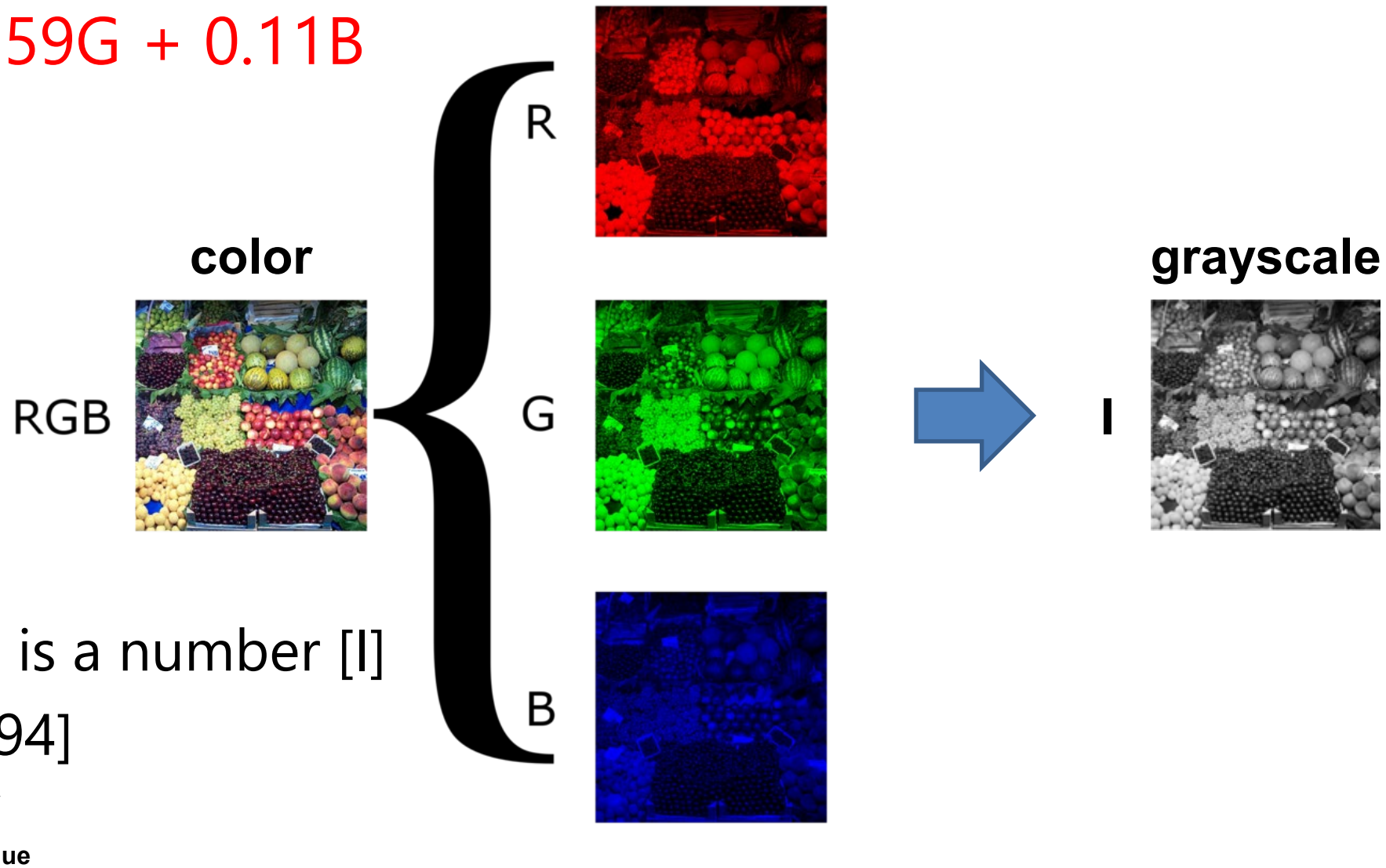
RGB information in separate “channels”

- We can match “real” colors using a mix of primaries R,G,B.
- Each channel encodes one primary. Adding the light produced from each primary mimics the original color.



Grayscale images (1 channel)

- $I \sim 0.30R + 0.59G + 0.11B$



- One pixel value is a number $[I]$
- E.g. $I(80,24) = [94]$

↑ ↑
Position in image I value

Other Image Acquisition Sources (Non-visible Light)

Image acquisition from non-visible light

Essentially any physical object that can be *radiated* and the response converted into a voltage range can be used to create an image (or if the object radiates itself)

- We just need an *illuminator* and a *detector* in such band

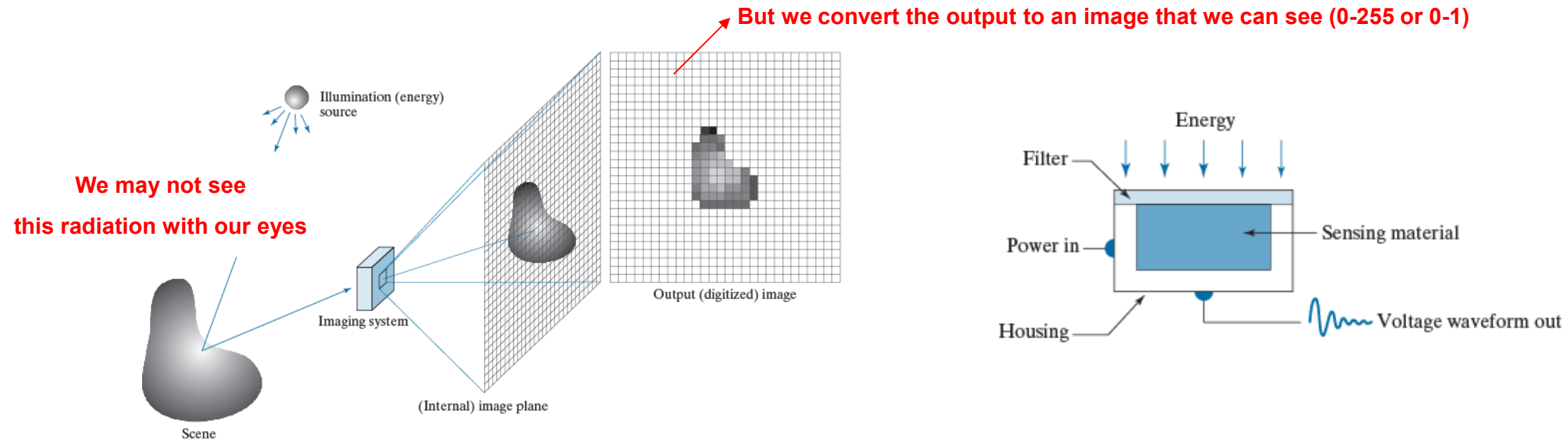
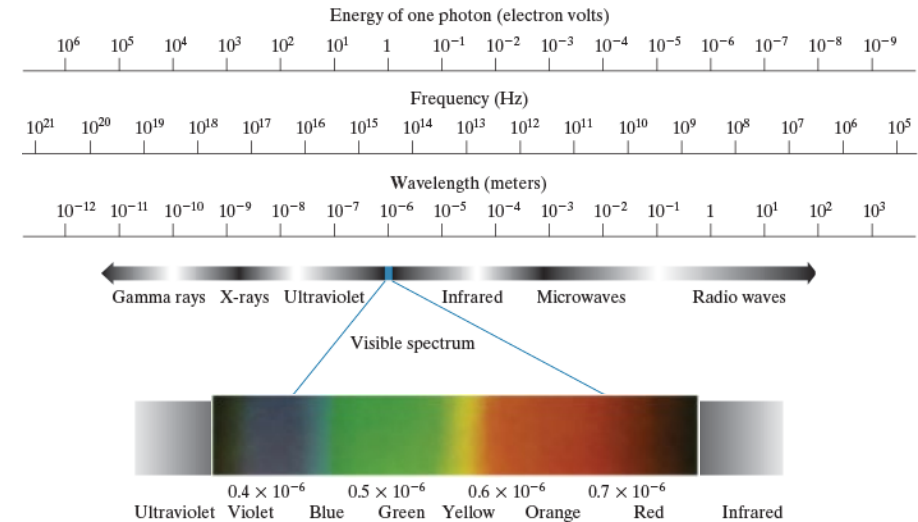


Image acquisition from non-visible light

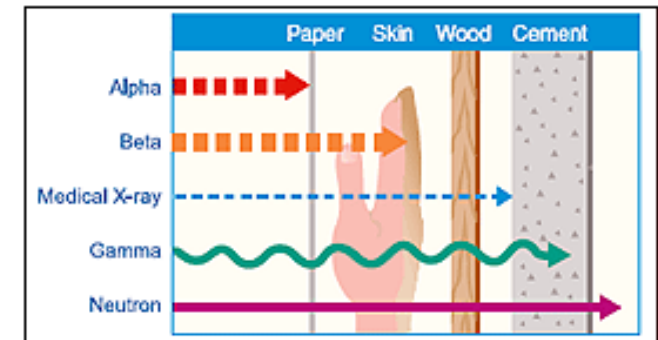
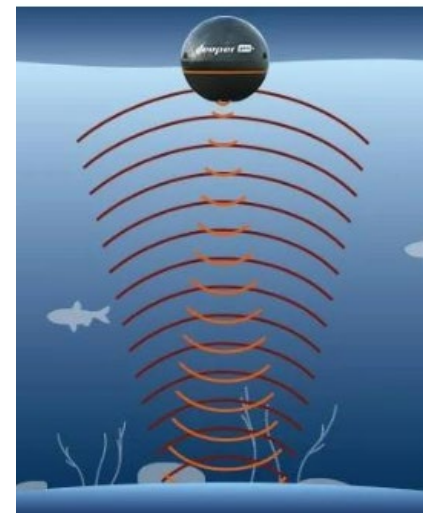
Sources of *radiation*:

- The EM spectrum 
- Acoustic
- Electron radiation
- Synthetic...



Use of the *appropriate* radiation depending on the properties of the object under such radiation

- E.g. penetrate or reflect
- Or the own object radiates in such band...



EM spectrum

- Gamma-ray

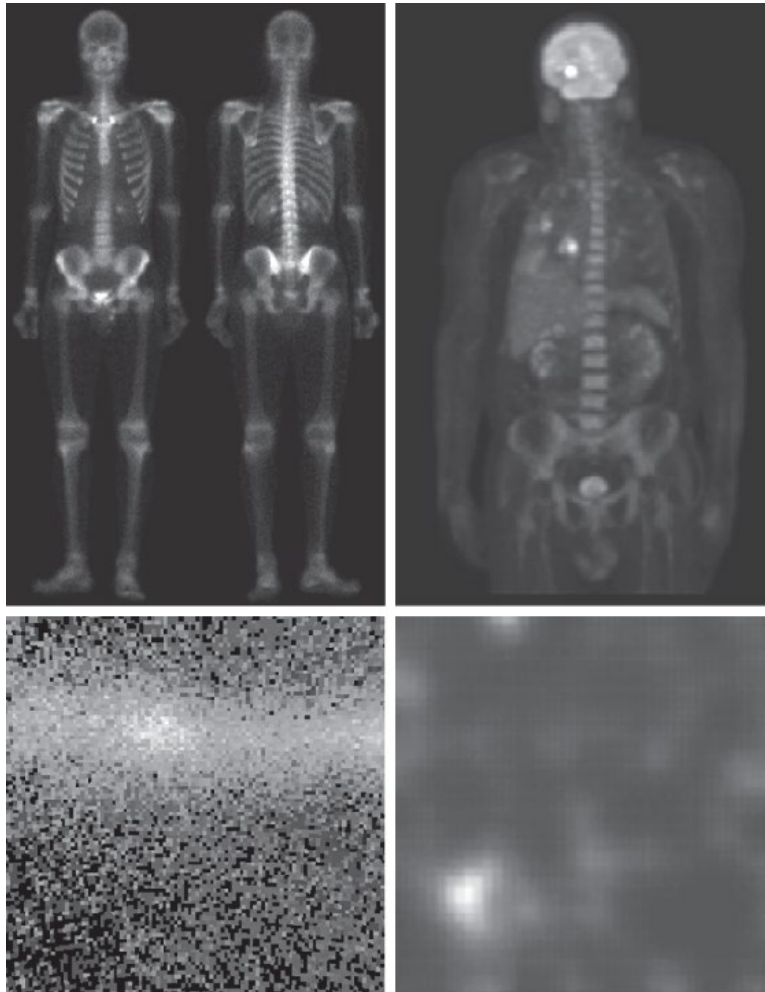
a b
c d

FIGURE 1.6

Examples of gamma-ray imaging.

(a) Bone scan.
(b) PET image.
(c) Cygnus Loop.
(d) Gamma radiation (bright spot) from a reactor valve.

(Images courtesy of (a) G.E. Medical Systems; (b) Dr. Michael E. Casey, CTI PET Systems; (c) NASA; (d) Professors Zhong He and David K. Wehe, University of Michigan.)



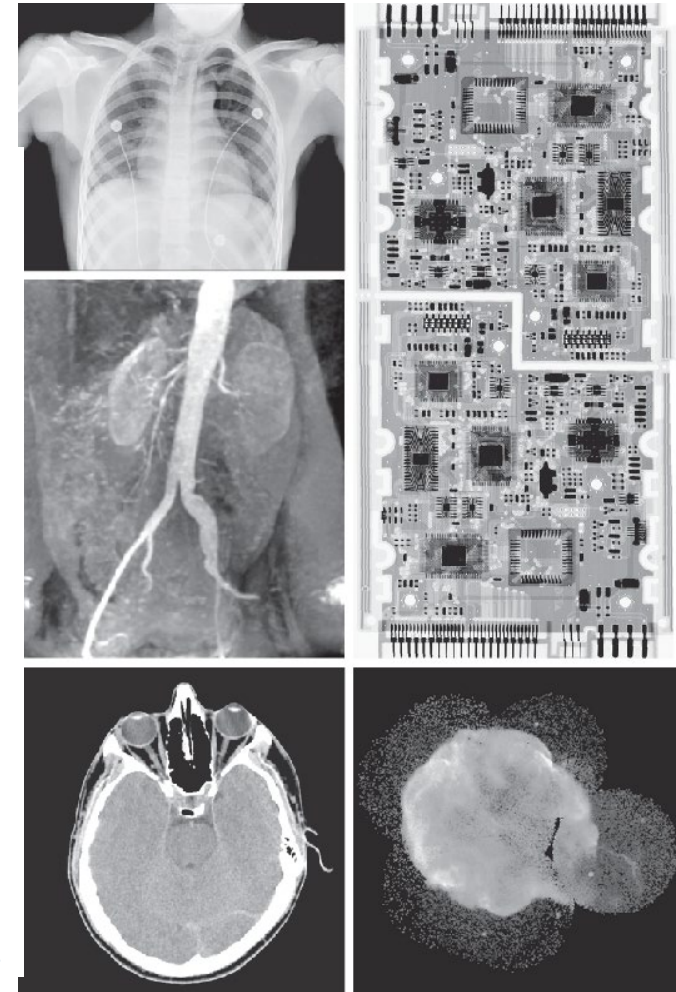
- X-ray

a d
c b e

FIGURE 1.7

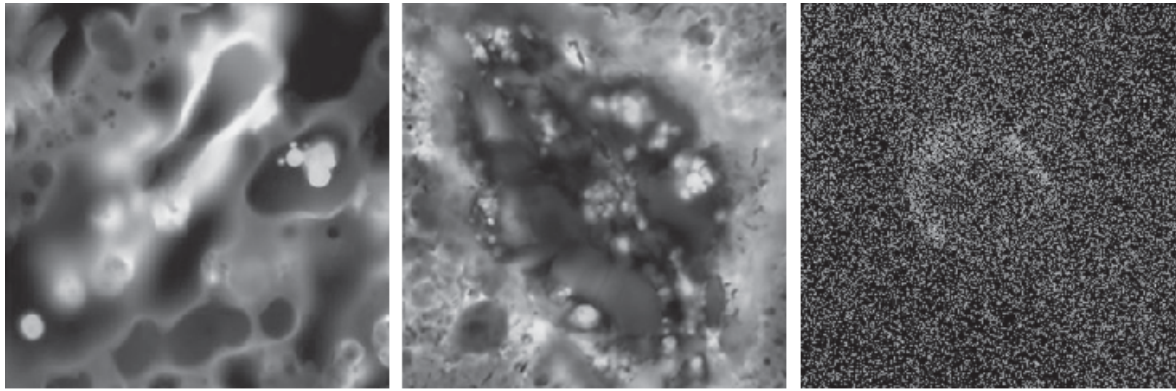
Examples of X-ray imaging.

(a) Chest X-ray.
(b) Aortic angiogram.
(c) Head CT.
(d) Circuit boards.
(e) Cygnus Loop.
(Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center; (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, Univ. of Michigan Medical School; (d) Mr. Joseph E. Pascente, Lixi, Inc.; and (e) NASA.)



EM spectrum

- Ultra-violet



a b c

FIGURE 1.8 Examples of ultraviolet imaging. (a) Normal corn. (b) Corn infected by smut. (c) Cygnus Loop. (Images (a) and (b) courtesy of Dr. Michael W. Davidson, Florida State University, (c) NASA.)

- Visible, infra-red

TABLE 1.1
Thematic bands
of NASA's
LANDSAT
satellite.

Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.53–0.61	Measures plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.78–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content: soil/vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Short-wave infrared	2.09–2.35	Mineral mapping

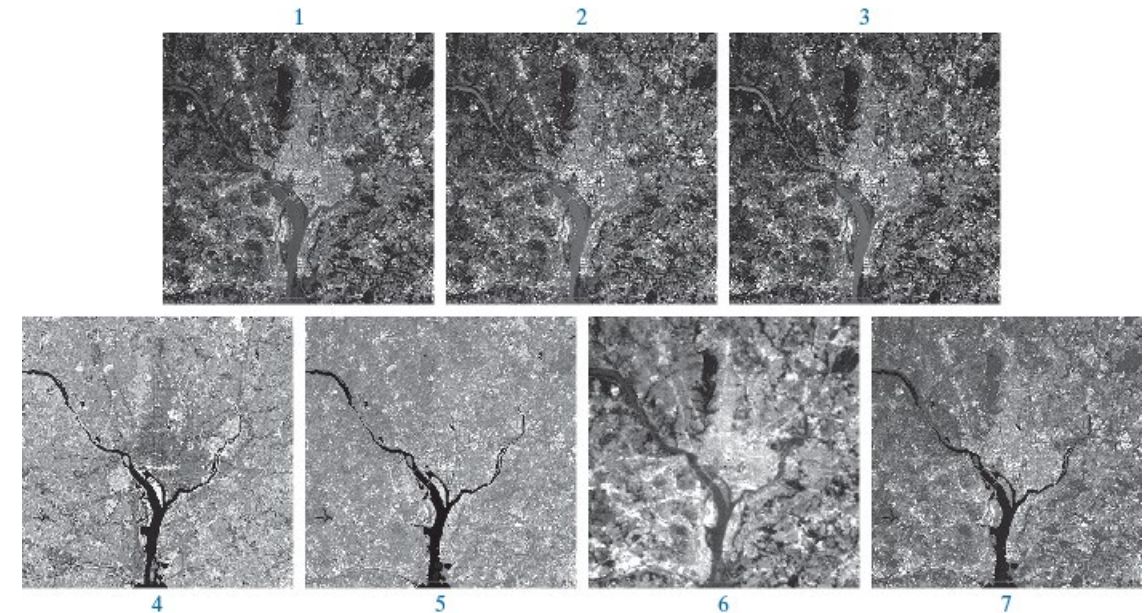


FIGURE 1.10 LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)

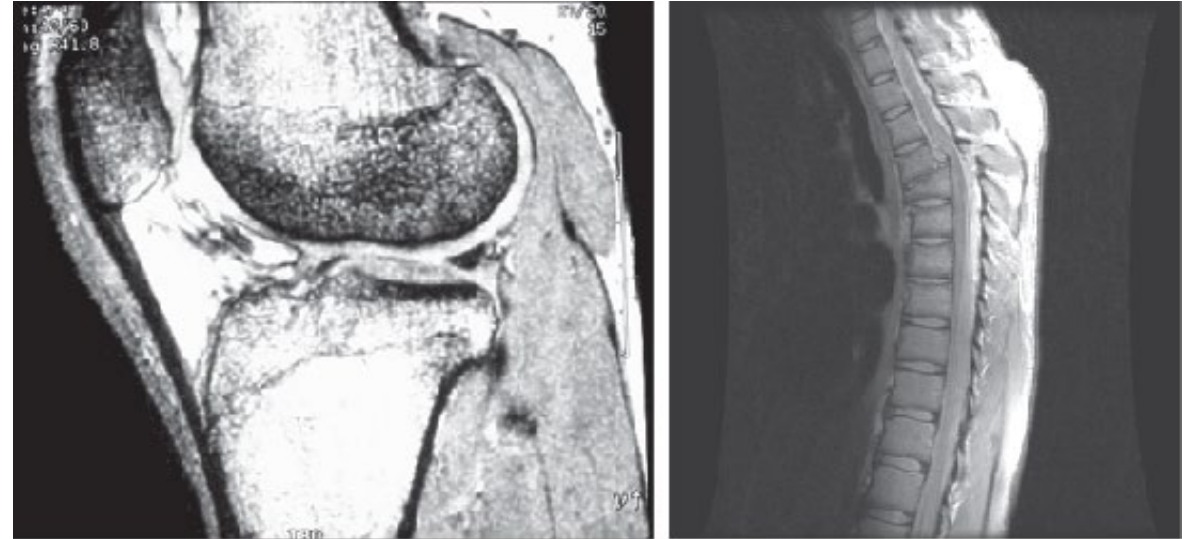
EM spectrum

- Microwaves



FIGURE 1.16
Spaceborne radar
image of
mountainous
region in
southeast Tibet.
(Courtesy of
NASA.)

- Radioband



a b

FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Figure (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

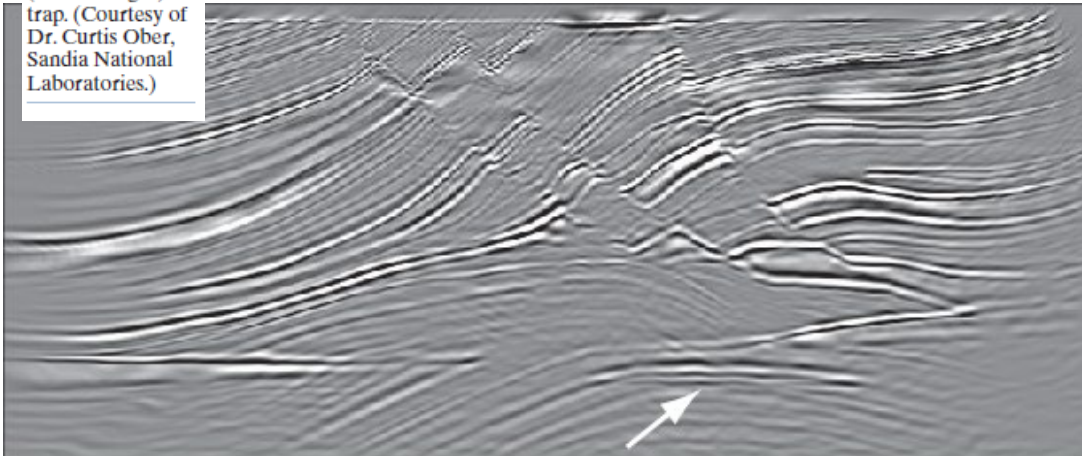
Other sources

- Sound sonar

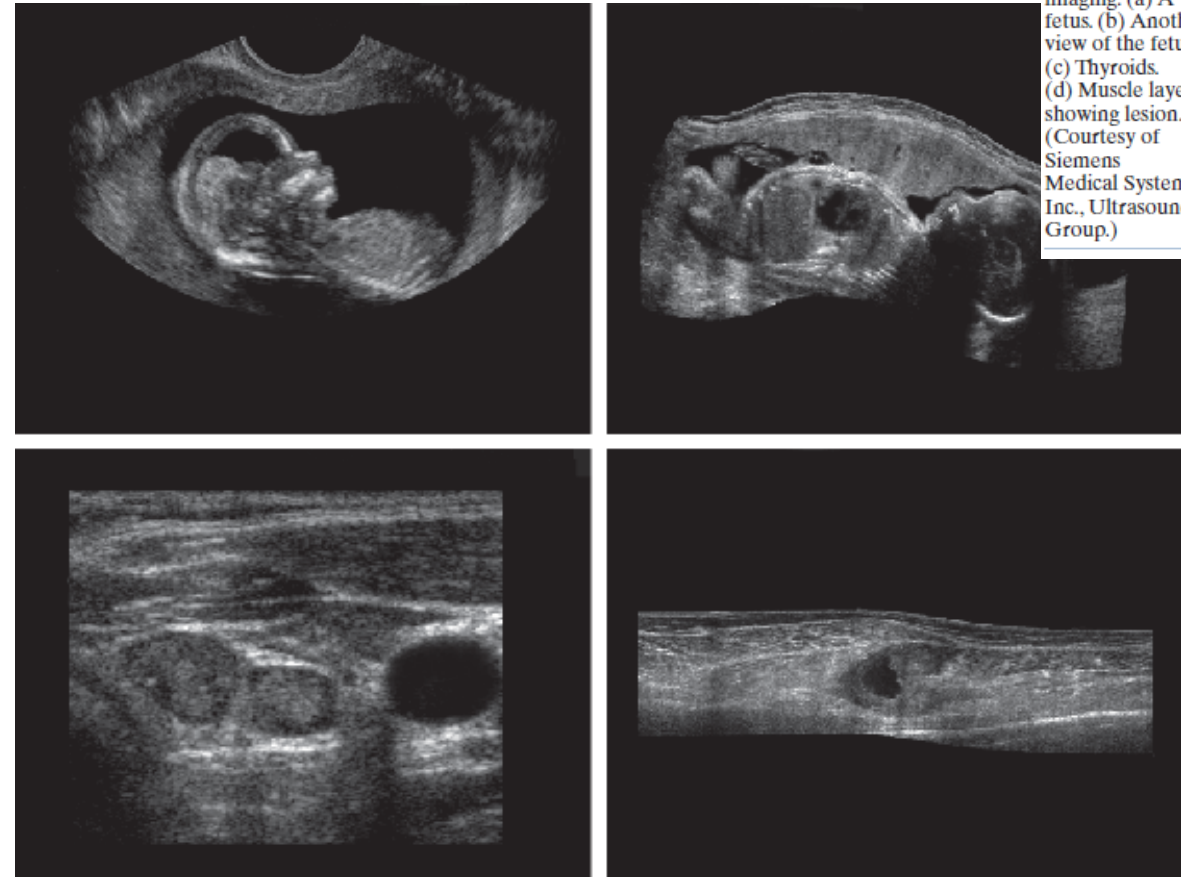


FIGURE 1.19

Cross-sectional image of a seismic model. The arrow points to a hydrocarbon (oil and/or gas) trap. (Courtesy of Dr. Curtis Ober, Sandia National Laboratories.)



- Ultrasound



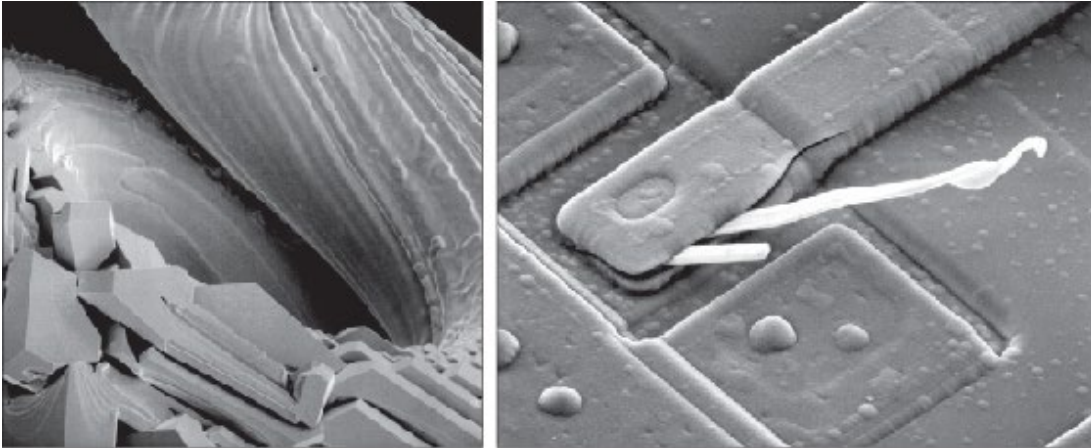
a b
c d

FIGURE 1.20

Examples of ultrasound imaging. (a) A fetus. (b) Another view of the fetus. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)

Other sources

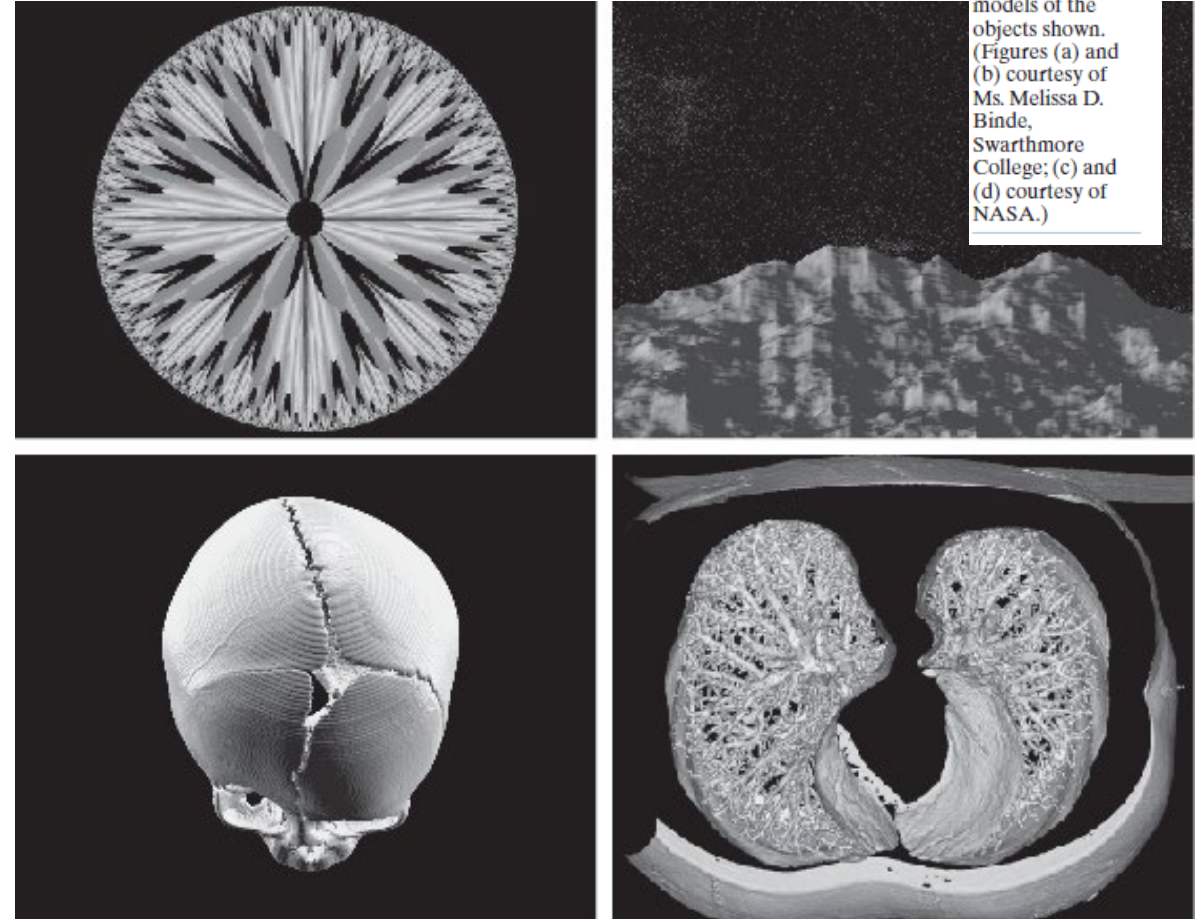
- Electron microscopy



a b

FIGURE 1.21 (a) $250\times$ SEM image of a tungsten filament following thermal failure (note the shattered pieces on the lower left). (b) $2500\times$ SEM image of a damaged integrated circuit. The white fibers are oxides resulting from thermal destruction. (Figure (a) courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene; (b) courtesy of Dr. J. M. Hudak, McMaster University, Hamilton, Ontario, Canada.)

- Synthetic

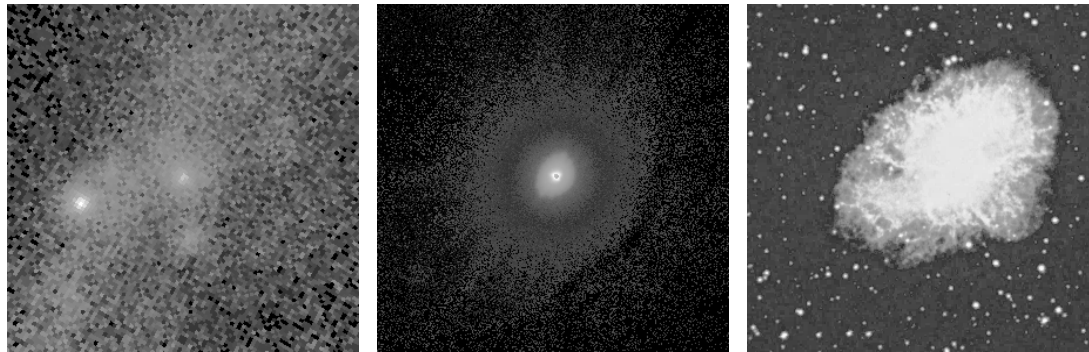


a b
c d

FIGURE 1.22 (a) and (b) Fractal images. (c) and (d) Images generated from 3-D computer models of the objects shown. (Figures (a) and (b) courtesy of Ms. Melissa D. Binde, Swarthmore College; (c) and (d) courtesy of NASA.)

Same object under different radiation

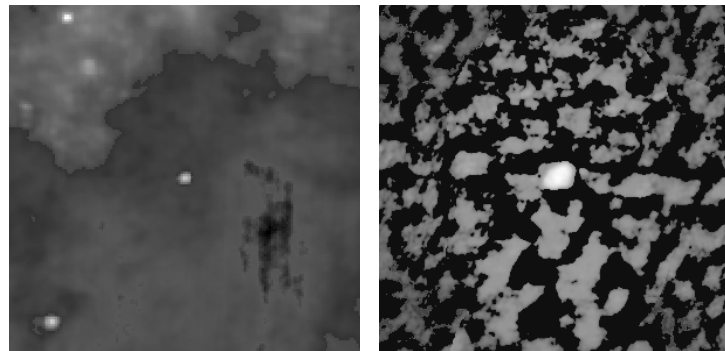
- Reveals different properties



Gamma

X-ray

Optical



Infrared

Radio

TABLE 1.1
Thematic bands
of NASA's
LANDSAT
satellite.

Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.53–0.61	Measures plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.78–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content: soil/vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Short-wave infrared	2.09–2.35	Mineral mapping

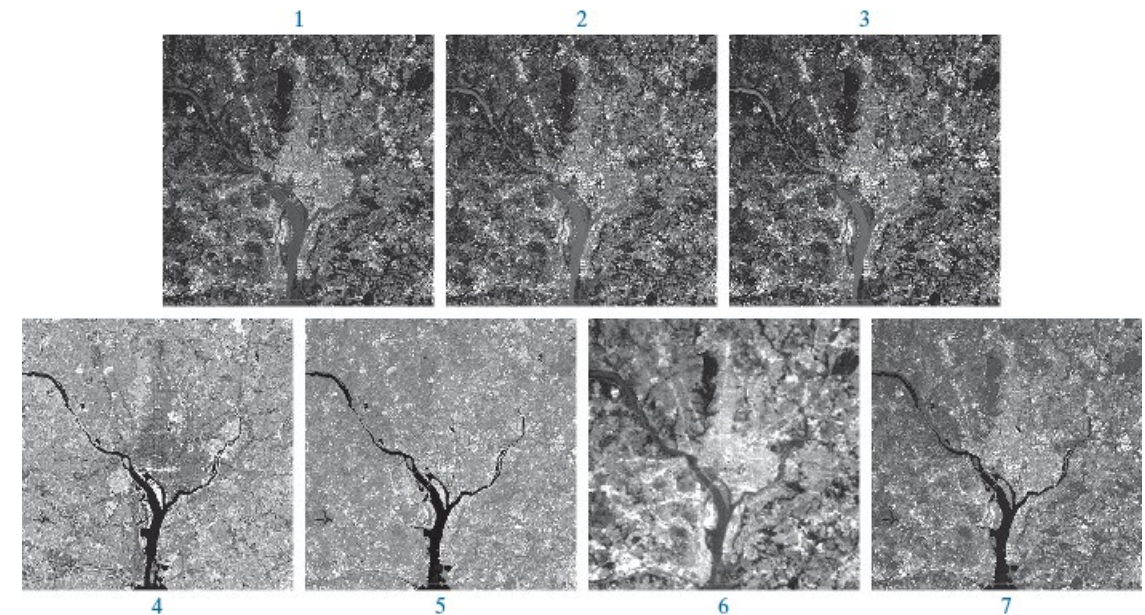


FIGURE 1.18 Images of the Crab Pulsar (in the center of each image) covering the electromagnetic spectrum. (Courtesy of NASA.)

Image Sensing Pipeline With Cameras

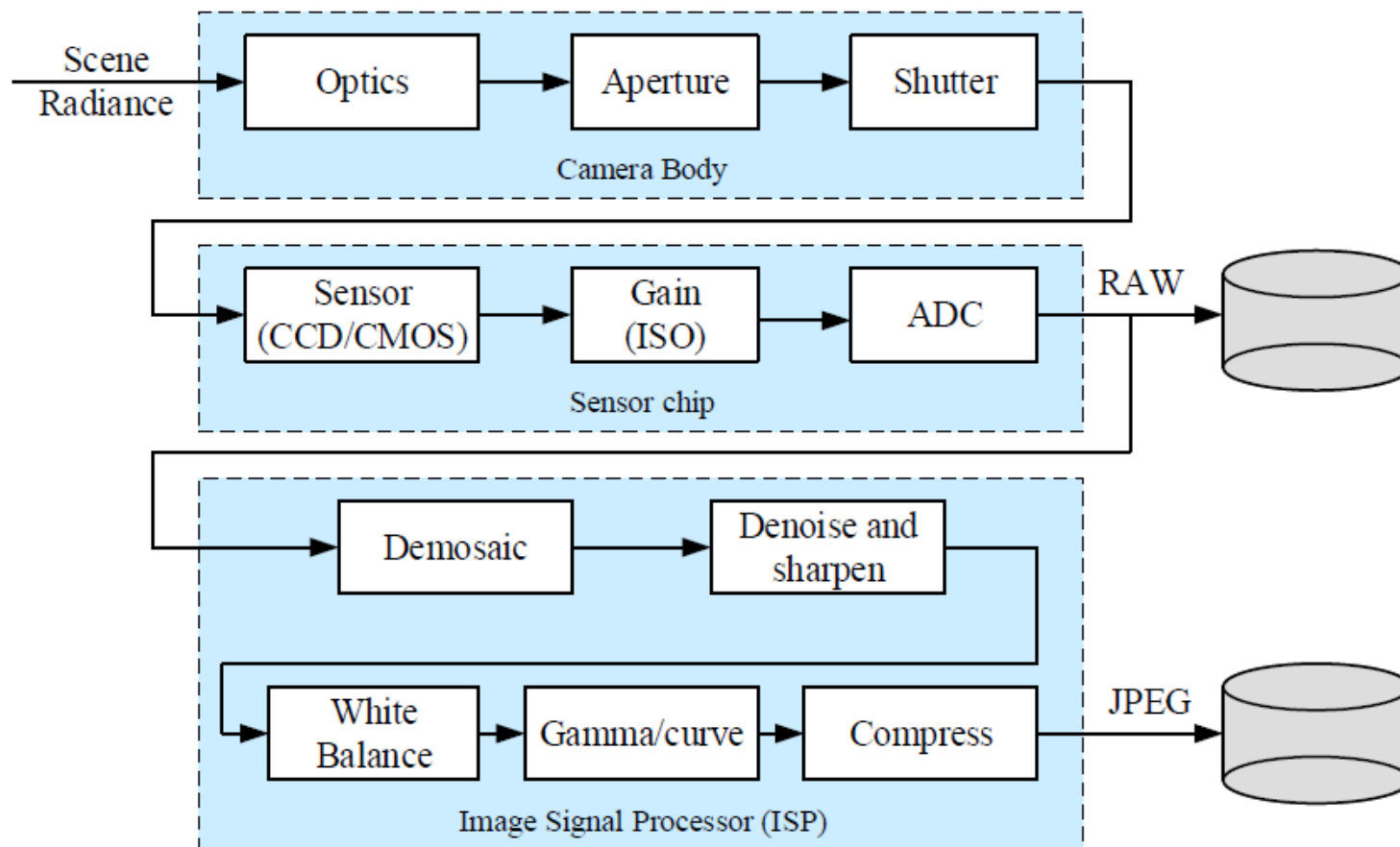
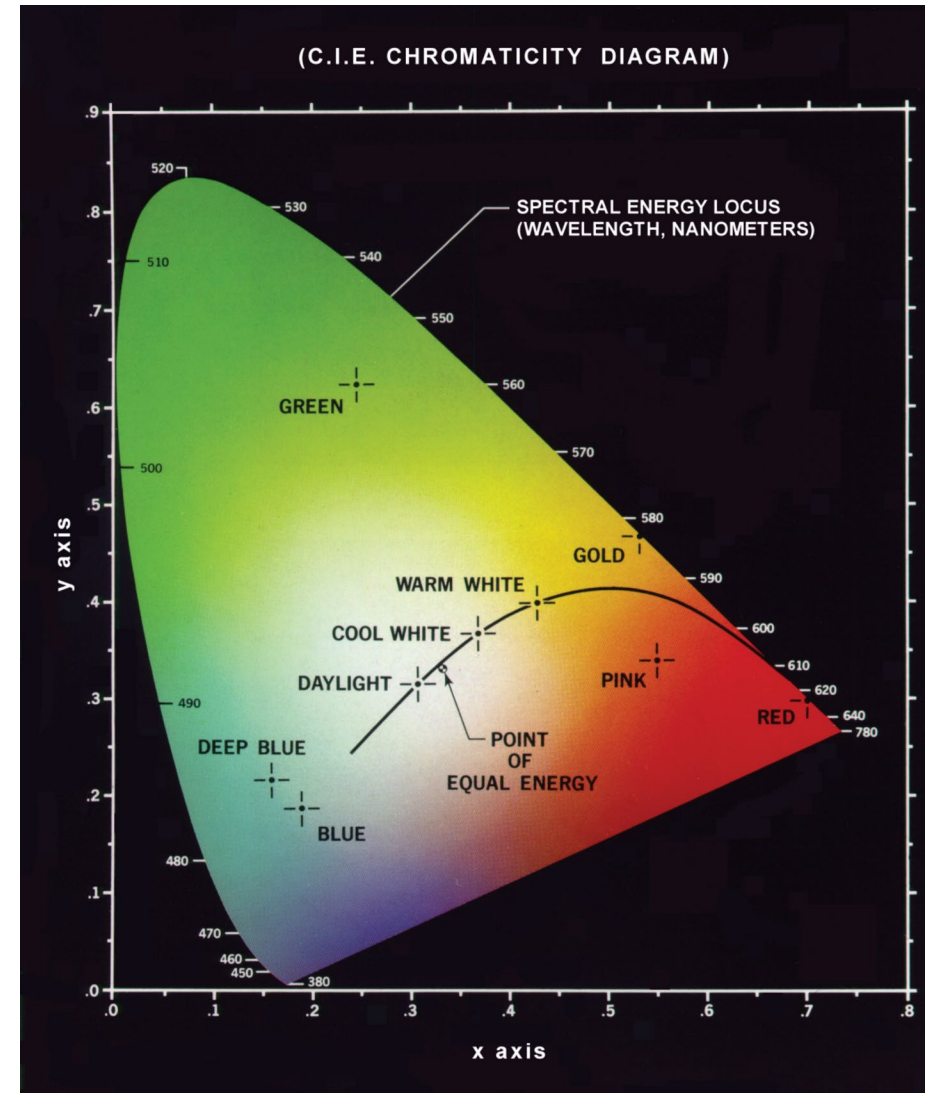


Figure 2.23 Image sensing pipeline, showing the various sources of noise as well as typical digital post-processing steps.

Color Spaces to Represent Images

Linear Color Space CIE XYZ

- Pick some primaries
- Can mix those primaries to match any color inside the triangle
- There is a commission that studies color! **Commission internationale de l'éclairage (CIE)** is a 100-year-old organization that creates international standards related to light and color



Linear Color Space CIE XYZ

FIGURE 6.6
Illustrative color gamut of color monitors (triangle) and color printing devices (shaded region).

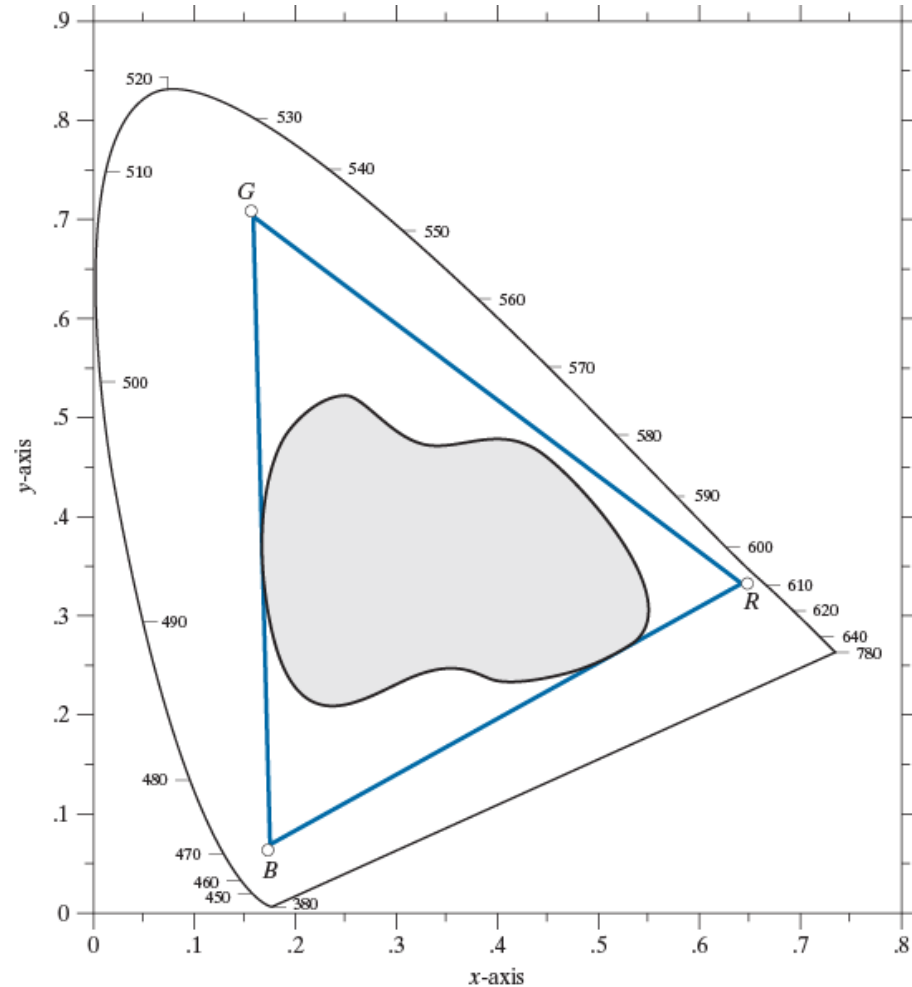
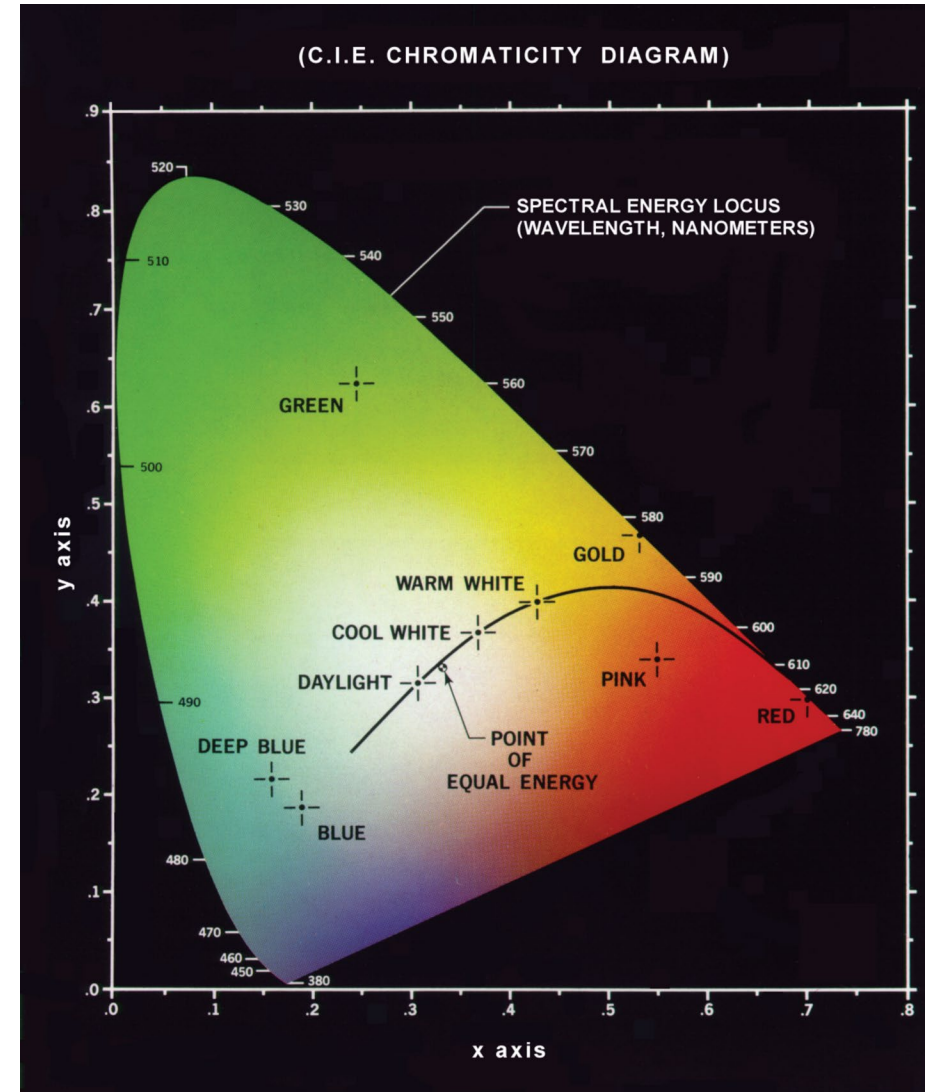


FIGURE 6.5
The CIE chromaticity diagram. (Courtesy of the General Electric Co., Lighting Division.)



RGB Cube

- Cartesian RGB space, used by **cameras, monitors, etc.**
- Represent a color with 3 numbers, e.g. $R,G,B=255,0,255$ or $R,G,B=1,0,1$ (normalized)... **WHAT COLOR IS THAT?**

(gray values: diagonal $R=G=B$)

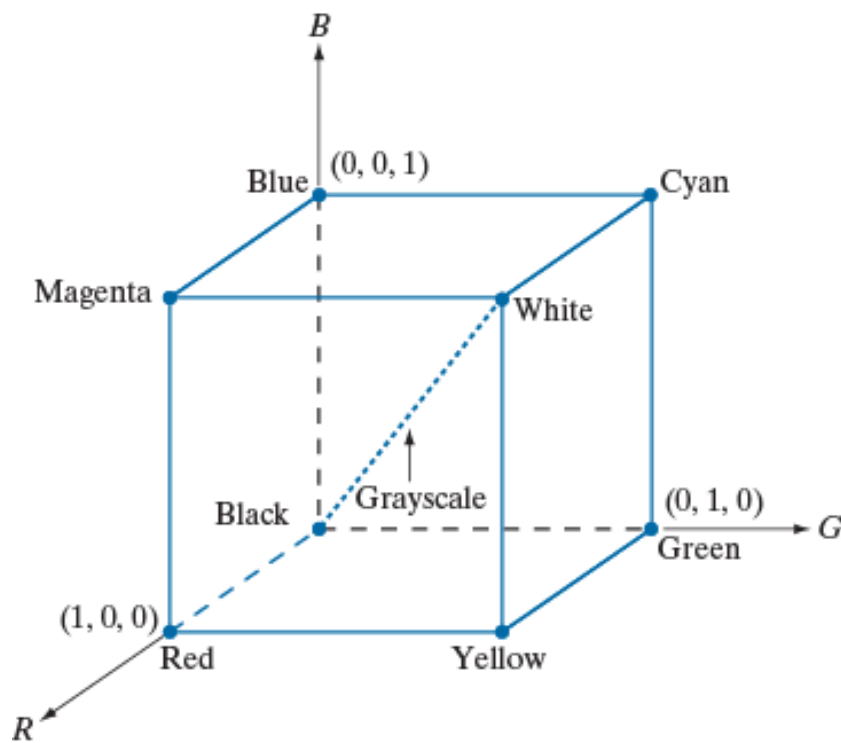


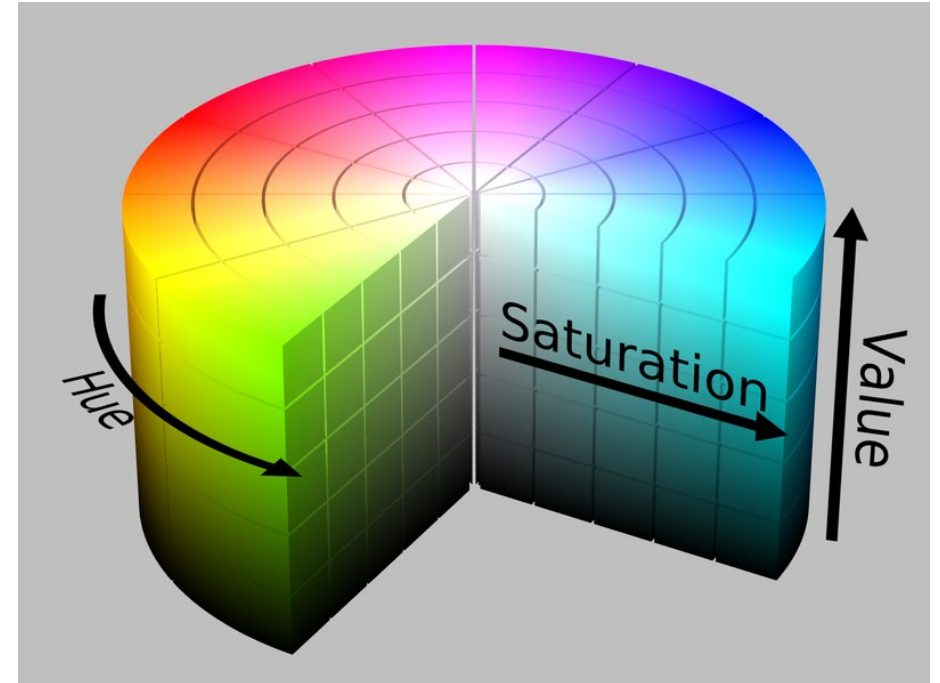
FIGURE 6.7
Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point (1, 1, 1).



FIGURE 6.8
A 24-bit RGB color cube.

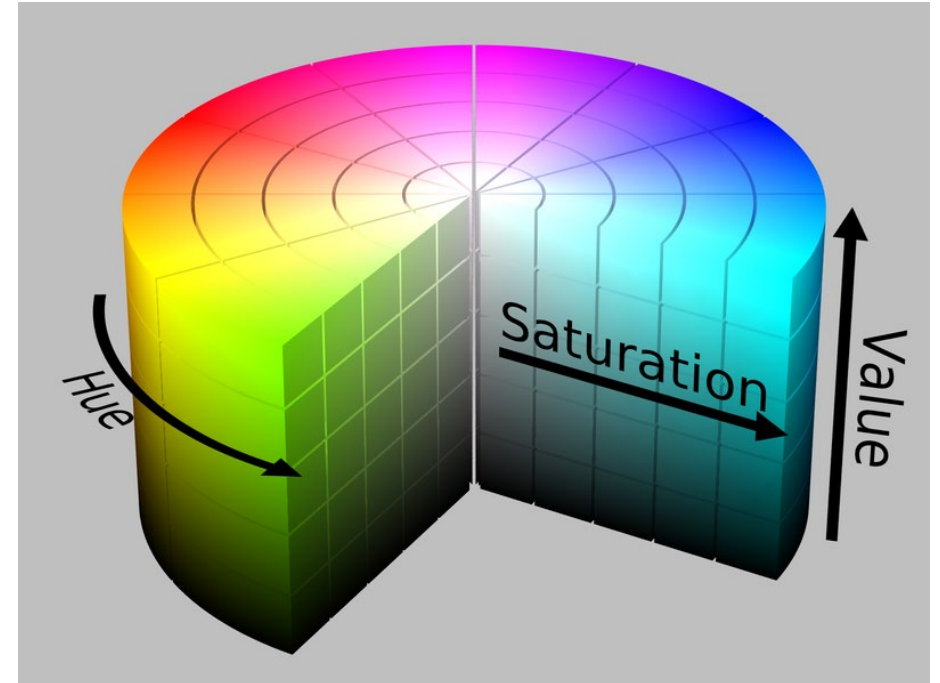
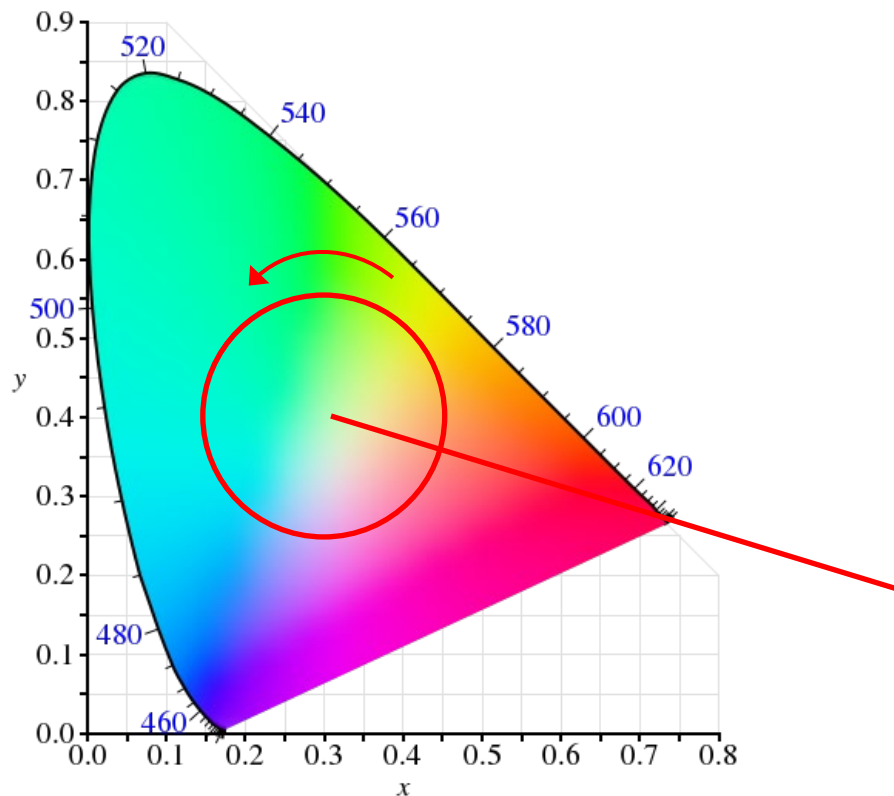
HSV/HSI Cylinder = Hue, Saturation, Value/Intensity

- Based on perception of light
- **Hue**: what color
- **Saturation**: how much color (diluted with white)
- **Value**: how bright (diluted with black)



RGB vs. HSV/HSI

- **Hue:** what color ->
Angle wrt the red color

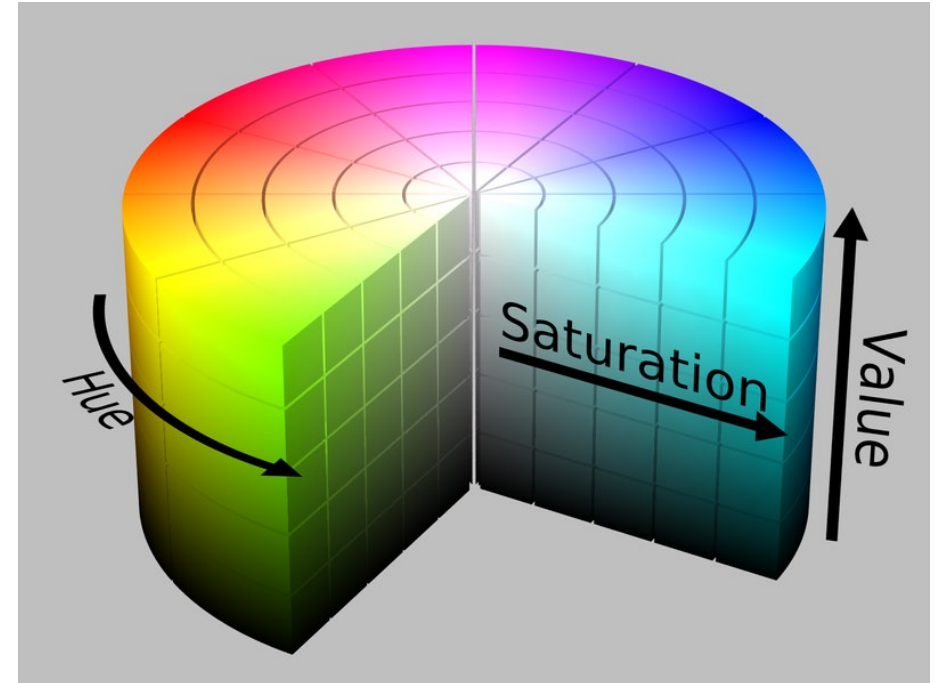
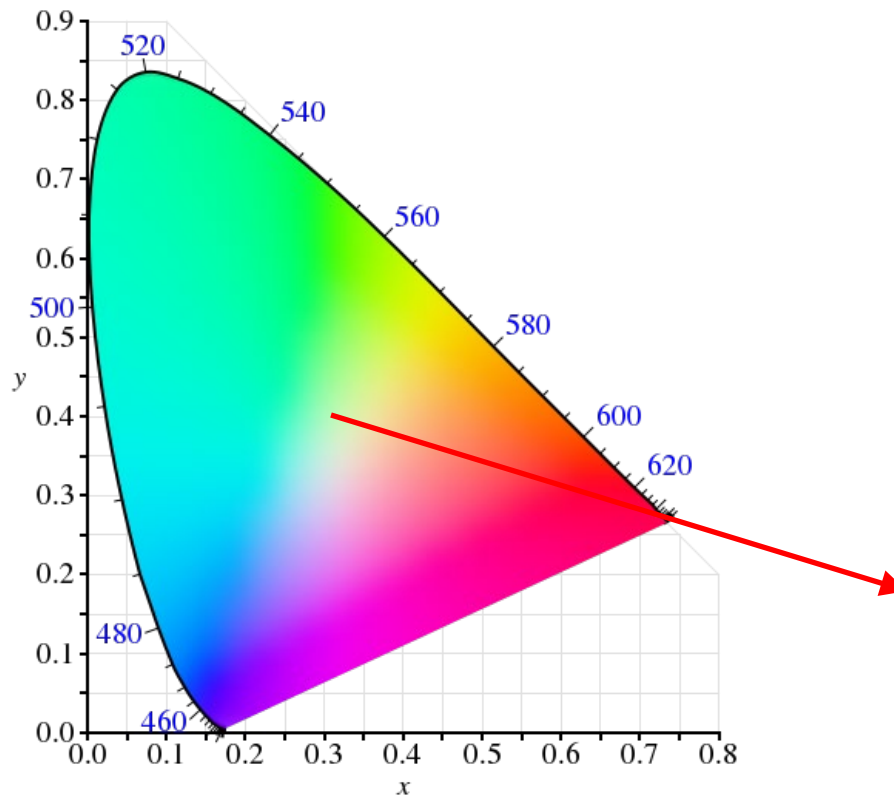


Notice that

- The CIE diagram (left) is ~like the top part of the HSV cube seen from above

RGB vs. HSV/HSI

- **Saturation**: how much color (diluted with white) -> Distance from white

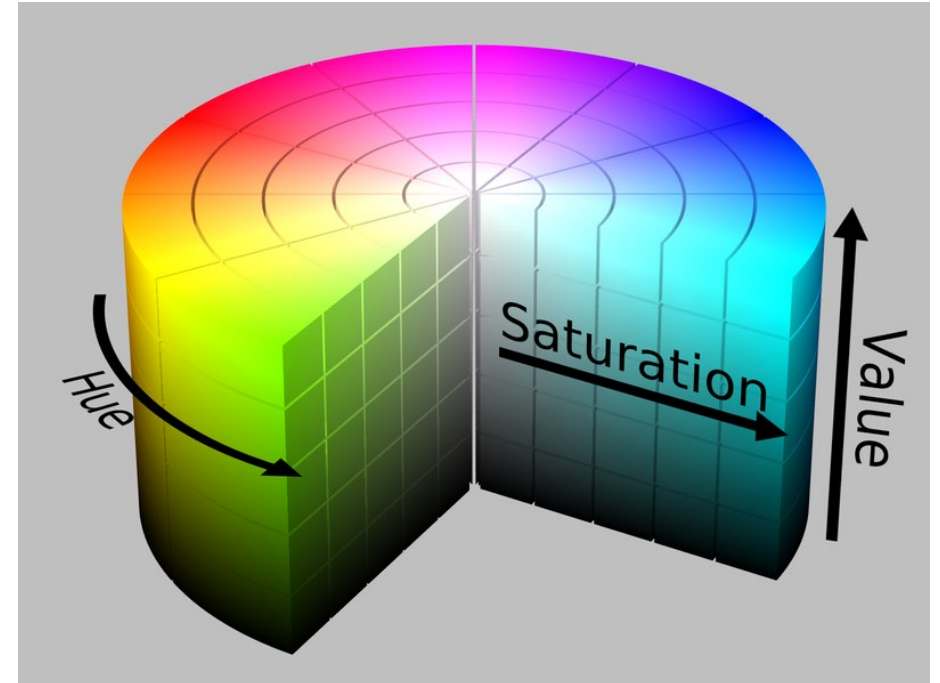
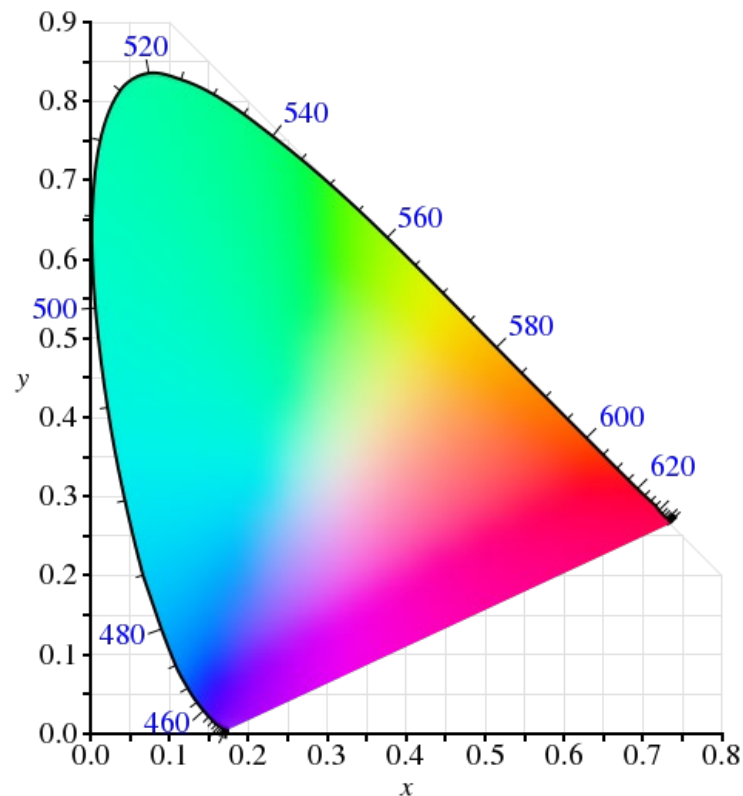


Notice that

- The center of the CIE diagram is the white "color"

RGB vs. HSV/HSI

- **Value:** how bright (diluted with black) -> ??

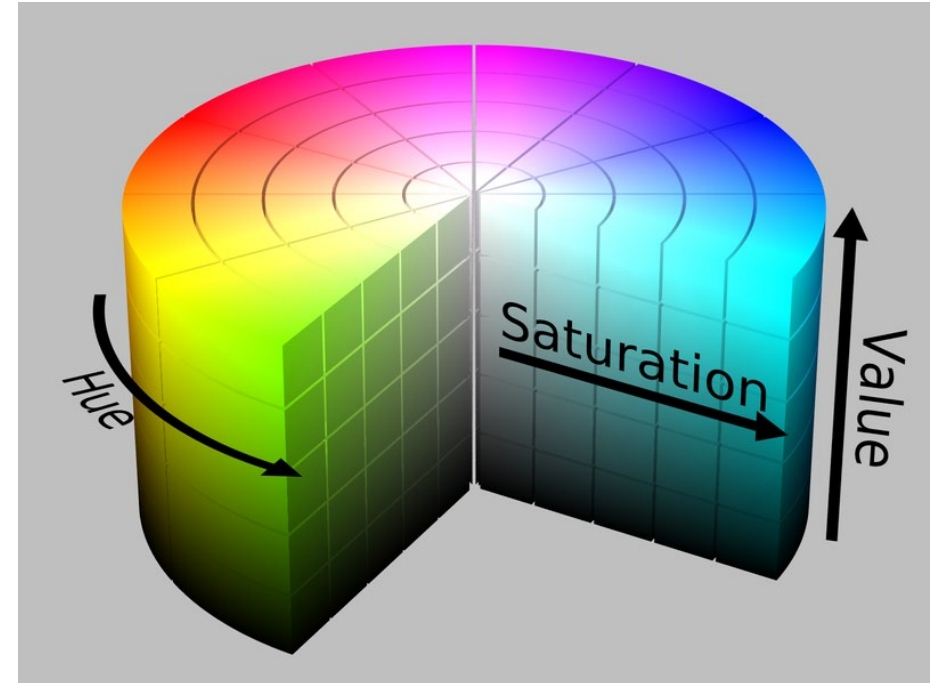


Notice that

- The CIE diagram only contains pure colours (no brightness info)

RGB vs. HSV/HSI

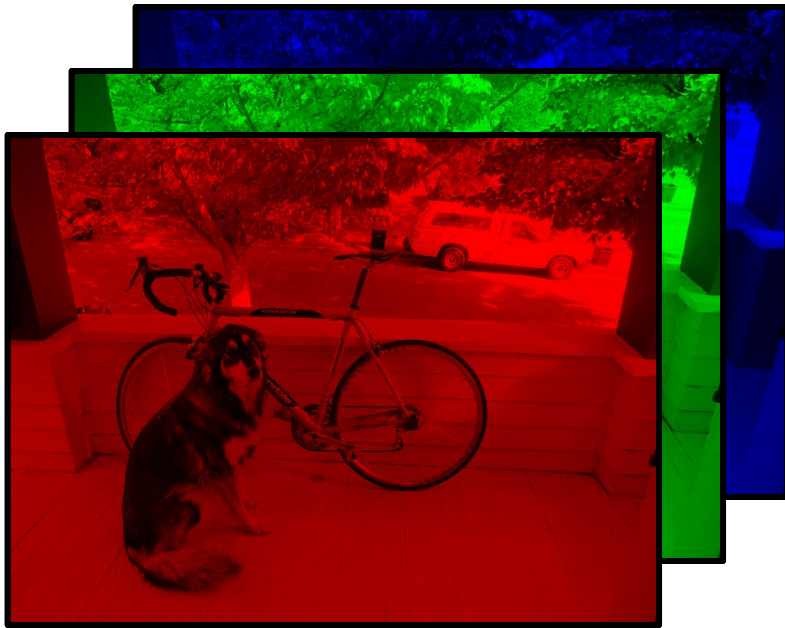
- **Value**: how bright (diluted with black) ->
- Approximation of how "bright" the image is without color information, i.e. **converting the image to grayscale**
- Typically $\sim .30R + .59G + .11B$



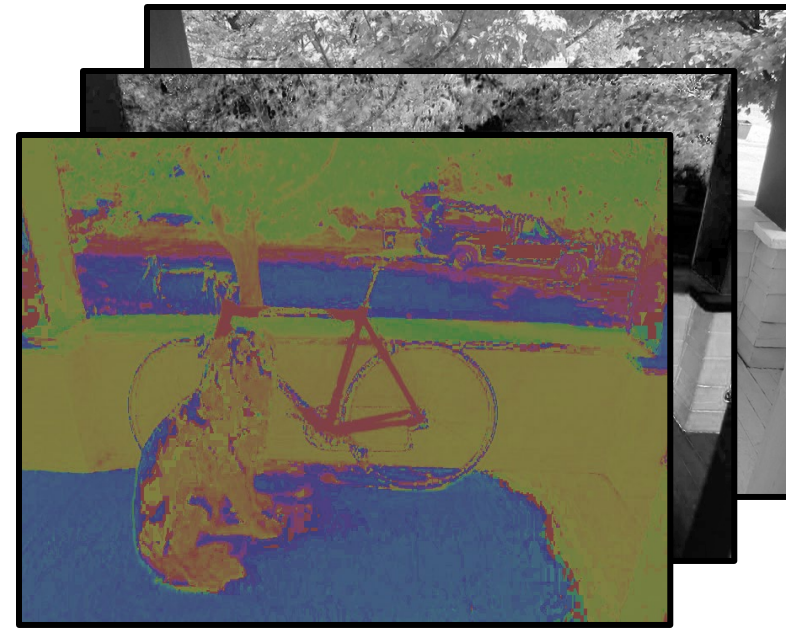
Easy transforms by manipulating HSV/HSI



3 channels, same info in different ways



R, G, B

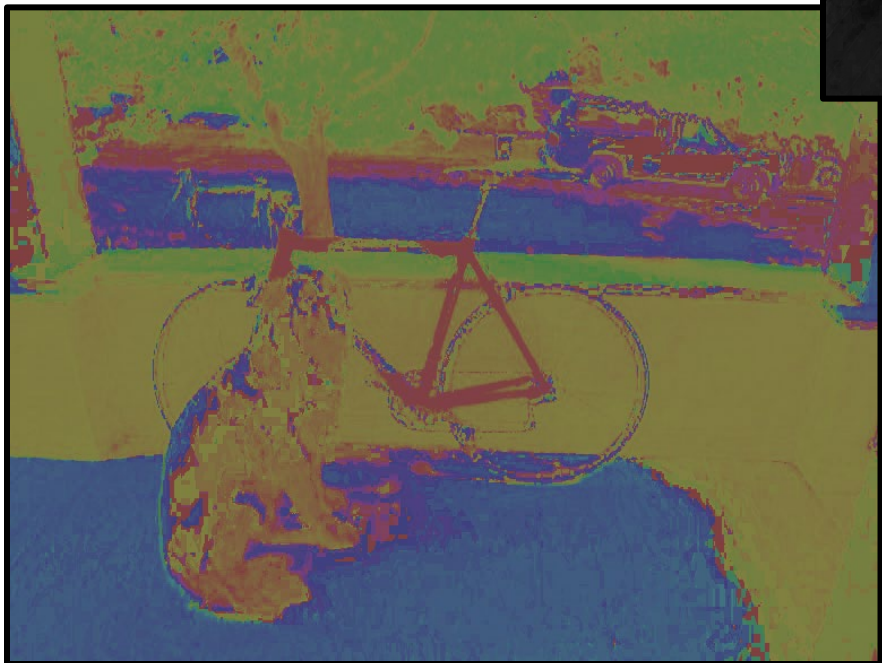


H, S, V

Hue

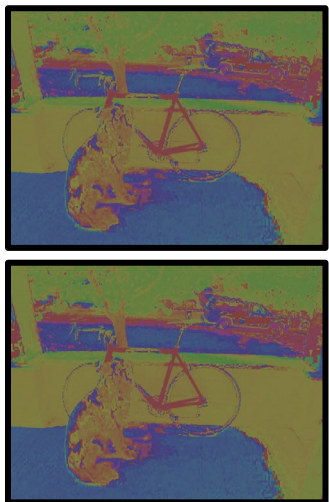
Saturation

Value



More saturation = intense colors

H Channel



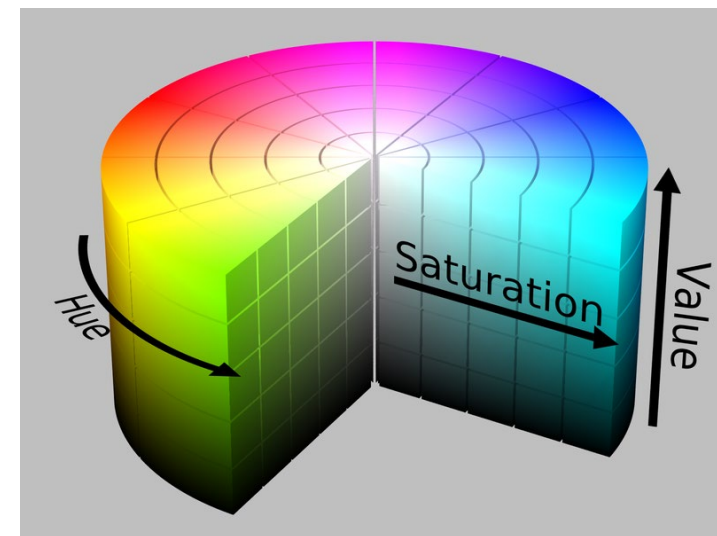
S Channel



V Channel

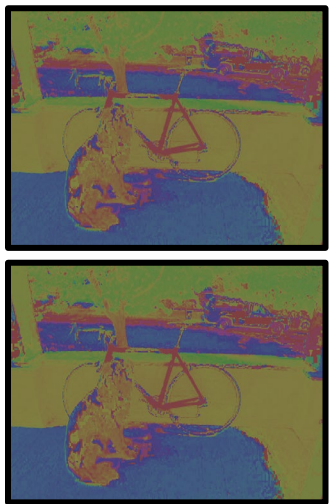


2x

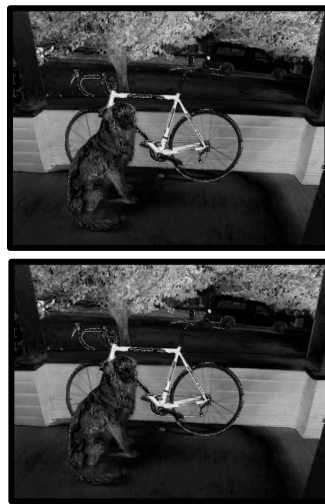


More value = lighter image

H Channel



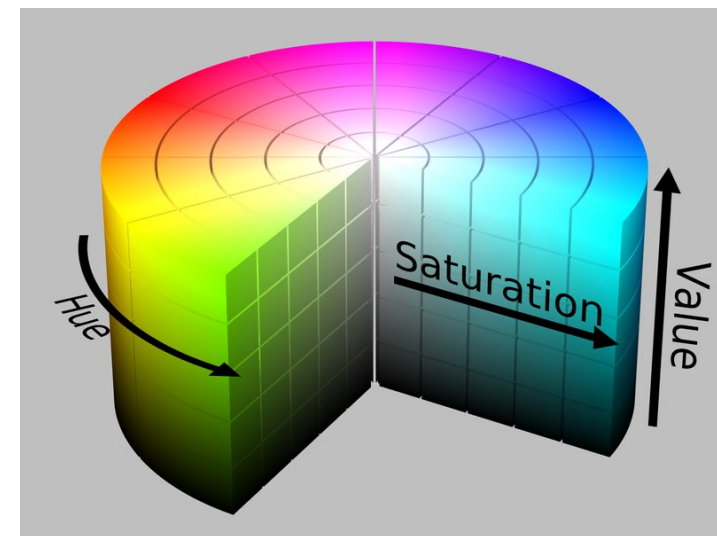
S Channel



V Channel

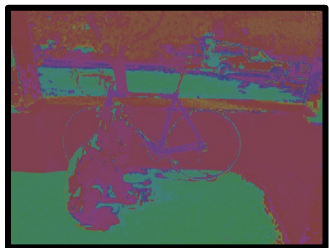
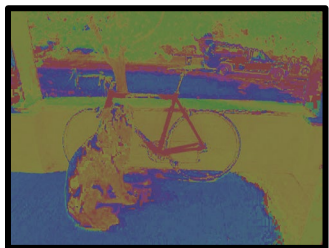


2x



Shift hue = shift colors

H Channel



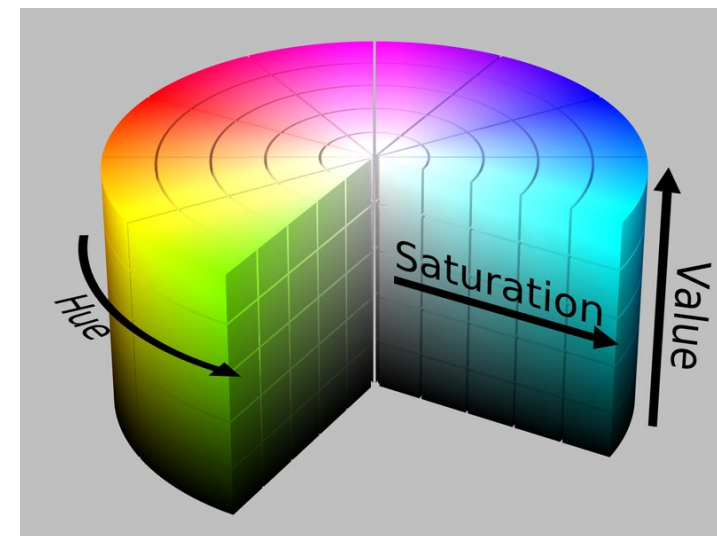
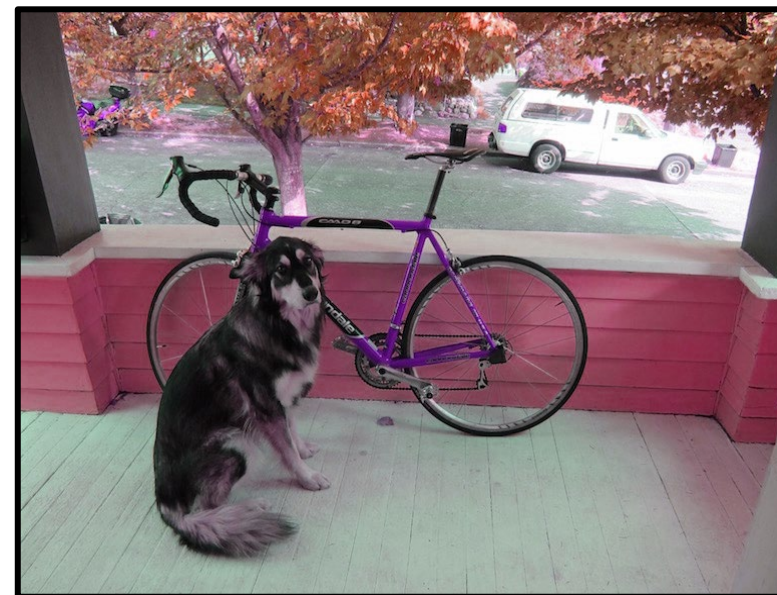
S Channel



V Channel

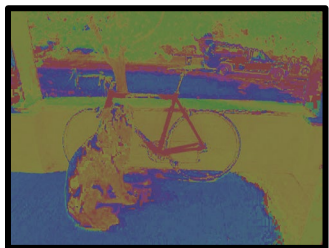


-0.2



Set hue to your favorite color!

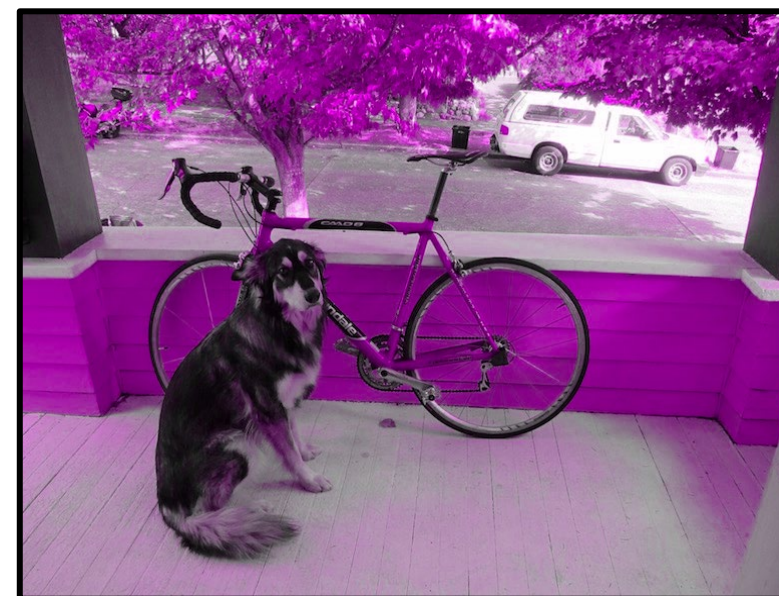
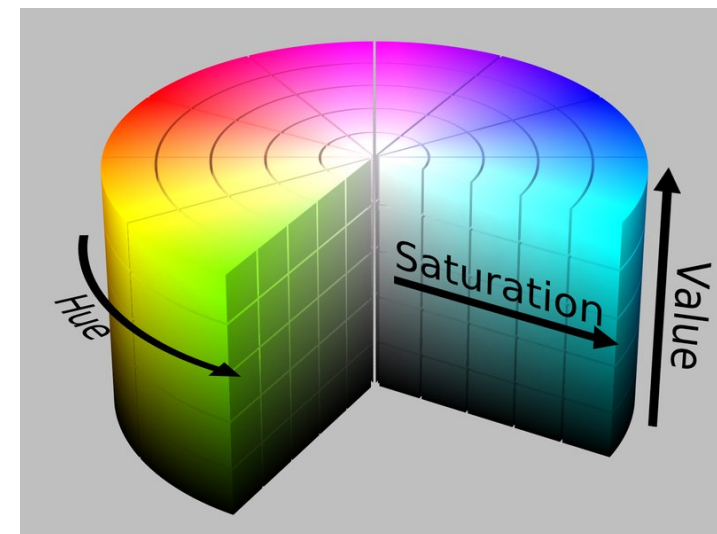
H Channel



S Channel

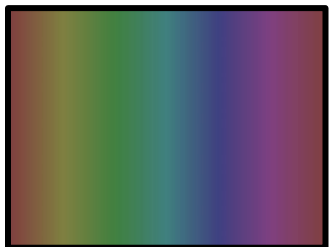
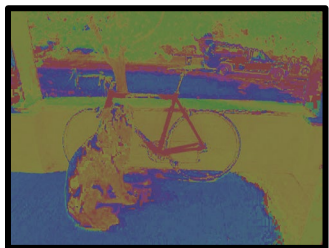


V Channel



Or pattern!

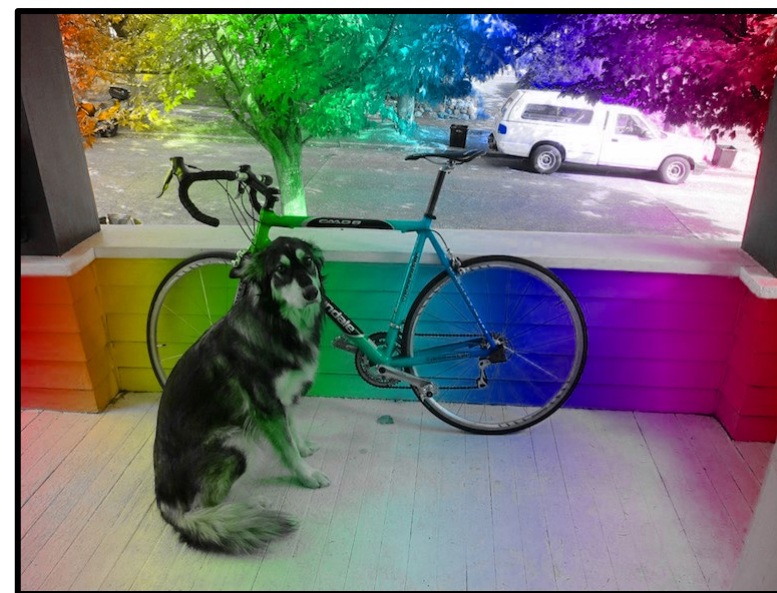
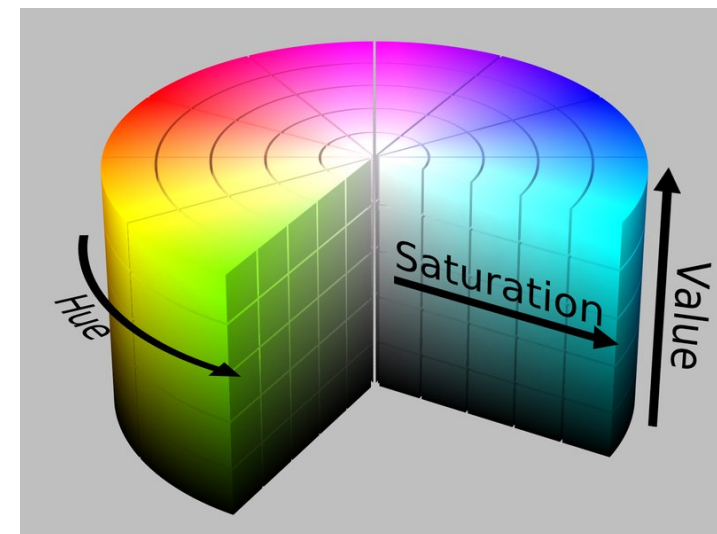
H Channel



S Channel



V Channel

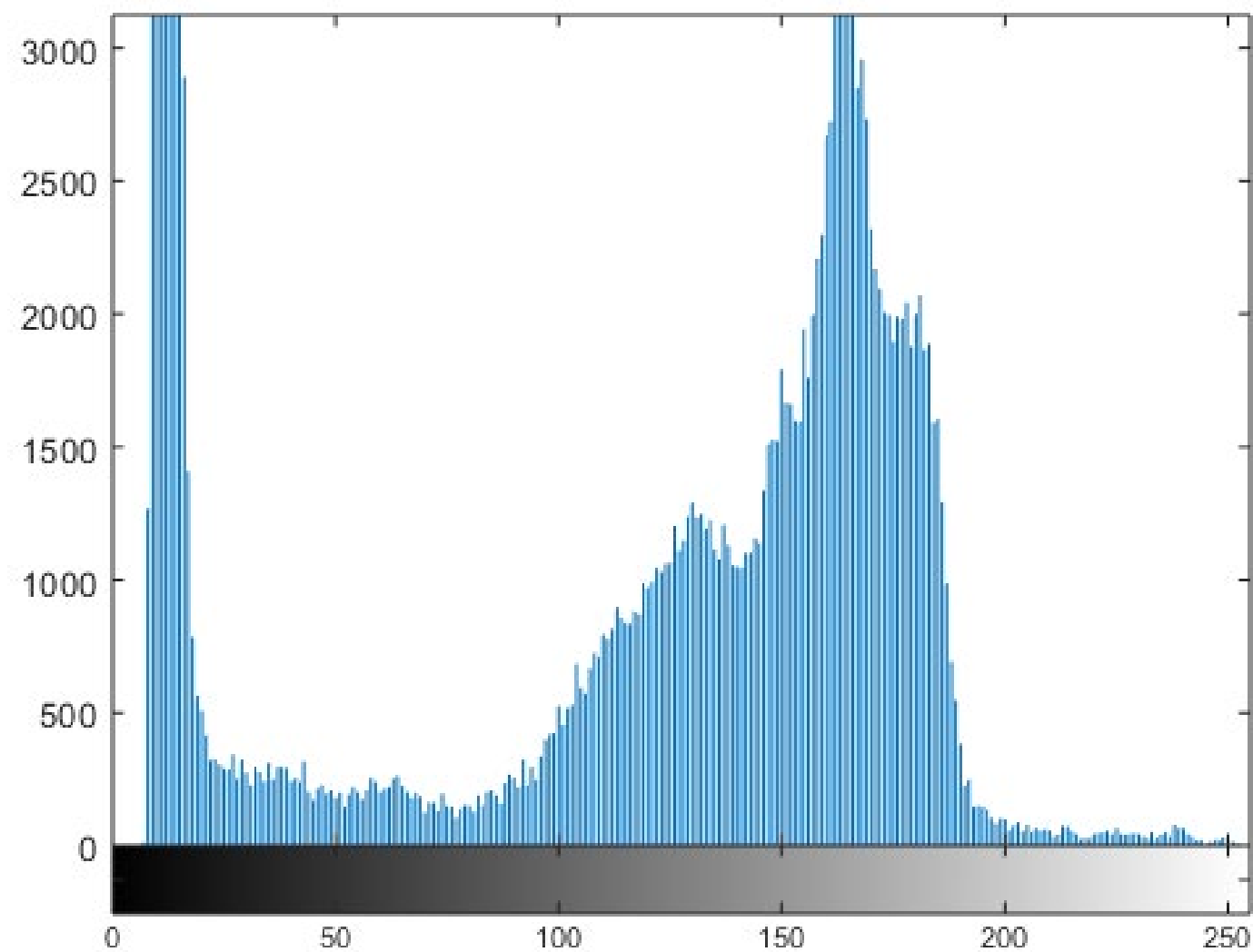


Histograms

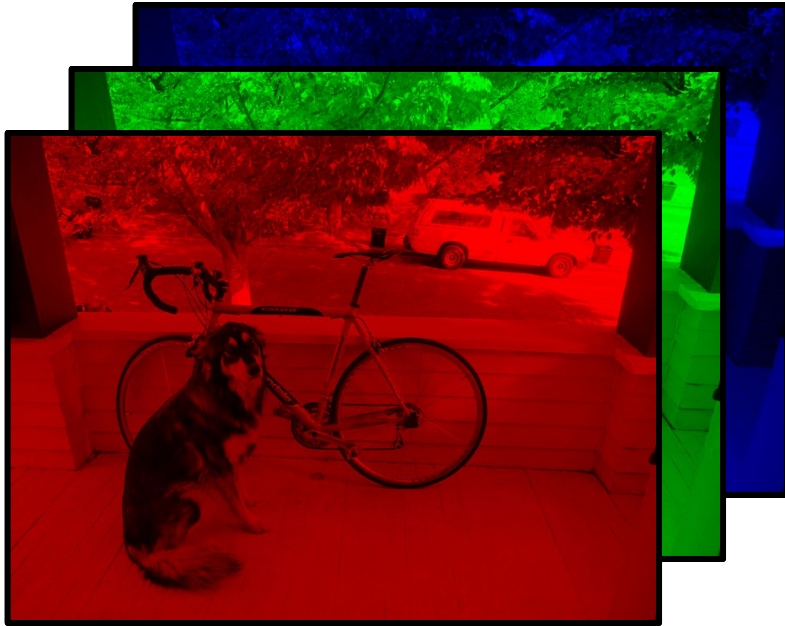
Histogram

- A histogram of a grayscale image is an array $H[*]$ of bins, one for each gray level
- $H[i]$ gives the count of how many pixels of an image have gray level i ($i=0\dots255$ or $i=0\dots1$)
- $P[i]$ (the normalized histogram) gives the percentage of pixels that have gray level i

Histogram



Histogram



R, G, B



R histogram

G histogram

B histogram



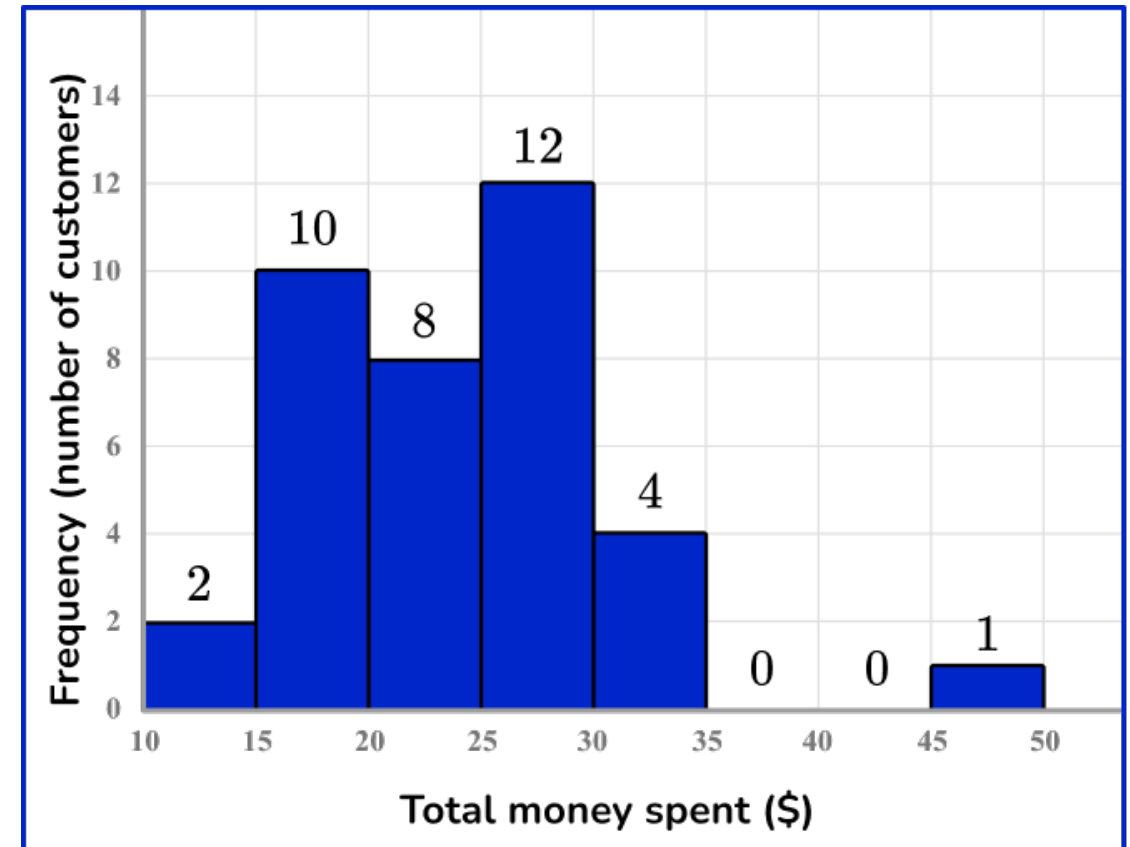
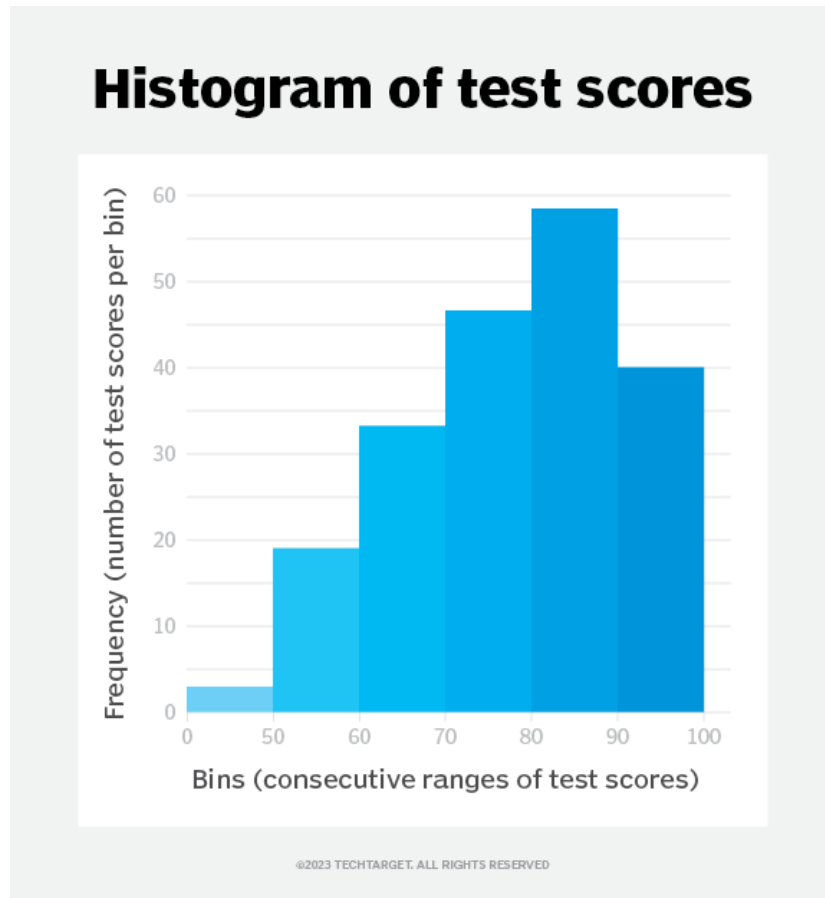
Gray image (V)



Histogram of gray values

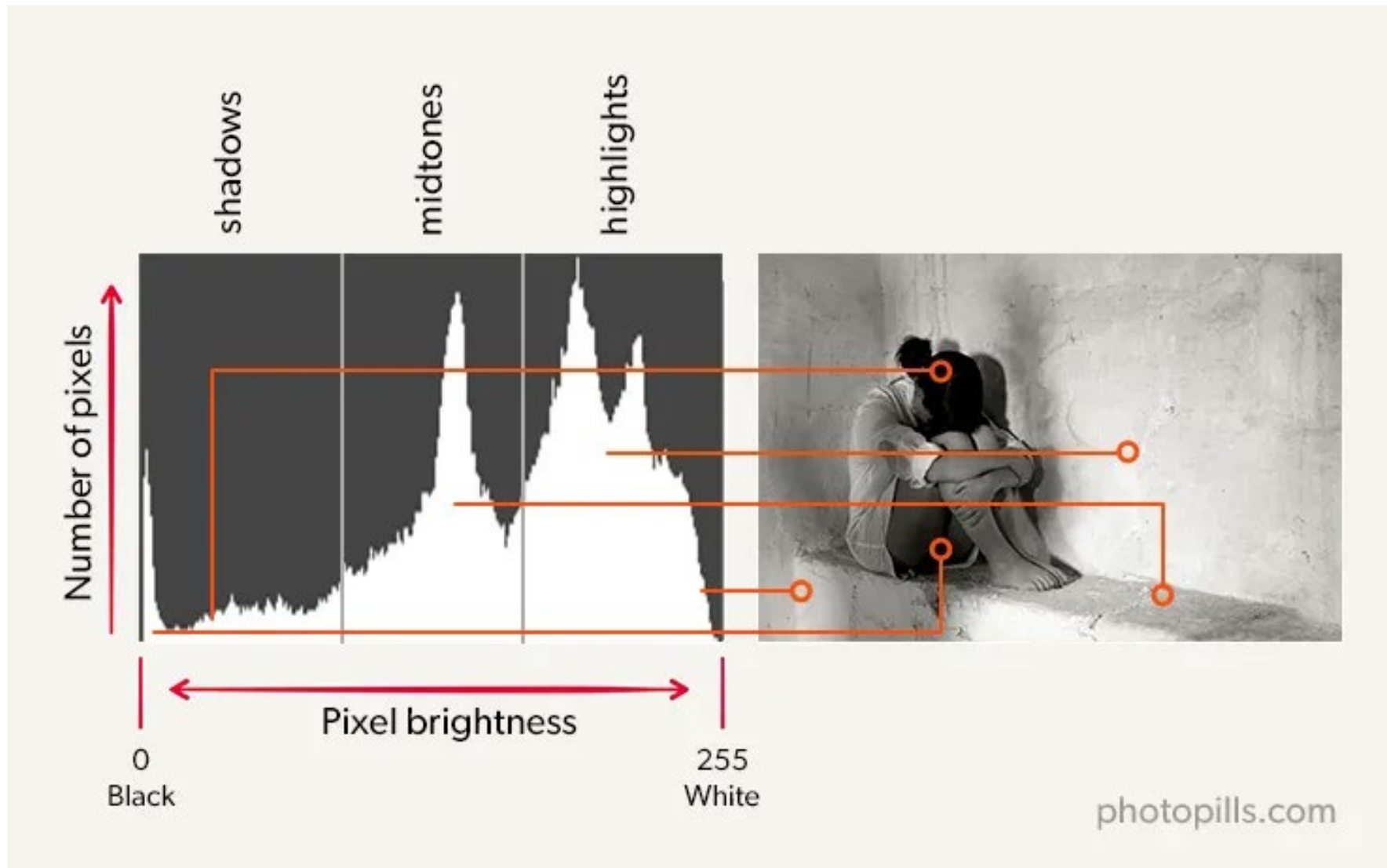
Histogram

- We can have histograms of many things, not just image intensities



<https://thirdspacelearning.com/>

Histogram: regions

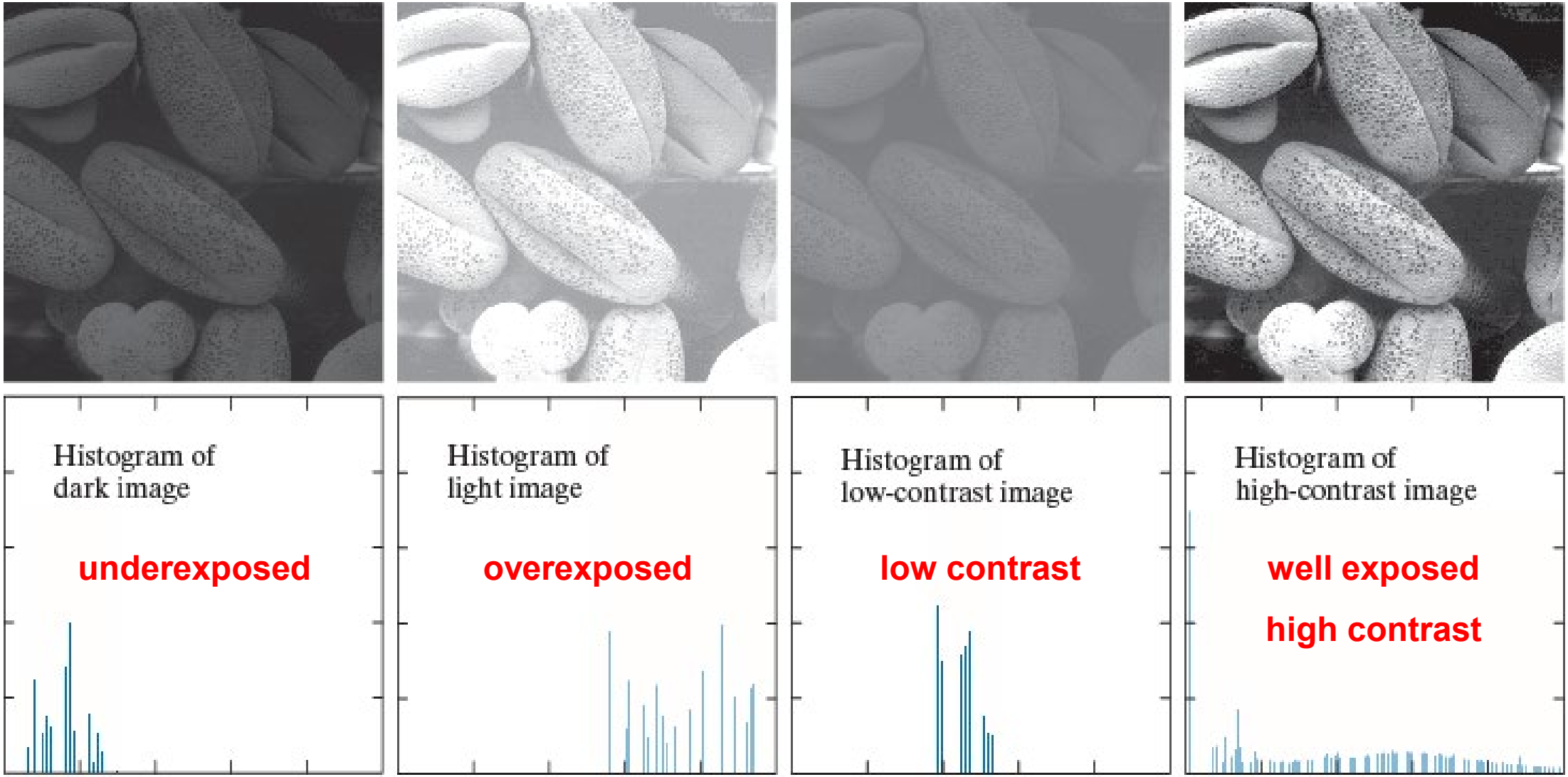


a b c d

FIGURE 3.16 Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.

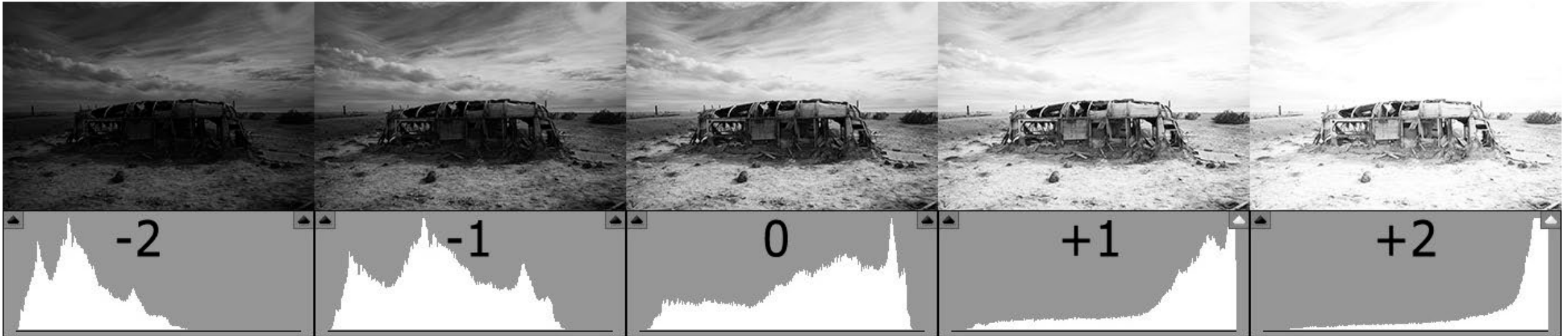
Histogram: how to read

- Goal: evenly spread-out histogram

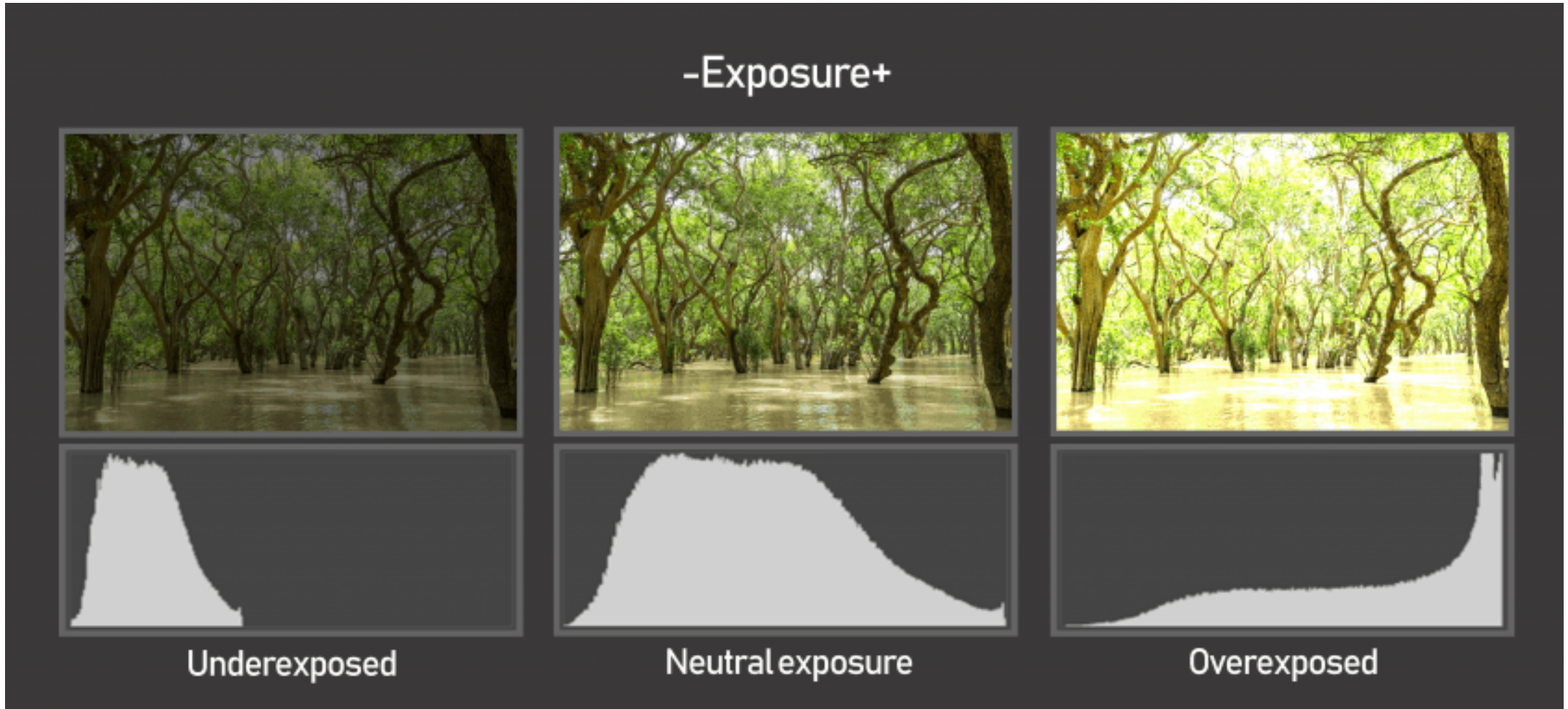


Histogram: how to read

EXPOSURE

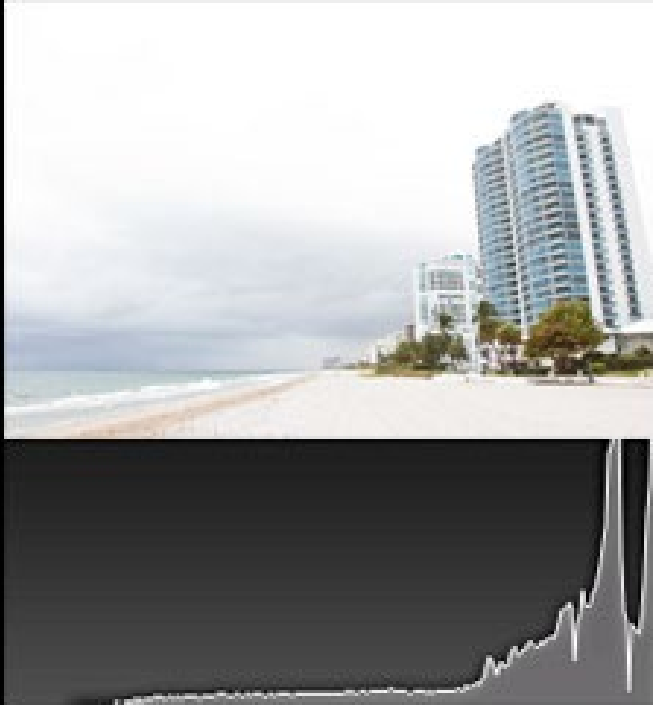


Histogram: how to read



Histogram: how to read

OVEREXPOSED



PROPERLY EXPOSED



UNDEREXPOSED



Questions?